# Essays in the Economics of Education

Damian Phelan

A thesis submitted in partial fulfilment of the requirements

for the degree of Doctor of Philosophy

Department of Economics

University College London

Primary Supervisor: Michela Tincani

Secondary Supervisor: Lars Nesheim

September 2026

# Declaration

## Part A: General Declaration

I, Damian Phelan, confirm that:

- This thesis is a presentation of original work.
- This work has not previously been presented for a degree or other qualification at this University or elsewhere.
- All input or assistance in the creation of the academic work other than from the supervisory team (including AI) has been acknowledged in Part B below.
- Any text or data in the thesis that has been presented for publication (including in review) elsewhere is declared in Part B below.

## Part B: Declaration of contribution from other sources

**Data sources, if not collected yourself.** Chapters 2 and 3 use published Ofsted inspection reports, school behaviour policy documents and school websites (all publicly available), the Department for Education's Get Information About Schools register, and published school performance data. The gold-standard school visit and headteacher interview data were collected by the candidate with a team of research assistants. Chapter 4 uses the Longitudinal Education Outcomes dataset, accessed through the ONS Secure Research Service, and published lifetime-returns estimates from Britton, Ogden, Ornadel and Waltmann (2026).

**Software code, instrumentation development, etc.** In Chapters 2 and 3, large language models are used as measurement instruments to score school documents; this methodological use, including the exact prompts and validation

against the gold standard, is documented in full in those chapters. AI assistance with the analysis code is declared under AI below.

**AI.** Generative AI (Claude, Anthropic) was used throughout the preparation of this thesis. It assisted with revising prose, writing analysis code in Python and Stata, preparing tables and figures, and checking the internal consistency of the numbers reported. All of this work was directed by the candidate, and every analytical decision, interpretation and piece of final wording was reviewed and approved by the candidate. For Chapter 4, the analysis was carried out by the candidate and co-authors inside the ONS Secure Research Service, where AI tools are not available. AI assistance for that chapter was therefore confined to revising the prose and preparing the tables and figures for the thesis.

**Other.** Not applicable beyond the contributions declared above.

**Publications.** All three papers are unpublished working papers. Chapter 4 is joint work with Arun Advani and Elaine Drayton; the candidate's contribution is stated in the Research Paper Declaration Form.

Date: 18 September 2026

# Abstract

This thesis measures two features of English secondary schools: their culture, and whether they offer a high-return subject.

The first paper measures culture. A research team that I led visited 103 schools. Each visit combined a day of observation with a headteacher interview, covering how warm and how strict the school is. Every school's inspection report, behaviour policy and website was then scored by a prediction model and by a language model applying a mark scheme. Scores from inspection reports agree modestly with the visit scores. Warmth is predicted from a report's ordinary wording. Strictness is found only by a language model applying a mark scheme that says which statements about behaviour to look for. What a headteacher says about the school agrees only weakly with what observers find, and a school's own documents agree with the head rather than the observers.

The second paper relates the visit scores to Progress 8. A standard deviation of warmth is associated with 0.31 of a standard deviation of Progress 8, and a standard deviation of strictness with 0.23. Headteachers' own ratings of strictness predict progress as well as the observers' scores do. Their ratings of warmth do not predict progress. The inspection-report scores exist for nearly every school in England, and the same models estimated with them give the same pattern.

The third paper, joint with Arun Advani and Elaine Drayton, shows that a school offers A-level economics only if it employs a suitably qualified teacher, and that during the period we study fewer than half of state schools did. When a school starts offering the subject, take-up rises by 5.7 percentage points, and taking it raises earnings at age 28 by around 14%. The extra tax raised from those earnings exceeds the cost of employing the teacher.

# Impact Statement

This thesis measures two features of English secondary schools that matter for pupils: the culture a school practises, and whether it offers a subject that leads to well-paid careers. The findings on each are relevant to policy.

The findings on school culture matter first for how schools are monitored. England introduced a new school inspection framework in 2025, and one question raised by that change is how much an inspection report tells parents about a school. A report's text carries some information about how warm and how strict a school is, but only a modest amount. A school's own behaviour policy and website carry no such information. They reflect what the headteacher believes, not what a visitor would see. Parents can therefore learn something about a school's culture from its inspection report, but not from the school's own documents. Inspection reports could also be written to say more about culture directly. The thesis also gives researchers a measure of culture for nearly every school in England, built from inspection reports and checked against the school visits.

The findings on school culture matter second for the public argument about how schools should manage pupils' behaviour. That argument usually assumes a choice between strict discipline and pupils' wellbeing. Warmth and strictness are separate features of a school, and pupils make the most progress at schools that have both. Warmth predicts pupil progress as strongly as strictness does, but only when it is measured by observers rather than by the headteacher. Headteachers describe their schools as warmer than observers find them. Judging a school's warmth therefore needs an outside observer, not the school's own account.

The third paper turns to A-level economics, the principal route into one of the highest-earning degrees. In the years the paper covers, fewer than half of English state schools offered the subject, usually because they had no qualified

teacher. The paper shows that hiring an economics teacher more than pays for itself. Pupils who take the subject go on to earn more, and so pay more tax. Over their careers, that extra tax comes to several times the full cost of recruiting, training and employing the teacher. Economics receives no teacher-training bursary, unlike other shortage subjects, and the paper provides a costed case for one. Schools without provision are concentrated outside London and the South East, so expanding provision would also broaden regional access.

All three papers have been presented at IFS and UCL seminars, and the third also at the Royal Economic Society Conference in 2024. Each will be submitted to a journal. Every participating school will receive a non-technical summary of the findings. All three papers were supported by the Institute for Fiscal Studies. The first two were also supported by the Stone Centre at UCL, and non-technical articles will appear on their websites. The third was also supported by the Discover Economics campaign and the Productivity and Diversity group at the London School of Economics.

# Research Paper Declaration Form

**Chapter 2:** Measuring School Culture: Warmth and Strictness in English Secondary Schools

*Publication status:* Working paper.

*Authors:* Damian Phelan

*Your contribution:* Sole author. I designed the study, managed the team of research assistants that helped to organise and conduct the headteacher interviews and school visits, developed the measurement framework, conducted all empirical analysis, and wrote the paper.

**Chapter 3:** School Culture and Pupil Progress: Evidence from English Secondary Schools

*Publication status:* Working paper.

*Authors:* Damian Phelan

*Your contribution:* Sole author. I constructed the linked school panel, specified and estimated all models, and wrote the paper.

**Chapter 4:** The Productivity and Diversity Costs of Constraints to Subject Choice

*Publication status:* Working paper.

*Authors:* Arun Advani, Elaine Drayton, Damian Phelan

*Your contribution:* Co-author. I contributed to the design of the study, conducted the empirical analysis, and wrote the paper with Arun Advani and Elaine Drayton.

# Acknowledgements

First, I have to thank my wife, Katherine. I am infinitely grateful to her for letting me scratch the itch of completing a PhD, and for sacrificing so much to help me get it over the finish line. I started this journey in September 2019, shortly after we married, and between then and now our household has expanded dramatically: first with the arrival of a beautiful and slightly crazy black labrador, Cleo, and then, more substantially, with the births of our two wonderful daughters, Orla and Maeve. Every day we spend together as a family brings me a joy I had rarely known before, and that is what has made it so hard to spend time away from all my girls working on this project. Through every holiday and weekend I spent beavering away, Katherine held the fort. I will be forever in her debt. Katherine, I love you so much.

I am also grateful to my parents-in-law, Sally Gaminara and Trevor Barnes, for all the hours they spent looking after Orla and Maeve so that I could work, and to my own family, my Dad, my brothers Garret and Jarlath and my sister Aisling, for their support and encouragement throughout.

I owe a great deal to the people who guided and supported the work itself. My primary supervisor, Michela Tincani, gave me calm, patient and thoroughly excellent feedback throughout, and I am very grateful to my secondary supervisor, Lars Nesheim, for his support. My co-authors on the third paper, Arun Advani and Elaine Drayton, guided me through that project so expertly, and I have learned a great deal from working with them. Several other academics at UCL have helped me along the way. Wendy Carlin's drive and clarity have been an inspiration. Parama Chaudhury, Cloda Jenkins and Peter Postl gave me brilliant guidance while I taught on ECON0002, the best module there is. At the Institute for Fiscal Studies I am grateful to Christine Farquharson, Tom Waters, Nick

Ridpath and Kate Ogden for their feedback and support. I am also grateful to Paul Johnson, Director of the IFS from 2011 to 2025, for his help in arranging headteacher interviews and school visits. Among my fellow PhD students, Hikaru Kawarazaki's unfailingly positive outlook and passion for research were a constant encouragement, and Matthew Nibloe's many helpful conversations got me to grips with econometrics.

The work in Chapters 2 and 3 was supported by the Institute for Fiscal Studies and by the Stone Centre at UCL. I am particularly grateful to Wendy Carlin and Imran Rasul, Co-Directors of the Stone Centre, for supporting and funding the data collection. The work in Chapter 4 was supported by the Institute for Fiscal Studies, the Economic and Social Research Council (ES/W010224/1), the Discover Economics campaign and the Productivity and Diversity group at the London School of Economics.

Finally, I am extremely grateful to the large number of research assistants who helped me with a range of tasks related to the work in Chapters 2 and 3: Abigail Lindon, Adiba Azman, Aken Li, Alannah Thornley, Amina Sheik, Ananya Pureti, Angel Chan, Anhelina Zavora, Anya Gray, Ashwin Ashwin, Benjamin Hannuksela, Cara Hayward, Clement Goy, Denis Balaj, Devansh Vyas, Devi Sankhla, Dina Asdaghi, Diya Daniel, Diya Karavdra, Ece Tumer, Edouard Mason, Ella Barry, Emily Glezer, Emma Spearman, Emma Wu, Florentia Loizidou, Hannah Evans, Jai Vasandani, Janice Goh, Judy Wang, Jun-Seo Kim, Kashvi Singh, Kavya Krishnan, Kym Anderson, Laboni Russell, Lena Curtet, Lottie Williamson, Lucy Sheppard, Luqa Ismail-Keyes, Lydia Whitehouse, Mahveen Ajid, Manav Mashru, Mariya Beletskaya, Maryam Hussain, Max Sears, Maya Munasinghe, Mehak Swani, Miella Hood, Millicent Cheung, Natalie Law, Neha Arvind, Nicolò Schütz, Noa Elovic, Oscar Wallington, Peggy Tang, Radha Patel, Raluca Rusu, Saahat Satyam, Saanvi Arya, Shagun Shrivastava, Shaurya Vinod, Shira Parnes, Shreya Rajesh, Shyam Patra, Siya Thakur, Tommaso Ongaro, Yanxi Liu, Yasmin Abdille, Yibo Wang and Yusra Binti Mohd Rosmadi.

*To my Mum. It is not quite a book, but I hope this is good enough.*

# Contents

**Declaration** **2**

**Abstract** **4**

**Impact Statement** **5**

**Research Paper Declaration Form** **7**

**Acknowledgements** **8**

**1 Introduction** **18**

1.1 Motivation . . . . . 18

1.2 Contribution to the Literature . . . . . 20

1.3 Overview of the Thesis . . . . . 22

**2 Measuring School Culture: Warmth and Strictness in English Secondary Schools** **24**

2.1 Introduction . . . . . 25

2.2 Data . . . . . 32

2.3 Measurement . . . . . 37

2.3.1 The gold standard . . . . . 37

2.3.2 Predicting the scores from text . . . . . 45

2.3.3 Scoring the text with mark schemes . . . . . 47

2.3.4 Judging both methods . . . . . 48

2.4 Results . . . . . 49

2.4.1 What direct measurement shows . . . . . 49

2.4.2 What the public text contains . . . . . 50

2.4.3 Which parts of a report the prediction models use . . . 55

2.4.4 What schools write about themselves . . . . . 57
2.4.5 A check against the parent survey . . . . . 59
2.5 Conclusion . . . . . 61
2A.1 Scoring Guide and Item Definitions . . . . . 64
2A.2 School Visit Protocol . . . . . 68
2A.3 Interview Guide . . . . . 78
2A.4 Inter-rater Reliability Tables . . . . . 85
2A.5 Sample Characteristics . . . . . 87
2A.5.1 Sample representativeness . . . . . 87
2A.6 Prediction Model Specification . . . . . 90
2A.7 What the Models Use: Examples . . . . . 91
2A.8 Further Parent-Comparison Checks . . . . . 99
2A.9 The Mark Schemes . . . . . 100

**3 School Culture and Pupil Progress: Evidence from English Secondary Schools 106**

3.1 Introduction . . . . . 107
3.2 Data . . . . . 111
3.3 Empirical Specification . . . . . 117
3.4 Results . . . . . 121
3.4.1 Estimates on the visited schools . . . . . 121
3.4.2 The national extension . . . . . 128
3.4.3 Robustness . . . . . 133
3.4.4 Exploratory results on traditional pedagogy . . . . . 140
3.5 Conclusion . . . . . 142
3A.1 How the Progress 8 Measures Are Calculated . . . . . 145
3A.2 Structural Stability of School Characteristics . . . . . 146
3A.3 Robustness in Detail . . . . . 149
3A.4 Benchmarks for the National Coefficients . . . . . 154
3A.5 The Traditional-Pedagogy Indicator . . . . . 155

3A.5.1 The rule the model was given . . . . . 156
3A.6 Additional Tables . . . . . 158

**4 The Productivity and Diversity Costs of Constraints to Subject Choice 170**

4.1 Introduction . . . . . 171
4.2 Data and Institutional Context . . . . . 177
4.3 Who gets to study economics? . . . . . 182
4.4 Effect on degree choice and lifetime earnings . . . . . 186
4.4.1 Empirical framework . . . . . 186
4.4.2 Earnings at age 28 . . . . . 191
4.4.3 Lifetime earnings . . . . . 193
4.4.4 The role of degree choice . . . . . 196
4.4.5 Who benefits most? . . . . . 198
4.5 The return on expanding teacher supply . . . . . 201
4.6 Conclusion . . . . . 207
4A.1 Additional Figures and Tables . . . . . 209

# List of Figures


2.1 Observed warmth and strictness in the visited schools . . . . . 50

2.2 The warmth model and the strictness scheme against the enacted scores . . . . . 51

2.3 Predicted warmth and parent-reported warmth, nationally . . . 60

2A.1 Regional distribution of interviewed and visited schools . . . . 88

3.1 Warmth, strictness and Progress 8 in the visited schools . . . . 127

3.2 The national distribution of strictness bands . . . . . 131

4.1 Pupil characteristics by school A-level economics provision . . . 188

4.2 Effect of A-level economics on earnings at age 28, by subgroup 199

4A.1 Patterns of A-level economics provision among switcher schools 210

4A.2 Share of schools offering A-level economics over time . . . . . 211

4A.3 Event study: take-up around provision changes . . . . . 212

4A.4 Event study: cohort attainment around changes in A-level economics provision . . . . . 213

# List of Tables


| | | |
|---|---|---|
| 2.1 | Distribution of schools across the three groups | 33 |
| 2.2 | Distribution of the sub-scores and scores | 39 |
| 2.3 | Variance components and score reliability | 41 |
| 2.4 | What each public text carries about enacted culture | 52 |
| 2.5 | Sentences that most move the models' scores | 58 |
| 2A.1 | Composition of the gold-standard sub-scores | 65 |
| 2A.2 | Inter-rater reliability: interview questions | 85 |
| 2A.3 | Inter-rater reliability: classroom observation items | 86 |
| 2A.4 | Inter-rater reliability: outside-of-lesson items | 86 |
| 2A.5 | Interviewed and visited schools against the national population | 89 |
| 2A.6 | The phrases with the largest weights | 98 |
| 3.1 | Progress 8 outcomes: summary statistics | 118 |
| 3.2 | Control variables: summary statistics | 119 |
| 3.3 | Four treatments of the inspection-grade control | 122 |
| 3.4 | Warmth and strictness alone and together | 123 |
| 3.5 | Stages 2 and 3 under the primary specification | 126 |
| 3.6 | Espoused culture and Progress 8 | 129 |
| 3.7 | The three national estimates | 130 |
| 3.8 | Robustness of the visited estimates | 135 |
| 3.9 | Timing checks on the visited-school estimates | 138 |
| 3A.1 | Structural stability of school characteristics | 148 |
| 3A.2 | Structural stability of the Progress 8 components | 149 |
| 3A.3 | Stage 1: total culture association, five outcomes | 159 |
| 3A.4 | Stage 2: teaching quality benchmark, five outcomes | 159 |

3A.5 Stage 3: culture net of teaching, five outcomes . . . . . 159
3A.6 Culture and curriculum entry . . . . . 160
3A.7 Progress 8 and Attainment 8 outcomes: summary statistics . . 161
3A.8 The national strictness bands on all five Progress 8 outcomes . 161
3A.9 The five visit sub-scores entered jointly . . . . . 162
3A.10 Item-level associations with Progress 8 . . . . . 163
3A.11 Linearity of the national strictness gradient . . . . . 164
3A.12 The authoritative-school typology . . . . . 164
3A.13 The espoused–enacted gap and Progress 8 . . . . . 165
3A.14 Culture coefficients on the 2024/25 stand-in outcome . . . . . 166
3A.15 Stability of Progress 8 and academisation . . . . . 167
3A.16 The SEMH composition mechanism . . . . . 168
3A.17 Pairwise correlations: culture scores and control variables . . . 168
3A.18 Sensitivity to unmeasured confounding . . . . . 169

4.1 Summary statistics by school economics offer behaviour . . . . 181
4.2 Access to and take-up of A-level economics by demographic group, 2011 cohort . . . . . 183
4.3 Effect of A-level economics on take-up, earnings, and lifetime income . . . . . 192
4.4 The role of degree choice . . . . . 197
4.5 Cost-benefit analysis of expanding economics teacher supply . . 205
4A.1 Effect of availability and school type on A-level economics take-up by subgroup, 2011 cohort . . . . . 214
4A.2 Access to and take-up of A-level economics by demographic group, 2003 and 2007 cohorts . . . . . 215
4A.3 Demographic composition of cohort, university students, and economics undergraduates . . . . . 217
4A.4 Access to and take-up of A-level economics by school type . . . 217
4A.5 School characteristics by A-level economics offer behaviour . . 218

4A.6 School transitions at age 16 and A-level economics access . . . 218
4A.7 Balance tests: effect of A-level economics availability on pre-treatment outcomes . . . 219
4A.8 Placebo test: IV estimates for students who do not take A-level economics . . . 219
4A.9 Robustness of IV earnings estimates to alternative specifications 220
4A.10 Effect of A-level economics availability on other educational outcomes . . . 221
4A.11 A-level subject combinations by economics availability . . . 222
4A.12 IV estimates by predicted propensity to take A-level economics 223
4A.13 Sensitivity of the main results to the availability threshold . . . 224
4A.14 School-cohorts contributing at each event time . . . 224

# Chapter 1

# Introduction

## 1.1 Motivation

Schools with similar pupils and similar budgets produce very different results. Part of the difference comes from what the schools do, not from which pupils they take. This thesis measures two of the things a school does: the culture of the school, and whether it offers a high-return subject. Culture is hard to measure, and the subject offer is rarely measured. The thesis measures both for English secondary schools, and asks what each one means for pupils.

The first two papers are about school culture. Nearly every school describes itself as warm, supportive and well ordered. The measures available to researchers and to policy do not settle whether that is true. An inspection grade reflects exam results as well as culture, so a good grade does not tell us whether a school has a good culture. A high exclusion rate could mean a school that enforces its rules firmly or one that has lost control, and the two cannot be told apart from the rate alone. Surveys record what people say rather than what a visitor would see.

The first two papers measure culture by observation instead. A research team that I led visited 103 schools, spent a structured day observing each one, and scored two dimensions of its culture: warmth, the relational quality of the school, and strictness, its behavioural structure. The two dimensions come from the parenting literature, where their combination — demanding and responsive at once — is the style that produces the best child outcomes. Separating them matters because the public debate about schools tends to treat them as opposites:

the case for strict schools is often put as a case against soft ones. Whether warmth and strictness are substitutes, complements or simply independent is an empirical question, and answering it requires measuring each on its own.

Direct observation produces credible scores, but it cannot be done for every school. Every school, however, produces text: an inspection report written about it, and a behaviour policy and a website written by it. The first paper asks how much of what an observer sees in a school can be predicted from the text that exists for every school, and the second asks what the observed scores predict. The second question depends on the first. The first paper checks the text-based scores against the visit scores. Only the scores from inspection reports agree with them. The second paper therefore uses the visit scores for the visited schools, and the inspection-report scores when it extends the analysis to every school in England. Scores from behaviour policies and websites are not used.

The third paper is about the subject a school offers. It is simpler to measure than culture. A school either offers the subject or it does not. Whether a pupil can take A-level economics — the usual route into one of the highest-earning degrees — depends on whether their school employs a teacher qualified to teach it, and during the period we study fewer than half of English state schools did. If a school does not offer the subject, its pupils cannot take it. The third paper measures what this constraint costs in lifetime earnings, establishes who it falls on, and asks whether the tax revenue from removing it would cover the cost of doing so.

The three papers share a method as well as a subject. Each measures something about schools for all, or nearly all, English secondary schools: the culture a school practises, how much of that culture appears in the documents written by and about the school, and whether a school offers a subject. The method combines direct measurement with administrative data. In the first two papers it also uses language models, whose scores are checked against the visits. The thesis shows that these features of schools can be measured for every school, and reports what

the measurements show.

## 1.2 Contribution to the Literature

The first paper contributes to two literatures. The school-climate literature has long used the parenting typology of demandingness and responsiveness to classify schools, and finds that pupils do best in schools that are both demanding and responsive; but it measures climate through surveys of pupils, parents or staff, which record perceptions rather than practice. The paper's first contribution is a gold standard built instead from direct observation. It draws an explicit distinction between what a school enacts and what its headteacher espouses, which the survey literature cannot draw. The distinction turns out to be large: headteachers' accounts of their schools differ from what observers record, and the difference is larger for warmth than for strictness.

The second contribution is to the text-as-data literature: the paper uses the documents that exist for every school as a source of measurement, scores them with a prediction model and with a language model working to a mark scheme, and validates both against the gold standard. The paper finds which documents contain information about a school's warmth and strictness, and which do not. Inspection reports contain information about both. Warmth is predicted by the words a report ordinarily uses. Strictness is found only in specific statements, such as that conduct standards are consistently enforced, which the model must be told to look for. Behaviour policies and websites contain no information about either that matches what observers record. Their scores agree with what the headteacher says about strictness, and with nothing about warmth. Of the three documents that exist for every school, only the one written by an inspector says anything about its culture.

The second paper contributes to the literature on school effectiveness. The most credible causal evidence on school culture comes from studies of high-expectation charter schools in the United States. Those schools combine several

practices at once, such as strict discipline, high expectations, extended time and intensive tutoring, so the studies estimate the effect of the whole package. This paper separates warmth from strictness and estimates each one's association with progress. In the visited schools the two scores are correlated at only about 0.36, so a school's warmth can be distinguished from its strictness. Both are associated with progress: a standard deviation of warmth with 0.31 of a standard deviation of Progress 8, and a standard deviation of strictness with 0.23. Schools in the top half on both dimensions make more progress than the rest, and their advantage is the sum of the two separate associations. Having both does not add anything beyond that sum. Controlling for the quality of the teaching observed on the visit does not reduce either association.

These estimates are associations, not causal effects. Schools that differ in culture also differ in other ways that are not observed: a headteacher who builds a warm culture may also be good at other things that raise progress, and parents who value a warm school may choose one for children who would progress well anyway. The estimates therefore show that pupils make more progress at warmer and stricter schools, not that changing a school's culture would change its pupils' progress. The paper's contribution is to measure culture by observation and to describe how it relates to progress. Warmth receives less attention in policy debate than strictness, but it predicts progress just as strongly, and this is only visible when warmth is measured by observation rather than self-report.

The third paper, joint with Arun Advani and Elaine Drayton, contributes to the literature on field-of-study returns. That literature measures the value of a subject by comparing pupils just above and below an entry cut-off. Our contribution is to measure it where the question is not who wins a place but whether the subject exists at the pupil's school at all. We use linked administrative data covering every pupil in England. Schools gain and lose the subject when a qualified teacher arrives or leaves, and we compare pupils at the same school in the years before and after such a change. We show that pupils at schools without the

subject do not take it, that those who miss it go on to lower-paying degrees and jobs, and that the schools without the subject are concentrated outside London and the South East.

The paper then costs the remedy: employing a qualified teacher at each school without the subject. One teacher is enough to let a school offer it. Pupils who take the subject go on to earn more, and so pay more tax over their careers. We compare that extra tax with the cost of employing the teacher. The return exceeds the cost several times over.

Together the papers make one argument in two settings. What a school does — the culture it enacts day to day, the subjects it puts on its timetable — is measurable, varies widely between otherwise similar schools, and predicts what its pupils go on to achieve.

## 1.3 Overview of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 develops the measurement framework. It constructs gold-standard warmth and strictness scores for 103 schools from school visits and headteacher interviews. It then asks how far those scores can be predicted from text that exists for every school: Ofsted inspection reports, behaviour policies and school websites. Two methods are used. A prediction model learns which words and phrases go with warmer or stricter schools, and a large language model scores each document against a mark scheme. Inspection reports predict both warmth and strictness, though only modestly, and in different ways. Scores from behaviour policies and websites match what headteachers say about their schools. They do not match what observers record on the visit.

Chapter 3 uses the observed scores as explanatory variables. It estimates the association between school culture and the progress pupils make over secondary school, measured by Progress 8, the Department for Education's official progress measure. Both warmth and strictness predict progress. One rival explanation is

that warm or strict schools simply have better teaching, and that the association is really with teaching quality. Adding the quality of the teaching observed on the visit as a control leaves both associations unchanged, so that is not what drives them. Warmth predicts progress only when it is measured by observation: replacing the observers' scores with the headteachers' own ratings removes the warmth association, whereas strictness is associated with progress under either measure. The text-based scores from inspection reports exist for nearly every English state secondary school. Estimating the same models with those scores, across all English state secondary schools, gives the same result: both warmth and strictness predict progress.

Chapter 4, joint work with Arun Advani and Elaine Drayton, turns from culture to curriculum. It shows that whether an English school offers A-level economics depends on whether it employs a suitably qualified teacher, compares pupils at the same school before and after the subject arrives or disappears to estimate the effect of taking the subject on degree choice and earnings, and computes the fiscal return to expanding the supply of economics teachers, which exceeds the cost of a teacher several times over.

# Chapter 2

# Measuring School Culture: Warmth and Strictness in English Secondary Schools

*Damian Phelan*[1]

***Abstract.*** *How warm a school is towards its pupils, and how strictly it runs, are widely thought to affect how much progress pupils make. This paper measures both, and asks how much of each can be inferred from the documents that every school produces. In 103 schools a research team that I led collected a gold-standard measure: a day of structured observation recorded what each school does, and an interview with its headteacher recorded what the head says it does. I then carried out textual analysis of every school's Ofsted inspection report, behaviour policy and website using two different approaches. Firstly, a prediction model learns, from the visited schools, which words and phrases are linked with warmer or stricter schools. Secondly, a large language model (LLM) is asked to use a one-page mark scheme to evaluate the document on each dimension on a 1 to 5 scale. There are four findings. First, inspection reports carry a modest signal of how warm a school actually is: the prediction model gives a correlation of* $+0.33$*. Second, we see a similar modest correlation of* $+0.25$ *for strictness measured by the LLM using the mark scheme. Third, what a headteacher says about their school is not what observers find there, and the gap is widest for warmth. The heads' ratings and the observers' scores correlate at* $0.25$ *for warmth and* $0.41$ *for strictness. Correcting both for measurement error raises them to* $0.42$ *and* $0.58$*. Fourth, schools' own documents match what their heads say about strictness and carry no signal about warmth.*

[1]Department of Economics, University College London, and Institute for Fiscal Studies. The school visits and headteacher interviews were approved by the UCL Research Ethics Committee (project 22979.001). I thank seminar participants at the Institute for Fiscal Studies and UCL for comments.

## 2.1 Introduction

How warm a school is towards its pupils, and how strictly it runs, are two of the things most often said to affect how much progress its pupils make. In this paper I construct measures of these two dimensions of school culture based on a structured day of observations at a school. I refer to these two dimensions as warmth and strictness. I then ask how well each dimension can be predicted from publicly available texts that exist for all 3,332 secondary schools in the country.

The two dimensions have precise meanings here. Warmth is the relational quality of a school: whether pupils are known, cared for and supported by the adults around them. In a lesson it shows in whether the teacher uses pupils' names, how praise is given and how the teacher and the class speak to one another. Between lessons it shows in how staff and pupils deal with each other in corridors and at break. It also shows in whether rewards are used consistently across the lessons of the day. Strictness is the school's behavioural structure: whether expectations of conduct are clear, enforced and met. In a lesson it shows in how much misbehaviour is tolerated before a sanction, how much disruption occurs and how pupils respond when they are corrected. Outside lessons it shows in how pupils arrive, how they move between lessons and how the dining hall runs. Strictness in this sense means clear rules applied consistently rather than harsh punishment. A school whose sanctions are applied by every member of staff in the same way is strict in this sense.

The two dimensions come from the parenting typology of Baumrind (1971) and Maccoby and Martin (1983), which classifies parenting by demandingness and responsiveness and finds that the combination of both, the authoritative style, produces the best child outcomes across cognitive and non-cognitive domains. Majumder (2016) and Zhang et al. (2017) find the same pattern in educational attainment: children of authoritative parents do best.

Pellerin (2005) applies the typology to schools, classifying each school as authoritative, authoritarian, permissive or indifferent based on its levels of demandingness

(academic and behavioural expectations) and responsiveness (support for pupils). She finds that pupils in authoritative schools have lower dropout and truancy rates. Gregory et al. (2010, 2011) use a similar typology and find that more authoritative schools have less bullying, less victimisation and lower suspension rates. Gottfredson et al. (2005) finds that a school's climate, the clarity and fairness of its rules and how consistently they are enforced (demandingness), predicts the level of disorder in the school even after accounting for the demographics of its pupils.

All of these studies rely on surveys of pupils, parents or staff and use slight variations on the demandingness and responsiveness language. This paper refers to the dimensions as warmth and strictness, as this is common terminology in the teaching profession, and measures them through direct observation.

What makes the two dimensions measurable in a school, as opposed to in a family, is that both are visible in public. A researcher sitting in a lesson can see whether names are used and how disruption is dealt with, and a researcher standing in a corridor can see whether pupils move calmly and whether staff speak to them. Neither requires access to anything private. I treat warmth and strictness as two separate dimensions rather than as the two ends of one scale, because a school can be warm without being orderly and orderly without being warm. Whether warm schools also tend to be strict, and strict schools warm, is an empirical question, and scoring the two separately lets the data answer it.

I also observed a third dimension, the quality of teaching practice. This is not part of culture as I define it, but it is the most obvious alternative channel for any association between culture and results, and it is kept in the analysis for that reason.

Warmth and strictness are not directly observed for schools in England. However, there is public text available for each school that may be informative about their culture. Every school has a report written by an external inspector, a behaviour policy written by the school, and a website written by the school. For

a smaller set of 103 schools this study adds data from direct observation of the school at work and from a structured interview with its headteacher. For a further 200 schools we have the interview alone. Those schools were not visited, so no observer scored them, but they widen the sample for one later test: whether what a school writes about itself matches what its headteacher says.

From the structured observation and interview data, I construct a gold-standard measure of each dimension. Because the interview and the visit disagree substantially about the same school, the gold standard is in fact two measures rather than one: an *enacted* score, from what the observers recorded, and an *espoused* score, from what the headteacher reported.

I then ask whether these scores could be predicted from a school's public text alone using two different methods. The first is a prediction model: using the visited schools, a statistical model learns which words and phrases of a document predict enacted scores, and it is validated only on schools that were not used to estimate it. Two things are done before estimation. First, identifying details are removed from the text. Second, each enacted score is regressed on three intake measures and the age of the inspection report, and the model predicts only the part of the score those variables do not explain. Without this step a model could predict the scores accurately just by recognising a school's intake from its report rather than by reading anything about its culture.

The second method uses a mark scheme approach. A large language model is provided with a one-page written mark scheme for each dimension — warmth, strictness, and teaching — and asked to score each document on a 1 to 5 scale. The first method assumes nothing about where culture shows up in text; the second assumes it is there and scores it.

Each source is read on its own rather than pooled with the others. The three documents have two authors: the school writes its behaviour policy and its website, and an inspector writes the report. Pooling them would mix what a school says about itself with what an inspector says about it, and this paper asks about the

two separately. There is a second reason. The language model applying a mark scheme reads one document at a time and gives it a band, so it cannot be given three documents at once. The prediction model is given one document at a time as well, so that both methods read the same input and their results can be compared. Pooling is reported later, for the prediction model on its own.

Each method gives every school a score. A set of scores is used for the national extension only if it agrees with what the observers recorded, and only if that agreement survives the checks described next. The two methods need different checks, because they can fail in different ways. A prediction model has many thousands of phrases to choose from, so it might reach some correlation by chance alone. A permutation test measures how much: the whole procedure is run again on scores shuffled between schools, which destroys any real link between a report and its score, and the real result has to beat what those shuffled runs produce. A mark scheme might succeed because of the particular language model applying it, or because it is following how long a document is rather than what it says, so its result has to survive a second scoring run, a different language model, and the removal of any effect of length. Across all the sources, methods and dimensions, many results are tested, so some would look convincing by chance alone. The significance levels are therefore corrected for the number of results tested, and a result counts only if it stays significant after that correction.

There are four main findings. The first is that inspection reports carry a modest but real signal of enacted warmth. A prediction model reading only the report's wording predicts the enacted warmth of schools it was not estimated on. Its predictions correlate with what the observers recorded at +0.33. When the model predicts the raw enacted warmth score, the correlation is +0.25. When it predicts only the part of the score that intake and report age do not explain, the correlation rises to +0.33. If the model were recognising intake from the report, removing intake from the target would lower the correlation. It does not, so the model's predictions rest on something other than intake. The phrases that

lower a school's predicted warmth differ in kind from those that raise it. The reports of schools with low enacted warmth use stock assurances such as "pastoral support is in place" and "pupils feel confident they have a trusted adult". The reports of warmer schools instead describe pupils and adults dealing with one another, spread across many small choices of phrase rather than concentrated in quotable lines. The same is true of observed teaching. Predicting each school from a model estimated without it, the model's predictions correlate with the observers' teaching scores at +0.28. The mark scheme applied to the same reports does not predict warmth: its bands correlate with enacted warmth at −0.07.

The second finding is that each method works for a different dimension: the prediction model for warmth, the mark scheme for strictness. The prediction model predicts enacted strictness poorly, at +0.19, where the mark scheme's bands correlate with it at +0.25. On warmth the order is reversed. The reason is that the two dimensions appear in the text differently. Warmth is associated with a report's general vocabulary, which a model can learn from the visited schools without being told in advance what warmth looks like. Strictness is associated with specific factual claims, such as "low-level disruption is rare". The mark scheme states in advance what a strict school's report should say, so the language model applying it is told which sentences to look for. The prediction model is given only the reports and the observers' scores, and must find the association from the reports alone. For strictness it fails to.

The third finding is that the enacted and espoused measures diverge, and that the divergence is stronger for warmth. The interview and the visit are correlated at only 0.25 for warmth, against 0.41 for strictness. Both measures contain random error. The error in one score is unrelated to the other score, so that part of each score cannot contribute to the correlation between them. The highest correlation the two scores could reach is therefore below one, and the raw figures understate how far the two agree. Correcting for the error in both raises warmth to 0.42 and strictness to 0.58. Two error-free measurements of the same quantity would

correlate near one. Both corrected figures fall well short of that, so the head's account and the observers' scores are measuring related but distinct quantities.

The fourth finding is that warmth and strictness divide the same way again in what schools write about themselves. Models estimated on a school's behaviour policy and website predict what its head says about strictness, correlating at +0.16, and more weakly what the head says about staff morale, at +0.13. On warmth they predict neither: +0.05 against the head's account, and no better against what the observers saw.

The paper contributes to various strands of the literature. The first is the school-climate literature, whose standard tool is the survey: pupils, teachers and parents are asked what their school feels like (Hoy and Miskel, 2001; Cohen et al., 2009; Thapa et al., 2013). Schools that come out better on these surveys have better wellbeing, attendance and attainment. But a survey records what people say about their school, not what the school does, and people tend to describe their own school kindly. Van Houtte (2005) makes the distinction directly: surveys measure climate, the sum of what members perceive, while what shapes behaviour is culture, the shared norms of the school. It is culture in that sense that this paper measures. The cheaper alternatives to a survey do not measure it either. Exclusion rates (Machin and Sandi, 2020) reflect who a school's pupils are as much as how the school treats them, and the inspection grade compresses everything a school does into a single verdict. What is missing is a measure of practice, taken by watching schools rather than asking them, that could in principle be extended to every school. That is what the paper builds.

The second is the school-effectiveness tradition that begins with Coleman et al. (1966) and Rutter et al. (1979). Its question is why some schools do better than others with similar pupils. Its answer was not resources but what Rutter called ethos: the expectations a school holds and the practice it keeps up day to day. Later studies agreed on what a good ethos involves, and the list pairs an orderly environment with a supportive climate (Edmonds, 1979; Sammons et al., 1995;

Teddlie and Reynolds, 2000). That pair is close to the strictness and warmth measured here. But the tradition has studied ethos through case studies, one school at a time, so its central idea has never been measured in a way that lets schools be compared. The strictness and warmth scores do that for its two main ingredients.

Structured classroom observation, in particular the CLASS framework (Pianta et al., 2012), measures emotional support and classroom organisation directly and predicts pupil gains at secondary level (Allen et al., 2013); observation instruments from the same family predict gains in English primary schools (Sammons et al., 2008). The observation forms I use draw on those two domains, but they aggregate across lessons and out-of-lesson settings so as to characterise a school rather than a teacher. To my knowledge no study has combined such observation with a leadership interview and public text in order to separate what a school does from what it says.

The literature on high-expectation, high-structure charter and academy models (Abdulkadiroğlu et al., 2011; Angrist et al., 2010; Dobbie and Fryer, 2013; Fryer, 2014) finds large effects of a bundle of practices in which strict behaviour norms and warm, individual relationships come together. It is the origin of the strictness dimension and the reason to expect it to matter, but it has not measured the two dimensions apart.

Finally, the text-as-data literature in economics (Gentzkow et al., 2019) supplies the tools for reading public text at scale. This paper uses both of that literature's modern tools, supervised text regression and mark schemes applied by a large language model (Gilardi et al., 2023; Eloundou et al., 2024). It tests each against direct observation and reports which dimension of culture each can read. Ludwig et al. (2026) argue that a language model's measurements cannot be trusted on their own and must be checked against a gold standard. This paper carries out exactly that check for one concrete measurement problem, and goes one step further by running the statistical alternative against the same gold standard.

Against those literatures, the paper contributes three things. The first is a measure of enacted culture built from observation rather than perception; the perception data from headteachers and parents are kept as comparisons against it. The second is a model that predicts enacted culture in every English state secondary school. The third is a comparison of the two ways economists now read text, prediction models and mark schemes, with both judged against the same enacted scores for the same schools. The comparison shows that neither method is better outright: which one works depends on what it is being asked to measure.

The next section describes the data. The paper then constructs the gold standard, describes how the two text measures are built and judged, reports what the measures show, and concludes.

## 2.2 Data

The paper uses four kinds of data on English state secondary schools: a structured observation from a visit and a headteacher interview that were carried out for this study, three collections of public text, a national survey of parents, and the Department for Education's school-level administrative data. The sources divide by whose account of the school they give. The visit records what trained observers saw the school do, and the inspection report records what an inspector saw, so these two are sources on *enacted* culture. The interview, the behaviour policy and the website record what the school says about itself, the first to a researcher and the other two in writing, so these three are sources on *espoused* culture. The parent survey belongs to neither group. Parents report their own family's experience of the school, and whether that experience follows what observers see or what the school says about itself is one of the questions the paper answers.

Schools fall into three groups by the data they carry. In 103 schools there is a visit, an interview and the public text. In a further 200 schools there is an interview and the public text but no visit, so 303 schools carry an interview score in all. Headteachers gave verbal consent to the interview and the visit, and schools

are not named. For the rest of the schools in the country there is the public text alone. Table 2.1 compares the three groups on intake, size and inspection grade.

Table 2.1: Distribution of schools across the three groups

| Group | *N* | FSM% | EAL% | Median size | Outstanding% | Good% |
|---|---|---|---|---|---|---|
| Visited and interviewed | 103 | 29.2 | 27.8 | 1,118 | 25 | 65 |
| Interviewed only | 200 | 27.2 | 17.9 | 1,052 | 15 | 75 |
| Rest of the national population | 3,028 | 30.1 | 18.2 | 1,050 | 14 | 70 |
| All schools | 3,332 | 29.9 | 18.4 | 1,051 | 15 | 70 |

*Notes:* The groups are mutually exclusive: the first received both a headteacher interview and a researcher visit, the second an interview only, and the third is the rest of the national population. FSM%: mean proportion eligible for free school meals. EAL%: mean proportion with English as an additional language. Size: median number of pupils on roll. Outstanding%/Good%: percentage rated Outstanding or Good at most recent inspection, among schools with a grade. The groups sum to one fewer than the total: one interviewed school whose head left the statement battery blank belongs to the total but to no group.

**Visits and interviews.** Between November 2023 and April 2025 the research team, working under my direction, interviewed the headteacher at 304 schools and visited 103 of them. In eight cases the head was unavailable and we interviewed the senior leader responsible for behaviour and culture instead.[2]

Schools were recruited by approaching headteachers directly by email and telephone, and by referral from schools that had already taken part. No quota was applied by region, school type or inspection grade. Three-quarters of the interviews and half of the visits took place in the 2024/25 academic year. At the end of every interview the headteacher was asked whether the school would be willing to host a visit, which means that the visited schools are a self-selected subset of the interviewed ones.

A visit lasted a school day. 71 of the 103 visits were made by a team of two or three researchers, and 32 by a single researcher. What single-researcher visits

[2]The counts that follow differ by a school or two from place to place, for four reasons which I state once here. One of the 304 heads left the statement battery blank, so 303 schools carry an espoused score and the interviewed-only group numbers 200; that school belongs to neither group in Table 2.1. Of the 103 visited schools, two have no scored inspection report and two have no published parent survey, so correlations and prediction models that use those sources run on 101 schools. The demographically adjusted correlations run on 100, because one school lacks a control variable. Among the interviewed schools, between 299 and 302 have each text source. Finally, one visited school and three interviewed schools could not be matched to the 2023/24 characteristics panel used in Section 2A.5, which therefore describes 102 and 300 schools.

mean for the scores is examined in the reliability section below and in the next chapter's robustness checks. The school was asked in advance to arrange four to six lessons spread across year groups and subjects, and it chose the particular lessons itself. The team observed those lessons, typically five of them, spread across Years 7 to 11. The researchers knew which school they were visiting but not its inspection grade, and the school knew it was being observed but not which items were being rated. Each school also allowed us to observe pupils moving between lessons, and pupils and staff interacting at break and lunch, which is what the outside-of-lessons form rates. Each observation covered either a whole lesson or half of one, and every researcher present rated it independently on the lesson observation form. The team also watched arrival to lessons, movement between lessons, break and lunch, and at the end of the day each researcher completed the outside-of-lessons form once, summarising the whole day.

The lesson form has twenty items; the mapping from the printed form to the scored items is in Section 2A.1. Ten of the twenty relate to warmth and strictness, for example the frequency with which the teacher uses pupils' names, and the level of misbehaviour tolerated before a first sanction. The other ten relate to teaching practice, such as questioning, feedback, explanation and structure. The outside-of-lessons form has thirteen items: nine on the school outside the classroom (sanctions and rewards, corridors, arrival, the dining hall, and interactions and relationships at break), two on the condition of wall displays, and two in which the observer compares the teaching seen across the day. Every item is rated from 1 to 5 and the item ratings are averaged across researchers and across lessons to give one value per school.

The observers saw the lessons the school chose to show them. What they scored may therefore be closer to the school's best lessons than to its typical ones. But the likely direction of the problem is known. A school will tend to show a visitor its better lessons. If every school does this, every school's enacted score is a little too high, and the scores are more alike than the schools really are. Scores

squeezed together make differences between schools harder to find, not easier, so this bias works against the paper's findings rather than for them. One version of the problem remains open: some schools may be better at putting on a show for a visitor than others. That difference could push the scores in either direction, and nothing in the data rules it out.

The interview lasted about an hour and followed a fixed guide. It had three parts. The first part consisted of twelve open questions on the school's approach to behaviour, relationships, wellbeing and teaching. This was followed by ten statements on which the headteacher rated their own school on a five-point agreement scale (for example "The sanction systems are applied consistently by all staff" and "Pupils feel like their teachers care"). The final part was a checklist of ten named behaviour systems (centralised detentions, an internal exclusion room, a mobile phone ban and so on) recorded as present or absent. Two researchers attended every interview, one asking the questions and one observing, and both scored the open questions independently. The observation forms and the interview guide are reproduced in full in Sections 2A.2 and 2A.3, and the agreement between researchers on the observation and interview items is reported together with the measures below.

**Public text.** Three collections of text cover the schools in the national population. A scored inspection report exists for 3,272 of them (98.2 per cent); the rest had opened too recently to have been inspected. A behaviour policy document exists for 3,294 (98.9 per cent), one document per school, taken from the school's website. About one document in ten is a trust-wide policy adopted word for word by the school. Website text exists for 3,284 schools (98.6 per cent). This text is taken from the home, values and welcome pages. The counts differ by a few dozen schools because a handful of schools have no report, no policy or no website; Section 2A.5 gives the reconciliation.

**Parent survey.** Ofsted's Parent View survey is open to the parents of every school and is published once enough parents have responded. I use the most

recent release for each school and the two items closest to warmth: the share of responding parents who agree that their child is happy at the school, and the share who agree that their child feels safe there. The two shares are averaged into one parent-reported warmth measure, which is available for 3,255 schools in the panel and for 101 of the visited schools. Parents choose whether to respond, and the typical release rests on about 120 responses at a response rate of about 12 per cent, so the measure is the view of the parents who chose to give one. Responses also accumulate over several years, so for some schools the parent data are older than the culture data. The survey has no item on strictness.

**Administrative data.** All schools are matched on their unique reference number to a school-year panel that I built from the Department for Education's published tables for 2015/16 to 2024/25. The panel carries pupil composition, prior attainment, Progress 8 (a value-added measure) and its components, workforce and finance. This paper uses the panel only to describe the samples; the following chapter uses it for outcomes and controls. The panel also carries each school's most recent overall inspection grade. Inspections from September 2024 onwards no longer award an overall grade, so the grade used throughout is the last one each school received before that date, on the scale 1 (Outstanding) to 4 (Inadequate). Schools inspected only since then, or never inspected, have no grade and are missing on that variable.

**Representativeness.** Table 2A.5 compares the visited and interviewed schools with the national population. Both samples resemble the population on school type, phase, admissions, size and urban location, and they differ from it on deprivation by at most two percentage points. Both over-represent London (45 per cent of visited schools against 15 per cent nationally), which shows up as a higher share of pupils with English as an additional language. Both also over-represent stronger schools: 25 per cent of visited schools held an Outstanding inspection grade as at August 2024 against 15 per cent nationally, and their mean Progress 8 in 2018/19 was +0.17 against +0.04. The mismatch on those dimensions is what

one would expect from recruiting through headteachers, since the schools most likely to agree to take part are those that are confident about what a visitor will find, and it has two consequences. The first is that estimates from the visited sample apply most directly to a population that is somewhat stronger than the national average. The second is a restricted range: the sample under-represents the weakest schools, so both culture and outcomes vary less than nationally.

## 2.3 Measurement

This section builds the measures starting with the gold-standard scores from the visits and the interviews, and reports how reliable they are. It then builds the prediction model and describes the mark scheme and LLM approach that were both used to produce scores for the schools that were not visited. It ends with how both approaches are judged.

### 2.3.1 The gold standard

The scores are built up in two steps. Every observation item and every interview statement is rated from 1 to 5. First, items are averaged into *sub-scores*, each of which covers one dimension in one setting: warmth in lessons, warmth outside lessons, strictness in lessons, strictness outside lessons, and so on. Second, the sub-scores for a dimension are averaged into that dimension's *score*, so that each school ends up with one warmth score and one strictness score on the same 1–5 scale as the items.

Two sets of scores are constructed and kept apart. The *enacted* scores use the observations alone. Enacted warmth is the average of in-lesson warmth (the use of names, praise, the quality of interaction, pupil participation and motivation) and out-of-lesson warmth (relationships and interactions during transitions and breaks, and how consistently rewards are used). Enacted strictness is the average of in-lesson strictness (misbehaviour tolerated, disruption, response to sanction,

respect and concentration) and out-of-lesson strictness (how consistently sanctions are used, corridors, arrival, the dining hall and recreational space). Enacted teaching is the average of the ten lesson items on teaching practice.

The *espoused* scores use the headteacher's statement ratings alone. Espoused warmth is the average of the three warmth statements (rewards applied consistently; pupils feel safe; pupils feel their teachers care) and espoused strictness is the average of the four strictness statements (sanctions applied consistently; behaviour in lessons; behaviour outside lessons; teachers feel respected). The two remaining scored statements, on staff turnover and morale, form a staff-climate score. The item definitions and the mapping of items to sub-scores are given in Section 2A.1.

To make the construction concrete, consider the in-lesson strictness sub-score of a single school. In each of five lessons, two researchers rated the level of misbehaviour tolerated before a sanction, the frequency of disruption, the pupils' response to correction, how respectfully they treated the teacher, and how well they concentrated. Misbehaviour and disruption are reverse-scored so that a higher value means a stricter lesson.

One item needs a rule: if no pupil was sanctioned during a lesson there is no response to correction to rate, and the researcher records a zero. That zero is not a low score, and dropping it would throw away what the lesson shows. It is therefore replaced by the reverse of the disruption rating, which means that a lesson with no disruption and no sanction is scored as if the pupils had responded fully, and a lesson with constant disruption and no sanction as if they had not responded at all.

The five ratings a researcher makes in a lesson are averaged to give that researcher's value for the lesson, the lesson values are averaged across lessons, and the researchers' values are averaged across researchers, giving one in-lesson strictness figure on the 1–5 scale. That figure is averaged with the out-of-lesson strictness figure to give the school's enacted strictness.

Espoused strictness is simpler: the headteacher's ratings on the four strictness

statements are averaged. A similar process produced espoused warmth. Every school that received a visit has an enacted score, every school that gave an interview has an espoused score, and 103 schools have both. Table 2.2 reports the distribution of every sub-score and score across the visited schools.

Table 2.2: Distribution of the sub-scores and scores

| Score | Description | $N$ | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|---|
| $W1$ | In-lesson staff–pupil warmth | 103 | 3.24 | 0.45 | 1.92 | 3.22 | 4.47 |
| $W2$ | Warmth in transitions and breaks | 103 | 3.13 | 0.63 | 1.75 | 3.12 | 4.25 |
| $W3$ | HT espoused warmth (statements) | 103 | 4.20 | 0.47 | 2.33 | 4.33 | 5.00 |
| $S1$ | In-lesson behaviour management | 103 | 3.69 | 0.49 | 2.12 | 3.70 | 4.75 |
| $S2$ | Out-of-lesson behaviour management | 103 | 3.33 | 0.50 | 2.10 | 3.30 | 4.60 |
| $S3$ | Behaviour management systems (count) | 103 | 3.11 | 0.73 | 0.62 | 3.12 | 4.38 |
| $S4$ | HT espoused strictness (statements) | 103 | 3.95 | 0.58 | 1.75 | 4.00 | 5.00 |
| $T1$ | In-lesson teaching quality | 103 | 3.48 | 0.42 | 2.49 | 3.51 | 4.51 |
| $SC$ | HT espoused staff climate (statements) | 103 | 3.91 | 0.78 | 1.50 | 4.00 | 5.00 |
| Warmth | *enacted* | 103 | 3.18 | 0.47 | 2.12 | 3.20 | 4.36 |
| Strictness | *enacted* | 103 | 3.51 | 0.43 | 2.42 | 3.52 | 4.48 |
| Teaching | *enacted* | 103 | 3.48 | 0.42 | 2.49 | 3.51 | 4.51 |
| Warmth | *espoused* | 103 | 4.20 | 0.47 | 2.33 | 4.33 | 5.00 |
| Strictness | *espoused* | 103 | 3.95 | 0.58 | 1.75 | 4.00 | 5.00 |
| Staff climate | *espoused* | 103 | 3.91 | 0.78 | 1.50 | 4.00 | 5.00 |

*Notes:* The visited schools with full data ($n = 103$); sub-scores and scores are on the same 1–5 scale as the items. The enacted scores are built from the visit sub-scores ($W1$, $W2$, $S1$, $S2$, $T1$) and the espoused scores from the interview statement sub-scores ($W3$, $S4$; $SC$ is the staff-climate score), as set out in Equations (2A.1) and (2A.2). The systems count $S3$ enters no score. No sub-score enters both sets.

There are two larger exclusions from the interview. The first is the ratings of how fully the headteacher answered each open question. They correlate with the enacted scores at only +0.09 for warmth, +0.01 for strictness and +0.03 for teaching, and no more strongly with the head's own ratings (+0.06 and +0.05). They therefore appear to record how fully the head talks rather than what the school is like, and they are excluded. The second is the count of the named behaviour systems (S3 in the tables), which counts eight of the ten and excludes the two reward systems. It correlates with enacted strictness at −0.01. In other words, the systems a school names are not a guide to how strictly it runs.

Agreement between researchers is moderate at the level of a single item, and the school-level scores built from many items, lessons and raters are reliable enough to use, as I show below. Of the 517 lessons observed, 292 were rated independently

by two researchers and 59 by three; in the remaining 166 only one researcher was available for that lesson, and it carries one rating. The outside-of-lessons form was completed by two or more researchers on 71 of the 103 visits, and by one researcher on the other 32. A researcher also left an item blank when they did not see what it asks about, so for any one item the number of visits with two ratings to compare is smaller than 71.

Where two raters saw the same lesson, the pairwise Pearson correlations across the classroom items range from 0.53 to 0.79; on the outside-of-lessons form, completed once per day, they range from 0.47 to 0.81; and the two raters are within one point of each other close to 90 per cent of the time. Tables 2A.3 and 2A.4 report the agreement item by item. Agreement is highest on the items whose scale points are described most concretely on the form, namely the consistency of the reward system (0.81, a day item) and the frequency of disruption (0.79), and lowest among the classroom items on the teaching-practice ratings that ask the observer for the most interpretation, such as the use of resources and the choice of methods (0.53 and 0.55).

However, the paper's claims do not rest on the agreement of two raters on one lesson. They rest on the reliability of a school's score once it has been averaged over its lessons and raters. Table 2.3 gives this. Take every rating that a researcher gave a lesson for in-lesson strictness: one number, for one lesson, from one researcher. Those ratings differ from one another for three reasons. The first is that the lesson was in a stricter or a less strict school, which is the difference the measure is meant to capture. The second is that, within the same school, that lesson was calmer or rowdier than the school's other lessons, because it had a different teacher, a different class or a different period of the day. The third is that, looking at the same lesson, one researcher scored it a little higher than the other.

The table divides the spread of the ratings into those three shares. For in-lesson strictness the shares are 21 per cent school, 61 per cent lesson and 18 per cent

rater. This means that well over half of the variation in single-lesson strictness ratings is lessons within the same school differing from one another, only about a fifth is schools differing from each other, and under a fifth is the two researchers disagreeing. For in-lesson warmth the shares are 21, 38 and 42 per cent. Lessons within a school vary less in warmth than in strictness, but the researchers disagree more about warmth, which matches the item-level pattern described above.

Table 2.3: Variance components and score reliability

| | *Share of variance (%)* | | | | |
|---|---|---|---|---|---|
| Sub-score | School | Lesson | Rater | Design | Reliability of school mean |
| In-lesson warmth ($W1$) | 21 | 38 | 42 | 4.7 lessons × 1.6 raters | 0.60 |
| In-lesson strictness ($S1$) | 21 | 61 | 18 | 4.7 lessons × 1.6 raters | 0.58 |
| Teaching practice ($T1$) | 27 | 40 | 33 | 4.7 lessons × 1.6 raters | 0.67 |
| Out-of-lesson warmth ($W2$) | 79 | — | 21 | 1.6 raters | 0.85 |
| Out-of-lesson strictness ($S2$) | 71 | — | 29 | 1.6 raters | 0.79 |

*Notes*: Lesson sub-scores are decomposed into between-school, between-lesson-within-school and between-rater-within-lesson components; the outside-of-lessons sub-scores, rated once per rater per school, into between-school and between-rater. Reliability is the share of variance in a school's mean score, over its own lessons and raters, that is between schools. Method-of-moments components from nested one-way analyses of variance on 517 lesson observations in 103 schools (lesson sub-scores) and 103 schools (outside-of-lessons sub-scores); design sizes are harmonic means. Reliability is $\sigma^2_{\text{school}}/(\sigma^2_{\text{school}}+\sigma^2_{\text{lesson}}/m_l+\sigma^2_{\text{rater}}/m_l m_r)$. The in-lesson and out-of-lesson halves of the enacted score, rated in different settings, are correlated at $r = 0.52$ for warmth and 0.53 for strictness across schools.

One implication of dividing the variation into those three shares is that averaging matters. A single lesson is a poor guide to its school, but when five lessons and the ratings of the researchers who saw them are averaged, the lesson-to-lesson and rater-to-rater variation is greatly reduced. The reliability of the in-lesson warmth score, averaged over a school's own lessons and raters, is 0.60, and that of the in-lesson strictness score is 0.58. The outside-of-lessons sub-scores, which summarise a whole day, are more reliable (0.85 and 0.79).

Each enacted score combines two scores: a lesson score, from the observed lessons, and an outside-of-lessons score, from the rest of the school day. The two are correlated at 0.52 for warmth and 0.53 for strictness across the 103 schools. Treating them as two readings of the same score, the reliability of the combined score is that correlation stepped up for the doubling of length. This puts the reliability of each combined enacted score at about 0.69 for warmth and 0.70 for strictness. The reliability is the share of the score's variance that is not error. A

score correlates with its own error-free value at the square root of that share, and no other measure can correlate with the score more highly than that. For enacted warmth this limit is $\sqrt{0.69} \approx 0.83$.

These figures rest on an assumption about why the lesson score and the outside-of-lessons score differ. They do not agree perfectly, and the figures above treat all of that disagreement as measurement error. That is the right treatment if lessons and the rest of the day are two readings of one and the same thing. But a school might genuinely be warmer in its lessons than in its corridors. If so, part of the disagreement is real rather than error, and the reliabilities are 0.82 and 0.79 instead. The corrections below use the lower figures. Dividing by a lower reliability gives a larger corrected correlation, so every corrected correlation in this chapter is the most generous version of that number: the true agreement could be lower than the corrected figure, but not higher.

One further limitation: each school was visited once. If a school is different on a different day, that difference cannot be separated from the school itself. These are reliabilities of the school as seen on its visit day.

The lessons that carry only one rating lower the reliabilities very little. Had every lesson been rated twice, the in-lesson reliabilities would have been 0.62 and 0.59 rather than 0.60 and 0.58. This is because most of the variation within a school is between lessons, not between raters, as the shares above show. A second rater on every lesson would reduce only the smaller of the two. They do, however, carry one caution. Single-rated lessons were rated lower on average than double-rated ones. The difference, single minus double, is $-0.24$ on warmth and $-0.15$ on strictness on the 1–5 scale. The 35 schools with such lessons have slightly lower scores as a result: the share of a school's single-rated lessons correlates with its in-lesson warmth at $r = -0.25$. The following chapter therefore checks that its estimates do not depend on those schools.

The division of the variation into a school share, a lesson share and a rater share also says something about the schools themselves, not just the measurement.

Behaviour varies more from classroom to classroom within a school than it does between schools. A school's enacted score is therefore an average over quite different lessons, and a visit of five lessons measures that average with error. Random error in a score adds to its variance without adding to its covariance with any other measure. Every correlation with the enacted scores is therefore lower than it would be without error, so the figures reported below are, if anything, too small.

Two corrections give a sense of how much the error lowers the correlations. The first correction divides the raw enacted–espoused correlation by the square root of the product of the two scores' reliabilities. The result is the correlation the two scores would have if neither contained error. It raises warmth to about 0.42 and strictness to about 0.58. Two error-free measurements of the same quantity would correlate near one, so even after the correction the head and the observers are far from describing the same school. The second corrects the mark scheme's strictness correlation with the enacted score, +0.30, for the enacted score's error alone, and it rises to about 0.36. In both cases the corrected number leads to the same verdict as the raw one, so the uncorrected figures are the ones reported throughout.

The two enacted scores are correlated at $r = 0.62$ across the 103 visited schools, and the two espoused scores at 0.55. Taken at face value, the enacted figure says that warm schools are usually strict ones. Part of that figure, though, comes from how the scores were made rather than from the schools. The same observers scored both dimensions at the same school on the same day, and an overall impression of a school formed on the day can move both ratings in the same direction.

The lessons that both researchers rated independently show where the correlation comes from. Suppose it arose within each observer, with the same person's judgement of warmth affecting their judgement of strictness. Then an observer's warmth rating would predict their own strictness rating much better than their colleague's. The two correlations are nearly the same: 0.28 and 0.26. So the

correlation reflects what both observers saw rather than the judgement of either one. The cross-observer figure still contains ordinary measurement error. Dividing it by the square root of the product of the two enacted reliabilities gives about 0.36. That, rather than 0.62, is the better estimate of how strongly warmth and strictness go together in schools. The two are related but clearly distinct.

The gold standard is two measures rather than one because the interview and the visit disagree about the same school. If the two were noisy readings of one quantity, the right thing to do would be to average them, because their errors would partly cancel. However, they agree far too little for that: the two warmth measures correlate at about 0.25, so each explains about one sixteenth of the other's variance, and the two strictness measures at about 0.41, so about one sixth. Averaging two measures of different things does not cancel noise. The average would be a mixture, partly what the observers saw and partly what the head said, and choosing different weights would give a different mixture.

I therefore keep the two measures separate throughout, and their divergence is itself a result. For strictness the corrected correlation reaches 0.58. The head's account and the observers' scores therefore share a substantial part of their variance, but well under all of it. For warmth the corrected correlation is 0.42, so the two share little. Two schools make this concrete, one where the two measures agree and one where they diverge. At the first, the headteacher rated the school 4.0 on warmth and 4.0 on strictness, and the observers, across their lessons and the day, recorded 3.9 and 4.3; the in-lesson and out-of-lesson sub-scores agree with one another to within half a point. At the second, the headteacher rated warmth 4.7 and strictness 3.9, near the top of the scale, while the observers recorded 2.3 and 3.0. In-lesson warmth at that school averaged 2.2 out of 5 and out-of-lesson warmth 2.4, while the head's ratings on the three warmth statements averaged 4.7 out of 5. The two interviews read alike: both heads placed their schools high on the same warmth statements. Only the visit separates them. Across the 103 schools, four in five have a warmth gap larger than half a point and half have a

gap larger than one point.

One espoused measure can be checked against an outside record. Heads rated how low their staff turnover was and how high morale was, and the staff-climate score built from those two statements is correlated with the school's actual teacher turnover, as recorded by the Department for Education, at $r = -0.52$. In a regression of teacher turnover on the staff-climate score, the other espoused scores, disadvantage and school size, the staff-climate coefficient corresponds to a partial correlation of $-0.48$. On the one question where a head's answer can be checked against the record, heads report accurately. This check concerns staff turnover, though, and says nothing about the warmth statements.

### 2.3.2 Predicting the scores from text

The first way of reading the documents uses no theory of culture at all. It asks only whether the words a document uses tend to go with the culture observers found in the school.

The method works as follows using warmth as an example. Each document is turned into a list of counts: how often each word, and each two-word phrase, appears in it. Phrases that appear in almost every document count for little; phrases that distinguish one document from another count for more. A statistical model is then estimated on the visited schools. It learns which phrases tend to appear in the reports of schools observed to be warm, and which in the reports of schools observed to be cold, and it attaches a small positive or negative weight to each. To score a new school, the model adds up the weights of the phrases in its document. No single phrase decides anything. A school's score is the sum of thousands of small weights, each attached to one phrase. In symbols, let $x_s$ be school $s$'s text vector, $y_s$ its enacted score and $C_s$ its four control variables. The target is the residual from the demographic adjustment,

$$\tilde{y}_s = y_s - \hat{c} - C_s'\hat{\gamma}, \tag{2.1}$$

with $\hat{c}$ and $\hat{\gamma}$ estimated by least squares on all schools. For each school $s$, the model is estimated on every other school,

$$(\hat{a}_{-s}, \hat{\beta}_{-s}) = \arg\min_{a,\,\beta} \sum_{t \neq s} \left(\tilde{y}_t - a - x_t'\beta\right)^2 + \lambda_{-s} \left\|\beta\right\|^2, \tag{2.2}$$

with the penalty $\lambda_{-s}$ chosen by leave-one-out error within the $t \neq s$ schools alone. The held-out prediction for school $s$ is $\hat{y}_s = \hat{a}_{-s} + x_s'\hat{\beta}_{-s}$, and every correlation reported for a prediction model is the correlation between $\hat{y}_s$ and $\tilde{y}_s$ across the held-out schools. Section 2A.6 gives the text representation and the remaining estimation detail.

Three safeguards stop the model from predicting the enacted scores accurately for the wrong reasons.

First, the model is never tested on a school it learned from. Every school's reported prediction comes from a model estimated with that school left out, so each number says how accurately the model predicts the score of a school that was not used to estimate it. As a further check, I ran a permutation test, re-running the whole procedure two hundred times on deliberately scrambled scores. Scrambling means shuffling the warmth scores across the schools, so that each report is paired with a score that belongs to some other school. The reports are real and the scores are real, but the pairing between them is destroyed, so there is nothing true left for the model to find. Whatever correlation it still reaches is what luck alone produces. A real result has to beat that, and the reported significance levels say how often the scrambled runs did.

The second safeguard is to ensure it cannot recognise the school. Before anything is counted, school and trust names, place names, dates, page numbers and standard boilerplate are removed from every document. Without this step, the model learns to spot which school or which era of report it is reading, rather than what the report says.

Finally, the model is prevented from reading demographics. A school's intake statistics predict part of its culture score, and a document often reveals the intake.

So each enacted score is first split into the part that intake statistics explain and the part they do not, and the model is asked to predict only the second part. A model that merely recognised a disadvantaged school from its report would predict none of the adjusted score, so its correlation would be near zero. Results are shown with and without this adjustment.

I also tried more elaborate ways of representing the text: longer phrases, topic groupings, and meaning-vectors from a neural network that scores passages as similar when they say similar things in different words. Section 2A.6 records the findings. None of them improved on counting, so that is the method reported throughout. The more complex methods failing to beat counting does not cast doubt on counting. The complex methods can read things counting cannot. A meaning-vector, for example, treats two sentences as alike when they say the same thing in different words. So if the reports held a signal that counting misses, the complex methods would have found it. The fact that they did not suggests that the signal in these documents is in the words themselves.

### 2.3.3 Scoring the text with mark schemes

The second way of reading the documents states what to look for. For each source and dimension I wrote a mark scheme of one page: a sentence saying what is being measured, five numbered bands, and two or three tie-break rules. To give the flavour, the inspection-report warmth scheme sets band 3, the default, as "a safe and generally positive school, described through labels, provision or outcomes, without describing how staff and pupils actually relate", and band 4 as "the report describes the relationships themselves". All seven schemes are printed in full in Section 2A.9.

In the same notation, a mark-scheme band is $B_s = f(x_s)$. Here $f$ is the fixed procedure of giving the one-page scheme and one document to a named version of the language model and taking the band it returns. The scheme and the model were fixed before any document was scored. Nothing in $f$ is estimated from the

schools, so there is no training step and no held-out distinction; the adjusted figure reported for a scheme is the correlation between $B_s$ and $\tilde{y}_s$.

A language model reads each document against the scheme and returns a band from 1 to 5, together with one quoted sentence as evidence (the models are named in Section 2A.9). The same fixed model reads every document.

Three checks are carried out to assess the robustness of the LLM scoring, and their outcomes are reported with the results. The first asks whether the language model gives the same answer twice: thirty documents per source were scored a second time, and the agreement between the two runs is reported. The second asks whether the bands depend on which large language model produced them. To assess this the whole exercise was repeated with a second, larger model. This check also settles a tie. When the prediction model and the mark scheme correlate with the enacted scores about equally, the mark scheme is used only if its correlation is about the same under both language models. If the correlation changes substantially with the larger model, the prediction model is used instead.

The third check asks whether the mark schemes are reading length rather than culture. A longer document gives the model more evidence to cite, so a scheme could award higher bands to longer documents rather than to warmer or stricter schools. The correlation between band and document length is therefore computed for every source and dimension, and each correlation is then recomputed after removing from both the band and the enacted score the part that document length predicts.

### 2.3.4 Judging both methods

Both methods are judged in the same way: a method is good if its scores agree with what trained observers recorded in the school, and it earns no credit for anything else. Comparing the two methods source by source and dimension by dimension then answers two questions at once. The first is how much the public text says about a school's culture at all. The second is what it takes to read

it: whether the words must be interpreted against a stated theory, as the mark schemes do, or whether counting them is enough, as the prediction models do.

One further rule applies throughout. Warmth, strictness and teaching are different things, and a school's three scores need not agree. So a signal that genuinely reads one of them should not help predict all three at once. When a feature of a document does help with all three, the likely reason is not that the school is better at everything. It is that the feature describes the document itself, so two schools with similar documents receive similar scores however much the schools differ. A signal of that kind is discarded. Document age, format and length were discarded for exactly this reason, and so were several changes that appeared to improve the models (Section 2A.6).

## 2.4 Results

The results come in three parts: what direct measurement shows on its own, what the public text contains and which method extracts it, and two checks from outside the project's own data.

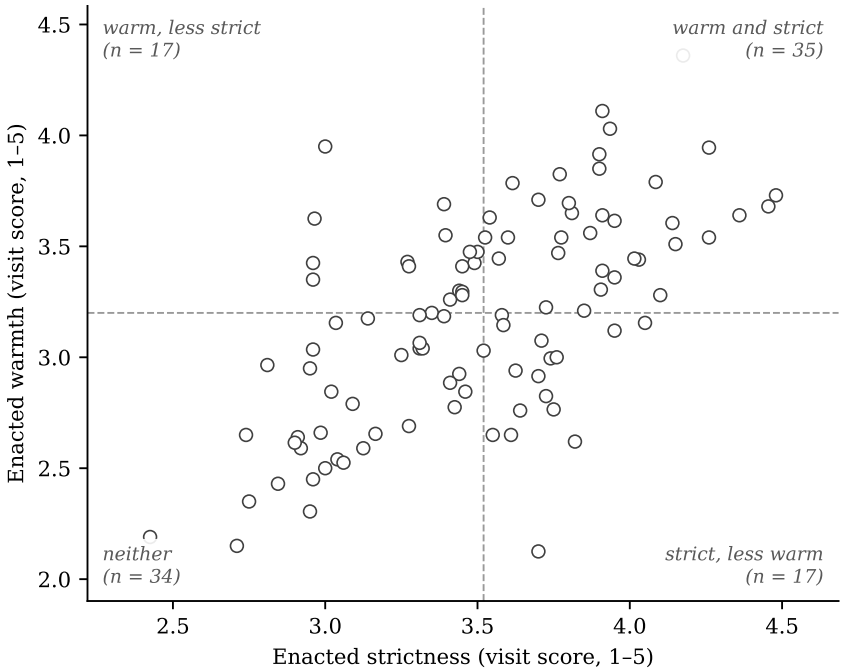
### 2.4.1 What direct measurement shows

Direct measurement shows two things. The first is that warmth and strictness are separate dimensions of a school. A school can be warm without being orderly, and orderly without being warm, and the visited sample contains both kinds: Figure 2.1 plots the schools on the two dimensions and every quadrant is occupied. The two dimensions are related but distinct: after the correction for shared observers and for measurement error, they correlate at 0.36.

The second is that what heads say and what observers see are different measures. The interview and visit scores are correlated at only 0.25 for warmth ($p = 0.011$) and 0.41 for strictness ($p < 0.001$), on the 103 schools with both.

Both scores are measured with error, so the highest correlation they could

Figure 2.1: Observed warmth and strictness in the visited schools




*Notes:* Each dot is a school. Dashed lines mark the sample medians, which divide the schools into the four labelled quadrants. All four are occupied: warm and strict; warm, less strict; strict, less warm; and neither.

reach is below one. Correcting for the error in both raises warmth to 0.42 and strictness to 0.58. Neither reaches the value two error-free measurements of the same quantity would give, and the shortfall is much larger for warmth: heads place their school's strictness with some accuracy and its warmth with very little. The disagreement on warmth is therefore real, not a product of measurement error.

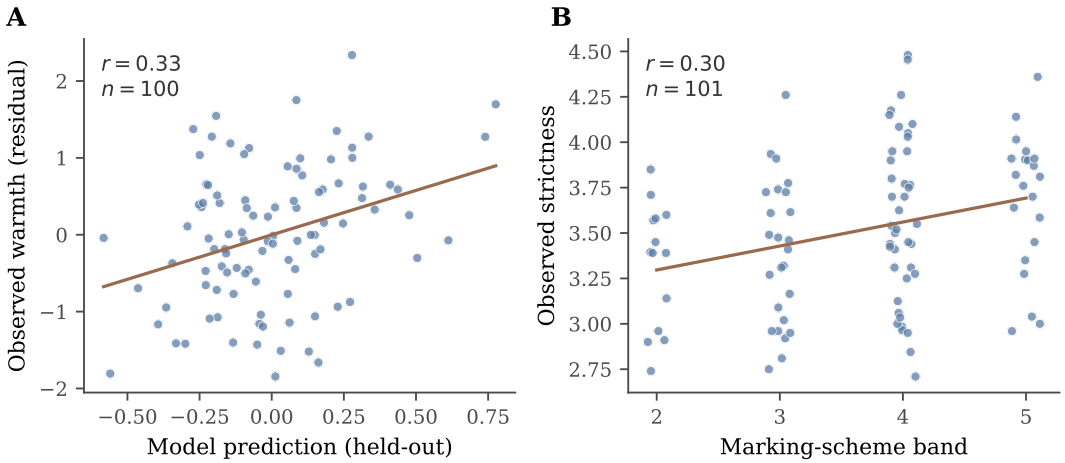
### 2.4.2 What the public text contains

This section shows three things. First, scores derived from public text do correlate with the enacted scores, but only when the text is an inspection report: every text measure that passes the tests comes from an inspection report, and the documents schools write about themselves carry no signal. Second, within the reports the two dimensions are read by different methods: the prediction model predicts warmth, the mark scheme scores strictness, and both work for teaching. Third, the measures the chapter goes on to use pass the three checks set out in

the measurement section.

Figure 2.2 plots the warmth scores from the prediction model and the strictness scores from the mark scheme against the enacted scores: the warmth model's held-out predictions on the left, the strictness scheme's bands on the right. In both panels the two rise together: schools the model reads as warmer are schools the observers scored as warmer, and reports the scheme places in higher bands come from schools the observers scored as stricter. The relationship is loose, with exceptions at every level, but it is not the work of a few outlying schools.

Figure 2.2: The warmth model and the strictness scheme against the enacted scores




*Notes:* Panel A shows the warmth prediction model's held-out predictions against enacted warmth, both after the demographic adjustment, that is, the part of each score that the four control variables do not explain; each school is predicted by a model that never saw it. Panel B shows the strictness mark scheme's bands against enacted strictness without that adjustment, so the correlation shown is the raw +0.30; the adjusted figure, +0.25, is the one in Table 2.4.

Table 2.4 reports fourteen text measures: seven pairings of a source with a dimension, each scored twice, once by the prediction model and once by the mark scheme. Each measure is summarised by one number: the correlation between the score it gives a school and the score the observers gave it. Each measure then faces two tests: it must beat luck, and it must stay significant once the number of tests is allowed for. Every measure that passes both tests comes from an inspection report, and none of the measures taken from the documents schools write about themselves passes either. Within the reports, warmth is read by the prediction

model and strictness by the mark scheme. The rest of this section works through the tests in Table 2.4 and then the three checks, one at a time.

Table 2.4: What each public text carries about enacted culture

| Source | Dimension | Model | 95% CI | $p$ | Scheme | 95% CI | Agree | Retest |
|---|---|---|---|---|---|---|---|---|
| Inspection report | Warmth | +0.33 | [+0.14, +0.49] | <0.005 | -0.07 | [-0.26, +0.13] | 68% | 97% |
| Inspection report | Strictness | +0.19 | [-0.01, +0.37] | 0.045 | +0.25 | [+0.06, +0.43] | 71% | 100% |
| Inspection report | Teaching | +0.28 | [+0.09, +0.45] | 0.010 | +0.27 | [+0.08, +0.45] | 72% | 97% |
| Behaviour policy | Warmth | +0.15 | [-0.05, +0.34] | 0.114 | +0.04 | [-0.16, +0.23] | 62% | 83% |
| Behaviour policy | Strictness | -0.15 | [-0.34, +0.04] | 0.771 | -0.03 | [-0.23, +0.17] | 72% | 100% |
| Website | Warmth | +0.14 | [-0.06, +0.32] | 0.100 | -0.13 | [-0.32, +0.06] | 31% | 97% |
| Website | Strictness | +0.17 | [-0.03, +0.35] | 0.104 | +0.13 | [-0.07, +0.31] | 39% | 97% |

*Notes:* Cells are correlations with the visit scores on the visited schools ($n = 100$ in every cell: 101 scored documents, one school lost to a missing control), with intake and report age partialled out of the target. "$p$" is the model's permutation $p$-value from 200 target-scrambled draws through the identical pipeline (floor 1/201, shown as <0.005). Each bracketed interval is a Fisher-$z$ 95% confidence interval for the correlation to its left (model and mark scheme respectively). "Agree" is exact band agreement between the two language models applying the same mark scheme; "Retest" is exact agreement on thirty documents re-scored by the primary model.

The scrambled scores say how easily each of the model's results could be luck. Its three inspection-report measures all beat luck (permutation test, warmth $p < 0.005$, teaching $p = 0.010$, strictness $p = 0.045$). Its measures from the behaviour policies and the websites all fail to.

The prediction model can be given several documents at once and the mark scheme cannot, so these pooled results are for the model alone. Each figure is again the correlation between the model's predictions and the observers' scores. Pooling the behaviour policy and the website, the two documents the school writes, gives a correlation with enacted warmth of +0.17; for strictness and teaching the pooled correlations are near zero. So pooling adds no signal that the separate readings missed. It can also lose one: the website on its own correlates with enacted strictness at +0.17, and adding the policy leaves −0.01. Pooling all three sources never improves on the best single source; on observed teaching the correlation falls from +0.28 for the inspection report alone to +0.05 for the three together.

Fourteen measures also give luck fourteen chances, so the measures in Table 2.4 are corrected for the number of tests as well. Four pass this test: the model's warmth and teaching, and the scheme's strictness and teaching (the strictness

scheme at $q = 0.036$).

The prediction model predicts enacted warmth from a report's wording, correlating at +0.33, and enacted teaching at +0.28, and both figures are essentially unchanged, or higher, when everything demographics could explain is removed. Strictness is different: no prediction model predicts it well. The best attempt reaches +0.19, it beats luck only narrowly ($p = 0.045$), and its confidence interval touches zero ($[-0.01, +0.37]$). Every word-count representation tried showed the same weakness.

For strictness the mark scheme does better than the prediction model, although the gap between the two is small enough to be chance at this sample size ($z = 0.65$, $p = 0.514$). The numbers alone therefore cannot separate the two methods on strictness. The three checks set out in the measurement section can: the retest, the change of language model, and the length check. The scheme's strictness bands pass all three.

The retest asks whether the language model gives the same answer twice. Each scheme was run twice on the same documents. Across the seven mark schemes, the two runs give the same band 96% of the time, and the report strictness scheme gave every school the same band on both runs.

The purpose of changing the language model is to check whether the reading is genuinely down to the mark scheme or whether the particular model that applies it makes a difference. If the scheme is doing the work, a different model applying the same mark scheme should find the same thing. For strictness, the second model's bands correlate with the enacted scores at +0.34, close to the primary model's +0.30, so the scheme is what determines the band, not the model. The warmth and teaching schemes fail the same check. Applying the same mark schemes, the second model's bands correlate with the enacted scores at essentially nothing (0.00 for warmth, +0.06 for teaching), so what the primary model found there came from the model rather than the scheme.

The length check asks whether the mark-scheme bands follow document length

rather than the school. For the inspection-report schemes the worry is small: the strictness bands correlate with length at −0.32 and the warmth bands at −0.05, and removing the effect of length moves the strictness correlation only from +0.25 to +0.24. The website schemes' bands follow length closely, one more reason their bands are not used.

Demographic adjustment is separate from the three checks, and it is applied to the mark-scheme bands in the same way as to the prediction models' predictions; Section 2A.6 gives the specification. The adjustment changes the strictness correlation little: the scheme's correlation with enacted strictness is +0.30 raw and +0.25 adjusted. The warmth scheme's bands, by contrast, correlate with enacted warmth at −0.07 and put three-quarters of schools in one band. The model, on the same reports, predicts enacted warmth at +0.33, and its advantage is itself significant (paired test, $p = 0.004$).

The two dimensions appear in the reports differently. Warmth is associated with many small differences in word choice across the whole report. A model that counts words can learn these from the visited schools, but a mark scheme cannot list them in advance. Strictness is associated with a few specific claims, for example that disruption is rare or that rules are applied consistently. A reader must be told to look for these claims. A word-count model cannot use them, because a report that says disruption is rare and one that says it is not rare use almost the same words. Teaching, kept in the analysis as the rival explanation for the next chapter, sits in between and is read about equally well by both methods. However, only the model's predictions pass the change-of-model check (the second model's teaching bands correlate at +0.06), so the prediction model scores are used for teaching.

Behaviour policies and websites tell neither method anything about enacted culture. The prediction models reach at most +0.17 on any pairing, too little to distinguish from zero at this sample size, and the bands from the mark schemes correlate less still. One of those two failures needs a caution. The two schemes

that read the behaviour policies, one for warmth and one for strictness, each put nearly every school in the same band, and a scheme that gives nearly every school the same score cannot correlate with anything, whatever the policies contain. Its zero is therefore not evidence that the policies say nothing. The evidence comes from the prediction models. They give different schools different scores, so a signal in the policies could have shown up, and none does.

### 2.4.3 Which parts of a report the prediction models use

The prediction models learn which words and phrases predict the enacted scores, and nothing in the method restricts them to any topic. A model can therefore predict the enacted scores accurately for two reasons. The first is that it has picked up the words inspectors use when they describe the dimension itself, such as the way staff and pupils treat one another in the case of warmth. The second is that it has picked up the positive tone that runs through a report on a good school whatever the topic. Warm schools tend to be good schools, so a model that has learned only the general tone would still predict warmth, but it would not be reading warmth. As an informal check on the method, I annotated the content of the reports and asked whether the sentences that raise each model's predicted score most are consistent with the dimension it is meant to measure. For example, are the sentences that predict warmth about relationships between staff and pupils? For the annotation, a language model read every sentence of every report and tagged what the sentence is about: relationships, behaviour, teaching, or something else. The tagging was produced without seeing the schools' scores or anything the models learned, so it can be used to check what they learned.

The check takes the hundred sentences that raise the warmth model's predicted score most, where a sentence's contribution is the sum of the weights the model has learned for the phrases in it, and asks what they are about. 21% of them are about staff–pupil relationships, against a base rate of 13% across all sentences, so the sentences that raise the warmth score are about relationships far more often

than chance would give. For teaching the corresponding shares are 40% against 45%, which is no enrichment at all. Inspection reports are mostly about teaching already, so the sentences that raise the teaching score cannot be about teaching much more often than the rest are. The teaching model therefore picks up the report's general tone rather than any one topic.

The topic shares, together with the demographic adjustment reported earlier, say what a model's score is made of. Part of it is language specific to the model's dimension: for warmth, the relational sentences the model seeks out. The rest is the general tone of a school doing well, which runs through a good school's whole report whatever the sentence is about. The demographic adjustment removes some of that general tone, but only the part that the four control variables (three intake measures and report age) can predict; the remainder stays in the score. A predicted warmth score therefore combines two things. Part is the relational language specific to warmth. Part is the general positive tone of a report on a good school. It is a measure of the report's whole account of the school, weighted towards warmth, not a measure of warmth alone.

Table 2.5 shows the sentences each model weighs most heavily, under selection rules stated with the table. The two listings differ. For teaching the strongest sentences name the topic directly. For warmth they do not.

For teaching, the strongest sentences are about pedagogy: choosing core knowledge, embedding learning in long-term memory, pupils focusing in lessons. Some of the strongest, though, are criticisms. A sentence saying that pupils are not remembering knowledge deeply raises the score as surely as one saying that they are. That happens for two reasons. The first is a flaw in the model. It counts words and cannot tell praise from criticism. The second is a fact about reports. Inspectors write about long-term memory, whether in praise or criticism, only when a school teaches in that way, so the vocabulary is informative either way. The appendix discusses both in more detail.

For warmth the pattern is different, and at first sight stranger. The sentences

that most lower a school's warmth score sound kind. "Many pupils feel confident they have a 'trusted adult' they can go to" and "pastoral support is in place" are among the strongest score-lowering sentences. The explanation for this is that an inspector writes about a warm school by describing adults and pupils dealing with one another. When there is nothing like that to describe, the inspector mentions support systems instead. Systems are what gets written about when the relationships are missing.

On the positive side, no single sentence carries much weight. The model's warmth evidence is spread across many small choices of phrase: attentive verbs, pupils' names used, care described in passing. Even its strongest whole sentences read as the tone of an engaged community rather than tenderness. The spread is a fact about warmth, not a weakness of the model. Warmth never concentrates in one sentence, so there is no quotable line to point to. The case that the model is reading warmth therefore rests on two things shown already. The first is the held-out correlation with the warmth scores for the visited schools. The second is the topic shares: the sentences that raise the model's predicted warmth most are about staff–pupil relationships far more often than their share of all sentences would predict. Section 2A.7 also lists the sentences about warmth towards staff that raise the predicted score: schools whose leaders listen to their teachers are predicted to be warmer. That part of the signal is easy to interpret, and it is not difficult to imagine that a school where leaders listen to staff is also one where teachers listen to the concerns of their pupils.

### 2.4.4 What schools write about themselves

A school's own documents, its behaviour policy and website, tell a reader what the head says about the school, not what the school does. The test is a rerun of the prediction models: they read the policy and website text as before, but this time they are trained to predict the head's interview answers rather than the enacted scores. This was done across the roughly 300 interviewed schools. On strictness

Table 2.5: Sentences that most move the models' scores

| *Panel A: Warmth* | |
|---|---|
| Raises the score | Lowers the score |
| "Sixth-form students are proud role models and make a highly positive contribution to the school and local community." | "Many pupils feel confident they have a 'trusted adult' they can go to." |
| "Pupils are encouraged to contribute to the school community." | "Pastoral staff work closely with these pupils and their families." |
| "Sixth-form students act as ambassadors for the school by volunteering in the local community." | "They check closely to ensure that disadvantaged pupils have the support they need." |
| *Panel B: Teaching* | |
| Raises the score | Lowers the score |
| "In lessons, most pupils focus on their learning and learn well." | "The personal, social and health education (PSHE) curriculum is designed to support pupils' wider development." |
| "They carefully choose the core knowledge they want pupils to know and remember in the long term." | "On other occasions, adults do not support pupils to complete work to a high standard." |
| "Teachers help pupils to embed learning in their long-term memory." | "When this is the case, these pupils do not learn as well as they should." |

*Notes:* A sentence's score is the sum of its phrases' learned weights. Selection: the highest-scoring sentences tagged as being about staff–pupil relationships and mentioning pupils (warmth) or tagged as positive teaching content (teaching); the lowest-scoring sentences tagged on-topic with no further restriction, except that the third teaching example is the tenth-lowest, shown in place of a second sentence about the personal, social and health education curriculum. Unfiltered listings appear in the appendix (Section 2A.7).

the documents match what the head says: the models predict the head's account at +0.16 (demographics removed), which survives the correction for the number of tests. On staff morale the relation is weaker, at +0.13: significant on its own, but not after that correction. On warmth there is no relation at all, at +0.05. What a school writes about warmth is unrelated even to what its own head says in the interview.

The bands from the website mark schemes do not correlate with any enacted or espoused score, on any dimension. That is weak evidence rather than contrary evidence: two language models applying the same website mark scheme agree on the band only 31% and 39% of the time, against 68% to 72% on inspection

reports, so the scheme is barely reliable on the documents schools write about themselves.

One caveat qualifies the warmth result from the prediction models. The espoused warmth score is built from just three interview statements, and heads bunch their answers near the top of the scale, so the score varies little from school to school. When the target score varies little between schools, any correlation with it is lower than it would be if the scores were spread out. Finding no correlation against such a target therefore rules out a strong link between public warmth language and the head's account, but not a modest one.

The inspection report at first seems to break this pattern. Its text correlates with the head's account of strictness at +0.31, as if the inspector somehow knew what the head thinks. There is a simpler explanation. The report describes what the school does, and what the school does is itself related to what the head says. If that is the route, the report is linked to the head only through the school, and the inspector never needed to know the head's views at all. The test removes from both measures the part that enacted strictness predicts and correlates what remains. The correlation falls from +0.31 to +0.19. So much of the link between the report and the head's account is explained by the school's actual behaviour, which both describe. The report describes the school. The head's account also describes the school. That is why the two agree, and the inspector did not need to know the head's views. So the two kinds of document line up with the two measures. What a school writes about itself matches what its head says, the espoused side. What the inspectorate writes about the school matches what the school does, the enacted side.

### 2.4.5 A check against the parent survey

The last check comes from parents. Everything so far has judged the prediction model's warmth scores against the visited schools, and there are only 103 of them. Parents offer a second, independent judge at national scale: Ofsted's Parent View

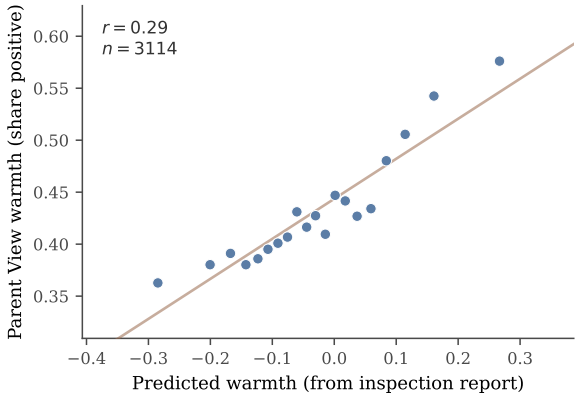
survey asks them, at nearly every school in England, whether their child is happy at the school and feels safe there, which is warmth as a parent experiences it. If the model's warmth scores are real, they should agree with what parents report. This section makes that comparison and reports two results: the model's warmth scores do agree with the parents, and the agreement is not explained by the inspection grade.

The measure is the parent-reported warmth score: the share of parents who say their child is happy at the school and feels safe there. There are two important limitations to this data that should be kept in mind. The first is that parents choose whether to respond, and the typical school's figure rests on about 120 replies, roughly one parent in eight. The second is a timing issue. Responses are taken over several years, so at some schools the parents' answers describe an earlier version of the school than the one the culture data describe.

The first result is the national agreement. The warmth prediction model was estimated on the visited schools alone. That leaves 3,114 schools the model has never seen. For each of these, the model reads the school's inspection report and turns it into a predicted warmth score. Those predictions can then be set against what parents say. Across the 3,114 schools, predicted warmth and parent-reported warmth correlate at +0.29; Figure 2.3 plots the two measures against each other. That agreement would mean nothing if the model had been given the answer: two measures agree trivially when one was built from the other, as when a model is tested on the data it was trained on. Neither is the case here. The model never saw these schools when it was estimated, and it never saw any parent data at all. The only way it can agree with the parents is if both are picking up something true about the school.

The second result concerns the inspection grade. The model reads inspection reports, and a report reflects the grade the school received. If the model's warmth score were the grade in different words, the parents' agreement with the model would add nothing beyond the parents' agreement with the grade. The test is to

Figure 2.3: Predicted warmth and parent-reported warmth, nationally




*Notes:* Correlation $r = +0.29$ on $n = 3{,}114$ schools. Each dot is a bin of schools (twenty equal-count bins); the model never saw any of these schools in training.

compare schools that share the same grade. Within such a group the grade is the same for every school, so it cannot explain why the model scores one school warmer than another. On the visited schools, 99 of which have a grade from an inspection before the pandemic, the 2019 grade used in the next chapter, the model's correlation with enacted warmth is +0.32, essentially its full-sample value of +0.33. Comparing only schools with the same grade leaves it at +0.29. The teaching model gives the same result. Its correlation with enacted teaching is essentially unchanged when only schools with the same grade are compared. The model's warmth scores are not the grade in different words. Section 2A.8 adds a stricter version of this check, together with what the visited schools show and a check on one possible route: some reports quote the Parent View survey, so the model could in principle be reading the parents' answers rather than independently agreeing with them.

The wider point is that the two measures, the parents' survey answers and the model's warmth scores, share nothing that could make them agree for any reason other than the school. There is a final point to consider, though. A school that is good in general tends to have both a report the model scores as warmer and happier parents, so part of the agreement may reflect general quality rather than warmth in particular. The agreement therefore stands as corroboration rather than proof.

## 2.5 Conclusion

Warmth and strictness can both be measured by observation and in public text. Warmth is the harder of the two to measure. In person it needs more lessons and more settings than strictness does to reach the same reliability. In public text it is detected only by a model that counts words and phrases. The words associated with warmth are many and each matters a little, so they are spread across an inspection report's ordinary wording, in whether the inspector writes about adults attending to pupils or falls back on stock assurances. A mark scheme asks the language model to find sentences that describe warmth, and no single sentence does.

Strictness is the opposite. In person it is easy to observe reliably. In text it sits in specific claims: whether disruption is rare, whether rules are enforced consistently. Checking claims is necessary because a report saying that disruption is rare and one saying that it is common use almost the same words. That is why the prediction model predicts strictness weakly but the mark scheme does better.

The documents schools write about themselves match what heads say about strictness and staff morale. On warmth they carry no signal detectable at this sample size, either against the observers or against what headteachers say when interviewed.

The parent surveys provide a useful check. At the national level their reports of happy, safe children agree with the model, which supports the claim that inspection reports carry a real warmth signal. In the visited schools, as the appendix shows, parents' answers are closer to the head's account than to the observers' scores.

There are four limitations that should be noted. The first is that the visit scores measure warmth and strictness with error. Two observers watching the same school agree well but not perfectly, so part of each score is error, and every correlation with the visit scores is lower than it would be without that error. The reported correlations therefore understate the true relations. The second is that inspection

reports are written in different years under different inspection frameworks, which adds noise. Removing dates, boilerplate and framework-specific wording reduces that noise but cannot remove it. This kind of noise pushes correlations towards zero.

The third limitation is that each school was visited once. A visit is a snapshot, and a school may have had an unusually calm or an unusually rough day; the scores cannot tell an unusual day from an ordinary one.

The final limitation is that the model's warmth scores cannot be fully separated from the inspection's overall verdict. The prose the model reads and the grade the inspection awarded come from the same visit and say related things, so part of what the model reads may be the inspection's judgement of the school rather than its warmth in particular. The checks in the results show that the model reads more than the grade alone. What they cannot show is how much of the rest is warmth, because the grade and the prose come from the same visit, so the grade already reflects some of the warmth the prose describes. Any comparison that removes the grade therefore removes part of the warmth signal as well.

With those limitations noted, the chapter provides three national measures, each validated against direct observation: prediction models for warmth and teaching, and a one-page mark scheme for strictness. The following chapter uses these to investigate the associations between warmth, strictness, teaching and pupil progress.

# Appendix to Chapter 2

This appendix collects the supporting material for the chapter, in the order in which the chapter uses it. It sets out how the visit and interview items combine into scores, reproduces the two observation forms and the interview guide as they were administered, gives the inter-rater reliability tables, describes the samples and the coverage of the text collections, specifies the prediction models, lists the sentences and phrases those models weigh most heavily, collects the further checks on the parent comparison, and finally reproduces the text of every mark scheme given to the language model.

## 2A.1 Scoring Guide and Item Definitions

This section sets out how the individual items on the forms combine into the nine sub-scores and the two sets of scores. The items themselves, that is, the question wording, the response format and the anchor description attached to each rating level, are reproduced in full in Section 2A.3 for the interview and in Section 2A.2 for the two observation forms.

Every item is recorded on a 1–5 scale. Two observation items can also take a zero, with a distinct meaning in each case. A zero on *response of pupils to sanctions* records that no sanction was issued during the lesson, and a zero on *effectiveness of pupil discussion* records that no discussion was initiated. For the reasons given in the chapter, the first of these is not a genuine low score. It is replaced by $6 - \text{disruption}$, which gives a calm unsanctioned lesson a high value and a disrupted unsanctioned lesson a low one. The second is treated as a genuine negative, since a lesson containing no discussion did not achieve effective discussion.

Observation item scores are averaged across lessons and across researchers to give one value per school before the sub-scores are formed. Table 2A.1 lists the

constituent items of each sub-score.

Table 2A.1: Composition of the gold-standard sub-scores

| | Sub-score | Constituent items |
|---|---|---|
| **Warmth** | | |
| W1 | Teaching warmth | *Lesson observation.* Frequency with which teacher uses pupils' names; Frequency of praise; Quality of interactions between teachers and pupils; Frequency of student-initiated participation; Motivation level of students |
| W2 | School warmth | *Outside-of-lessons observation.* Quality of relationships between pupils during transition; Quality of non-behaviour related interactions between staff and pupils at break; Quality of relationships between pupils at break; Consistency of use of reward system |
| W3 | Espoused warmth | *Interview statements.* St2 Reward systems applied consistently; St7 Pupils feel safe at school; St8 Pupils feel their teachers care |
| **Strictness** | | |
| S1 | In-lesson strictness | *Lesson observation.* Level of misbehaviour tolerated before a first sanction (reversed); Frequency of low-level disruption (reversed); Response of pupils to sanctions; Extent to which pupils address the teacher respectfully; Concentration level of students |
| S2 | Out-of-lesson strictness | *Outside-of-lessons observation.* Consistency of use of sanction system; Calmness of corridors; Arrival time to next lesson; Organisation of canteen/dining hall; Organisation of recreational space |

| | Sub-score | Constituent items |
|---|---|---|
| S3 | Systems count (not a score component) | *Interview.* Centralised detentions; Internal exclusion room; Behaviour points/demerits; Line-ups; Silent corridors; After-school revision; Weekend revision; Ban on mobile phones |
| S4 | Espoused strictness | *Interview statements.* St1 Sanction systems applied consistently; St5 Pupil behaviour in class is generally good; St6 Pupil behaviour outside class is generally good; St9 Teachers feel respected by pupils |
| **Teaching practice** | | |
| T1 | Teaching practice (observed) | *Lesson observation.* Frequency of questioning/checking pupil progress; Quality of verbal feedback; Effectiveness of pupil discussion; Quality of differentiation; Clarity of teacher explanation; Extent to which learning outcomes are achieved; Effectiveness of teaching methods; Clarity of lesson structure; Effectiveness of resource use; Frequency of teacher-initiated participation |
| SC | Espoused staff climate (there is no espoused teaching score) | *Interview statements.* St3 Teacher turnover is low; St4 Staff morale is good |

*Notes:* Item wording is given in full in Section 2A.3 and Section 2A.2.

Every other sub-score is the mean of its items, computed over the items present so that a missing item reduces the number of contributing items rather than the school's score.

The sub-scores combine into two sets of scores, grouped by source, with

observed behaviour and stated philosophy kept apart:

$$\text{Enacted:} \quad W^{\text{enac}} = \overline{(W1, W2)}, \quad S^{\text{enac}} = \overline{(S1, S2)}, \quad T^{\text{enac}} = T1 \tag{2A.1}$$

$$\text{Espoused:} \quad W^{\text{esp}} = W3, \qquad S^{\text{esp}} = S4, \qquad \text{SC} = \overline{(\text{SC}_1, \text{SC}_2)} \tag{2A.2}$$

There is no espoused teaching score. The two interview statements under the teaching heading concern staff turnover and morale, and they form the staff-climate score SC instead. The systems count S3 enters no espoused score.

Schools with an interview but no visit have the espoused set of scores alone (every visited school was also interviewed). No imputation is attempted across the two sets: as the chapter reports, the espoused and enacted scores are too weakly correlated for either to stand in for the other.

The mapping in Table 2A.1 is generated directly from the scoring code, so it cannot drift from what the scoring code does.

The assignment of items to dimensions is theoretical, but it can be checked against the data. A two-factor analysis of the school-level item means (principal components with varimax rotation, on the 98 schools with every culture item observed) recovers most of the structure Table 2A.1 imposes: fourteen of the nineteen culture items load most heavily on their assigned dimension, with the in-lesson strictness items and the relational items separating most cleanly. The remaining five load more on the other dimension. Those five are either near-ties or weakly loaded items about order in transitions. When the ten teaching items are added they form the dominant factor, and eighteen of the nineteen culture items then sort with their assignment. The analysis forces the two factors to be uncorrelated, while the chapter estimates the two dimensions correlate at about 0.36. Items that belong to both are therefore harder for this method to assign, and the check is if anything conservative.

## 2A.2 School Visit Protocol

Two proformas were used on each visit. The lesson observation form was completed once per lesson observed, and the outside-of-lessons form once per school, covering transitions, break and lunch. Every item is rated 1–5 against the anchor descriptions below, with anchors specified at 1, 3 and 5 and even values used for intermediate judgements. Visits were made by teams of two or three researchers, who completed the forms independently; item scores are averaged across researchers and across lessons to give one value per school, and Tables 2A.3 and 2A.4 report agreement between observers.

### Lesson observation

**Section 1 — Strictness and Warmth**

**Level of misbehaviour that results in a first sanction/level of misbehaviour which is tolerated** Rated 1 (minor) to 5 (serious).

**1** Isolated instances of misbehaviour such as whispering or fidgeting.

**3** Repeated disruptions, e.g., repeated talking or distraction of peers.

**5** Aggressive or defiant behaviour

**Response of pupils to sanctions** Rated 1 (very poor) to 5 (very good). A score of 0 is recorded separately (see below).

**0** If no sanctions are observed

**1** Students ignore sanctions or react defiantly, showing no change in behaviour.

**3** Noticeable change in behaviour in response to sanctions, but not consistently maintained.

**5** Immediate and sustained positive response to sanctions e.g. apologises for actions.

**Frequency of low-level disruption** Rated 1 (very low) to 5 (very high).

**1** Classroom is calm with disruptions rare or not occurring at all.

**3** Regular disruptions, sometimes affecting the flow of the lesson but generally manageable.

**5** Constant disruptions, creating a challenging environment for teaching and learning.

**Concentration level of students** Rated 1 (very low) to 5 (very high).

**1** Students are consistently off-task, showing little interest or engagement with the lesson.

**3** Moderate levels of concentration, with students generally engaged but susceptible to distractions.

**5** Exceptional concentration, with students deeply engaged and consistently focused throughout the lesson.

**Frequency of teacher-initiated participation** Rated 1 (very low) to 5 (very high).

**1** Teacher rarely encourages student participation, leading the class in a predominantly lecture-style format.

**3** Regular attempts by the teacher to involve students in the lesson, but some parts where teacher talks continuously.

**5** Teacher consistently involves students in the learning process.

**Frequency of student-initiated participation** Rated 1 (very low) to 5 (very high).

**1** Students rarely or never take the initiative to participate, showing passivity in the learning process.

**3** Moderate frequency of student-initiated participation. Sometimes active but sometimes passive.

**5** Students consistently participate in the lesson demonstrating high levels of engagement in the learning process.

**Motivation level of students** Rated 1 (very low) to 5 (very high).

**1** Students exhibit a lack of interest and motivation, showing little effort in classroom activities.

**3** A moderate level of motivation, with students showing interest at times but being disengaged at times.

**5** Exceptional motivation and enthusiasm, with students consistently demonstrating a strong desire to learn and participate.

**Extent to which students address the teacher respectfully** Rated 1 (not at all) to 5 (always).

**1** Disrespectful interactions, use of inappropriate language or tone.

**3** Respectful interactions most of the time

**5** Exceptionally respectful and considerate interactions at all times.

**Frequency with which teacher uses pupils' names** Rated 1 (very low) to 5 (very high).

**1** Rarely addresses students by name; impersonal interactions.

**3** Regularly uses student names, but some students are less frequently addressed.

**5** Consistently uses every student's name, fostering a sense of individual attention.

**Frequency of praise** Rated 1 (very low) to 5 (very high).

**1** Teacher provides little to no praise, no positive reinforcement.

**3** Regular praise, but it is not always meaningful or specific.

**5** Regular and specific praise which highlights what students have done well.

**Quality of interactions between teachers and pupils** Rated 1 (impersonal) to 5 (warm).

**1** Interactions are formal, distant, and lack warmth.

**3** Interactions are warm, but still quite formal in nature.

**5** Exceptionally warm and supportive interactions throughout the lesson.

**Section 2 — Teaching practice**

**Frequency of questioning/checking pupil progress** Rated 1 (very low) to 5 (very high).

**1** Teacher rarely asks questions or checks on pupils' understanding.

**3** Teacher regularly asks standard questions, adequately assessing pupils' basic understanding but may not probe deeper comprehension.

**5** Teacher consistently integrates questioning seamlessly into lessons, skilfully using them to assess understanding and encourage higher-order thinking.

**Quality of verbal feedback** Rated 1 (very poor) to 5 (very good).

**1** Feedback is rare, nonspecific, or irrelevant, offering no guidance or improvement for pupil learning.

**3** Feedback is regularly given, is clear, and somewhat constructive.

**5** Feedback is exceptionally well-tailored to each pupil and highly constructive.

**Effectiveness of pupil discussion** Rated 1 (not effective) to 5 (very effective). A score of 0 is recorded separately (see below).

**0** No teacher-initiated pupil discussion is observed.

**1** Pupil discussions are rarely initiated or if they are, then they are poorly managed, resulting in minimal pupil engagement.

**3** Regularly encourages discussions which have a moderate impact on learning but may not fully engage all pupils.

**5** Regularly encourages discussions and structures them effectively to ensure that pupils engage with them and benefit from them.

**Quality of differentiation (adaptation to students' needs)** Rated 1 (very poor) to 5 (very good).

**1** Teaching lacks adaptation to different learning needs, often leading to disengagement or unmet learning objectives for various students.

**3** Regularly employs differentiation strategies that address the needs of most students but not all.

**5** Exceptional in adapting teaching to meet all students' needs. Challenges high ability pupils and/or supports low ability pupils appropriately.

**Clarity of teacher explanation** Rated 1 (very poor) to 5 (very good).

**1** Explanations are often unclear, leading to widespread confusion or misconceptions among students.

**3** Usually provides explanations that are clear and aid in student understanding, though some concepts may not be understood by all students.

**5** Provides exceptionally clear and comprehensive explanations, making complex concepts accessible to all or almost all students.

**Extent to which learning outcomes are achieved** Rated 1 (not at all) to 5 (completely).

**1** Learning outcomes are rarely met, with most students failing to achieve the set objectives. Unclear what learning objectives are.

**3** A majority of learning outcomes are met, though some objectives may not be fully achieved by all students.

**5** All learning outcomes are fully met, with evidence of deep understanding and application of knowledge by all or almost all students.

**Effectiveness of teaching methods** Rated 1 (not effective at all) to 5 (very effective).

**1** Teaching method(s) are not appropriate to the content and tasks.

**3** Teaching method(s) are mostly appropriate to the content and tasks.

**5** Chosen teaching method(s) are very well matched to the content and every task is clearly well thought out.

**Clarity of lesson structure** Rated 1 (unclear) to 5 (very clear).

**1** Lessons lack a clear structure, resulting in disorganized content delivery. Structure is not communicated to students.

**3** Generally clear lesson structure with a logical flow, though minor issues in transition or clarity may occasionally arise. Structure is communicated quite well to students.

**5** Lessons are exceptionally well-structured, with seamless transitions, and a well-thought-out progression. Structure is regularly communicated to students.

**Effectiveness of resource use** Rated 1 (low) to 5 (high).

**1** The resources do not allow most students to achieve the learning objectives and/or engage with the topic.

**3** The resources allow most students to achieve the learning objectives and/or engage with the topic.

**5** The resources allow all or almost all of the students to achieve the learning objectives and/or engage with the topic.

## Outside-of-lessons observation

**Section 1 — Comparison across lessons**

**Consistency of use of sanction system** Rated 1 (not consistent at all) to 5 (completely consistent).

**1** Teachers are using different sanction systems in every classroom.

**3** Teachers are generally taking the same approach to misbehaviour, but there is no shared language.

**5** All teachers use the sanction system in exactly the same way and there is a clear shared language.

**Consistency of use of reward system** Rated 1 (not consistent at all) to 5 (completely consistent).

**1** Teachers are using different reward systems in every classroom.

**3** Teachers are generally taking the same approach to positive behaviours, but there is no shared language.

**5** All teachers use the reward system in exactly the same way and there is a clear shared language.

**Consistency of verbal feedback** Rated 1 (not consistent at all) to 5 (completely consistent).

Should be calculated as the average of the scores from individual lessons.

**Quality of differentiation (adaptation to students' needs) across lessons** Rated 1 (very poor) to 5 (very good).

Should be calculated as the average of the scores from individual lessons.

**Section 2 — Transitions**

**Calmness of corridors** Rated 1 (chaotic) to 5 (calm).

**1** The corridors feel unsafe. There is a lot of boisterous behaviour and students do not seem concerned with getting to their next lesson.

**3** Most students move between lessons in a calm fashion. There is an occasional loud voice and a small amount of boisterous behaviour. Most students move purposefully to their next lesson.

**5** All students move between lessons in a calm fashion with no loud voices or boisterous behaviour at all. All students move with purpose to their next lesson.

**Arrival time to next lesson** Rated 1 (lots of lateness) to 5 (all on time).

**1** The majority of students arrive late to lessons.

**3** Most students arrive on time with a few late arrivals.

**5** All or almost all students arrive on time.

**Quality of relationships between students during transition** Rated 1 (very poor) to 5 (very good).

**1** Lots of students speak to each other disrespectfully. There is clear evidence of bullying and/or intimidation.

**3** Most students are polite in their interactions with each other. There is evidence that a small amount of bullying/intimidation is taking place.

**5** All or almost all students are polite in their interactions with each other. There is no evidence of bullying or intimidation of any kind.

**Presentation of wall/corridor displays** Rated 1 (poor condition) to 5 (excellent condition).

**1** There are no displays, or if there are, then they are old and tattered.

**3** Most displays are in good condition, but some are a bit tattered and look old.

**5** All displays are in pristine condition with no damage. They all look as if they are new.

**Extent to which wall/corridor displays which are not student work align with school values** Rated 1 (random) to 5 (complete alignment).

**1** There are no displays which convey messages about the school values.

**3** There are some displays which convey messages about the school values.

**5** All or almost all displays convey messages about the school values.

**Section 3 — Break and lunch**

**Quality of non-behaviour related interactions between staff and students** Rated 1 (very poor) to 5 (very good).

**1** In general, staff do not engage with students.

**3** Most staff actively engage with students and interactions are mostly warm and personal.

**5** All or almost all staff actively engage with students and interactions are warm and personal.

**Quality of relationships between students** Rated 1 (very poor) to 5 (very good).

**1** Lots of students speak to each other disrespectfully. There is clear evidence of bullying and/or intimidation.

**3** Most students are polite in their interactions with each other. There is evidence that a small amount of bullying/intimidation is taking place.

**5** All or almost all students are polite in their interactions with each other. There is no evidence of bullying or intimidation of any kind.

**Organisation of canteen/dining hall** Rated 1 (chaotic) to 5 (calm).

**1** Students get their food in a haphazard fashion. They leave food behind on tables and on the floor. They do not take care of their environment.

**3** There is some organisation to the way that students get their food. There is a little bit of mess left behind.

**5** Students get their food in an organised fashion. They leave the table and floor where they eat clean and tidy.

**Organisation of recreational space** Rated 1 (chaotic) to 5 (calm).

**1** Activities are interfering with each other and there is no supervision from staff. There is a general sense of disorganisation. There is lots of rule breaking.

**3** There are clear areas for different activities, and some are supervised by staff. There is some rule breaking.

**5** There are clear areas for different activities, and these are all supervised by members of staff. All or almost all students abide by the rules.

## 2A.3 Interview Guide

The schedule below is the instrument as administered. Each open-ended question was scored 1–5 against the anchor descriptions shown. Each question was asked by one researcher and scored by both; Table 2A.2 reports agreement between them, and the analysis uses the final recorded score for each question. A checklist of key words and phrases was also completed for each question; it enters no score reported in the chapter and is omitted here. Item-to-sub-score mapping is given in Section 2A.1.

**Open-ended questions**

**Q1. How long have you been in post?** Recorded verbatim; not scored on the 1–5 scale.

**Q2. How would you describe your school's culture?** Rated 1 (weak answer) to 5 (strong answer).

**1** A vague or generic description is provided, lacking specific details or examples.

**3** A detailed description of the school's culture, including some specific examples, but no explanation of the rationale for this culture.

**5** A detailed description of the school's culture with specific examples that also includes a clear explanation of the rationale for this culture.

**Q3. How do you communicate this culture to your staff?** Rated 1 (weak answer) to 5 (strong answer).

**1** The strategies used to communicate the culture to the staff are not clearly explained.

**3** The strategies used to communicate the culture to the staff are clearly explained, but there is a lack of detail.

**5** There is a detailed and clear explanation of the strategies used to communicate the culture to the staff.

**Q4. What do you think is the most important characteristic of an effective school? Why?** Rated 1 (weak answer) to 5 (strong answer).

**1** Various characteristics are mentioned but there is no clear explanation of why these characteristics lead to a school being effective.

**3** One or multiple characteristics are mentioned and there is a clear explanation of why these characteristics lead to a school being effective.

**5** One or multiple characteristics are mentioned and there is a clear explanation of why these characteristics lead to a school being effective. There is also a discussion of how the school culture links to these characteristics.

**Q5. How does your reward and sanction system work? If a pupil misbehaves what should happen? If a pupil makes a positive contribution what should happen?** Rated 1 (weak answer) to 5 (strong answer).

**1** The explanation of the reward and sanction system is unclear, lacking specific details or a coherent structure.

**3** The response outlines the reward and sanction system to a decent extent, but there may be gaps in the explanation, and details might be limited.

**5** A clear and detailed explanation is provided on how the reward and sanction system operates; the response demonstrates a deep understanding of the procedures for addressing both misbehaviour and positive contributions.

**Q6. What do you think are the key characteristics of an outstanding classroom teacher?** Rated 1 (weak answer) to 5 (strong answer).

**1** Various characteristics are mentioned but there is no clear explanation of why these characteristics make a teacher outstanding.

**3** Various characteristics are mentioned and there is a clear explanation of why these characteristics make a teacher is outstanding.

**5** Various characteristics are mentioned and there is a clear explanation of why these characteristics make a teacher is outstanding. There is also a discussion of how these characteristics link to the school's culture.

**Q7. How important do you think extracurricular provision is? What kind of extracurricular provision have you put in place?** Rated 1 (weak answer) to 5 (strong answer).

**1** The importance of extracurricular provision is not clearly articulated, and the response lacks specific details about the types of activities available.

**3** The response acknowledges the importance of extracurricular provision, but the explanation may be somewhat basic or lacking in depth; details about the types of activities implemented could be more specific.

**5** The response shows a clear understanding of the importance of extracurricular provision and provides specific details about the types of activities in place.

**Q8. What is your view on the importance of marking and feedback? Is there a school-wide policy on frequency and type?** Rated 1 (weak answer) to 5 (strong answer).

**1** The explanation of the importance of marking and feedback is unclear. If there is a school-wide policy, then it is also not explained clearly.

**3** The explanation of the importance of marking and feedback is clear. If there is a school-wide policy, then it is also explained clearly but without specific details.

**5** The answer lays out a clear and comprehensive view on the importance of marking and feedback. It provides specific details about a school-wide policy, including the frequency and type of marking and feedback.

**Q9. What is the structure of your staff team?** Rated 1 (weak answer) to 5 (strong answer).

**1** The explanation is unclear and difficult to follow.

**3** There is a clear explanation of the staff structure from Senior Team through to Middle Leaders.

**5** The answer clearly lays out the staff structure from Senior Team down to classroom teachers and support staff with some explanation of the rationale for the structure.

**Q10. How do you hold staff to account?** Rated 1 (weak answer) to 5 (strong answer).

**1** A brief answer which mentions the use of performance management structures but does not go into detail.

**3** A clear explanation of the performance management process with discussion of informal and formal steps.

**5** A clear explanation of the performance management process with discussion of informal and formal steps, as well as an explanation of an underlying philosophy.

**Q11. How do you manage the wellbeing of your pupils?** Rated 1 (weak answer) to 5 (strong answer).

**1** An imprecise answer which emphasises the importance of managing pupil wellbeing but does not lay out any clear strategies for doing so.

**3** An answer which lays out strategies for managing pupil wellbeing, but some strategies are not very precise.

**5** An answer which lays out clear strategies for managing pupil wellbeing alongside an explanation of why this is important.

**Q12. How do you assess pupil progress?** Rated 1 (weak answer) to 5 (strong answer).

**1** An answer which mentions no specific strategies for monitoring the progress that pupils are making.

**3** An answer which outlines strategies for assessing pupil progress in a summative manner only.

**5** An answer which outlines a range of strategies for assessing pupil progress formatively and summatively.

**Yes-or-no questions**

Does your school make use of the following systems:

- Centralised detentions
- An internal exclusion room
- Reward assemblies/prize-giving ceremonies
- Behaviour points/demerits
- Reward points/merits
- Line-ups
- Silent corridors
- After-school revision
- Weekend revision
- Ban on mobile phones in school

Would you like to say anything about your school's use of any of these systems?

**Statement ranking**

For each of the following statements circle the appropriate number:

Each statement is rated 1 (strongly disagree) to 5 (strongly agree).

- The sanction systems are applied consistently by all staff
- The reward systems are applied consistently by all staff
- Teacher turnover is low
- Staff morale is good
- Pupil behaviour in class is generally good
- Pupil behaviour outside of class is generally good
- Pupils feel safe at school
- Pupils feel like their teachers care
- Teachers feel respected by all students
- In general, parents are engaged with their child's education

Would you like to say anything about any of your responses to these questions?

Would you be willing to allow a few members of our team to come and observe the normal functioning of your school for a day?

**To be completed after the interview**

**Rate the confidence with which the headteacher responded to the questions** Rated 1 (not confident at all) to 5 (confident).

**1** There was a lot of hesitation throughout the interview and a lack of clarity in almost all the responses.

**3** The answers were generally clear and delivered without hesitation.

**5** Every answer was clear and delivered without hesitation. There were no wasted words.

**Willingness to reveal information** Rated 1 (very reluctant) to 5 (totally willing).

**1** There was a reluctance to go into specific details throughout the interview.

**3** The headteacher went into specific detail with most answers, but not all.

**5** The headteacher was happy to go into very specific detail in every answer.

**Interviewee patience** Rated 1 (little patience) to 5 (lots of patience).

**1** The headteacher rushed through all their answers.

**3** The headteacher gave detailed responses to most questions but did not add any detail to the yes/no or statement questions.

**5** The headteacher was happy to give detailed responses to all questions and wanted to add detail to their answers to the yes/no and statement questions.

## 2A.4 Inter-rater Reliability Tables

Table 2A.2: Inter-rater reliability: interview questions

| Item | $N$ | $r$ | MAD | Exact% | W1% | $\kappa_w$ |
|---|---|---|---|---|---|---|
| Q2 *How would you describe your school's culture?* | 292 | 0.709 | 0.56 | 40 | 90 | 0.537 |
| Q3 *How do you communicate this culture to your staff?* | 293 | 0.698 | 0.56 | 33 | 94 | 0.499 |
| Q4 *Most important characteristic of an effective school?* | 293 | 0.712 | 0.51 | 41 | 93 | 0.533 |
| Q5 *How does your reward and sanction system work?* | 291 | 0.726 | 0.43 | 44 | 97 | 0.512 |
| Q6 *Key characteristics of an outstanding teacher?* | 293 | 0.613 | 0.52 | 40 | 93 | 0.424 |
| Q7 *How important is extracurricular provision?* | 294 | 0.617 | 0.50 | 41 | 95 | 0.448 |
| Q8 *Your view on marking and feedback?* | 294 | 0.691 | 0.48 | 42 | 94 | 0.552 |
| Q9 *What is the structure of your staff team?* | 294 | 0.623 | 0.45 | 48 | 94 | 0.491 |
| Q10 *How do you hold staff to account?* | 291 | 0.707 | 0.52 | 37 | 92 | 0.547 |
| Q11 *How do you manage the wellbeing of your pupils?* | 292 | 0.681 | 0.45 | 45 | 93 | 0.550 |
| Q12 *How do you assess pupil progress?* | 293 | 0.740 | 0.40 | 53 | 95 | 0.606 |

*Notes:* For each item, a questioner and an observer scored independently on a 1–5 scale. $r$: Pearson correlation; MAD: mean absolute difference; Exact: proportion of pairs with identical scores; W1: proportion agreeing within one scale point; $\kappa_w$: weighted Cohen's $\kappa$ (linear weights).

Table 2A.3: Inter-rater reliability: classroom observation items

| Item | $N$ | $r$ | MAD | Exact% | W1% | $\kappa_w$ |
|---|---|---|---|---|---|---|
| Misbehaviour | 439 | 0.702 | 0.45 | 51 | 93 | 0.566 |
| Response | 277 | 0.669 | 0.59 | 43 | 88 | 0.545 |
| Disruption | 466 | 0.788 | 0.42 | 53 | 93 | 0.638 |
| Concentration | 467 | 0.647 | 0.51 | 39 | 94 | 0.474 |
| Teacher | 465 | 0.636 | 0.58 | 41 | 89 | 0.450 |
| Student | 457 | 0.637 | 0.63 | 37 | 88 | 0.456 |
| Motivation | 467 | 0.610 | 0.53 | 40 | 93 | 0.454 |
| Respectfully | 467 | 0.654 | 0.53 | 44 | 94 | 0.473 |
| Names | 463 | 0.698 | 0.62 | 43 | 88 | 0.524 |
| Praise | 456 | 0.688 | 0.59 | 39 | 90 | 0.525 |
| Interactions | 463 | 0.616 | 0.62 | 35 | 88 | 0.420 |
| Questioning | 468 | 0.585 | 0.61 | 38 | 89 | 0.440 |
| Verbal | 454 | 0.562 | 0.68 | 37 | 85 | 0.417 |
| Discussion | 217 | 0.722 | 0.59 | 41 | 91 | 0.542 |
| Differentiation | 437 | 0.637 | 0.65 | 38 | 84 | 0.476 |
| Explanation | 454 | 0.578 | 0.51 | 44 | 93 | 0.439 |
| Outcomes | 432 | 0.620 | 0.52 | 44 | 91 | 0.464 |
| Methods | 467 | 0.552 | 0.54 | 41 | 92 | 0.424 |
| Structure | 461 | 0.560 | 0.58 | 40 | 90 | 0.398 |
| Resource | 463 | 0.528 | 0.54 | 43 | 91 | 0.378 |

*Notes:* Each row pools all pairwise comparisons across lessons observed by two or three researchers simultaneously. $r$: Pearson correlation; MAD: mean absolute difference; Exact: exact agreement; W1: within-one-point agreement; $\kappa_w$: weighted Cohen's $\kappa$ (linear weights, categories 1–5).

Table 2A.4: Inter-rater reliability: outside-of-lesson items

| Item | $N$ | $r$ | MAD | Exact% | W1% | $\kappa_w$ |
|---|---|---|---|---|---|---|
| Sanctions | 87 | 0.714 | 0.48 | 52 | 91 | 0.611 |
| Rewards | 83 | 0.805 | 0.40 | 63 | 90 | 0.691 |
| Verbal feedback | 76 | 0.485 | 0.45 | 32 | 91 | 0.400 |
| Differentiation | 70 | 0.552 | 0.48 | 27 | 93 | 0.440 |
| Corridors | 93 | 0.657 | 0.49 | 39 | 96 | 0.486 |
| Arrival | 91 | 0.474 | 0.58 | 42 | 91 | 0.334 |
| Students (transitions) | 93 | 0.705 | 0.42 | 47 | 100 | 0.534 |
| Displays | 91 | 0.548 | 0.51 | 54 | 88 | 0.504 |
| Alignment | 84 | 0.578 | 0.62 | 44 | 85 | 0.451 |
| Interactions (break) | 86 | 0.684 | 0.58 | 36 | 92 | 0.524 |
| Relationships (break) | 89 | 0.724 | 0.38 | 54 | 96 | 0.519 |
| Canteen | 91 | 0.727 | 0.38 | 51 | 99 | 0.530 |
| Recreational | 91 | 0.532 | 0.49 | 45 | 92 | 0.388 |

*Notes:* Each row pools all pairwise comparisons across school days observed by two or three researchers simultaneously. $r$: Pearson correlation; MAD: mean absolute difference; Exact: exact agreement; W1: within-one-point agreement; $\kappa_w$: weighted Cohen's $\kappa$ (linear weights, categories 1–5).

## 2A.5 Sample Characteristics

### 2A.5.1 Sample representativeness

Table 2A.5 and Figure 2A.1 compare the interviewed and visited schools (300 and 102 of them matched to the characteristics panel; 303 and 103 in all) to the national population of open secondary and all-through schools in England, across a range of structural characteristics drawn from the school panel spine and the Get Information About Schools register, together with the inspection grade as at August 2024 and the pre-pandemic Progress 8 of 2018–19. The characteristics panel, which is the 2023/24 school-year cut, holds 3,312 of the 3,332 schools in the analysis panel used in the chapter; the grade and Progress 8 rows are computed on the full analysis panel and the full groups, as the table note records.

The interviewed and visited samples are broadly representative of the national population on most structural dimensions. School type (academy, maintained, free school), phase of education, gender composition, admissions policy, and urban–rural classification are all closely matched to national proportions, with no statistically significant differences. Mean school size and free school meals percentage are also close to the national averages: interviewed schools average 1,109 pupils (national mean: 1,079) and a free school meals share of 27.9% (national mean: 29.9%), a difference of two percentage points that is statistically detectable at this sample size but substantively small.

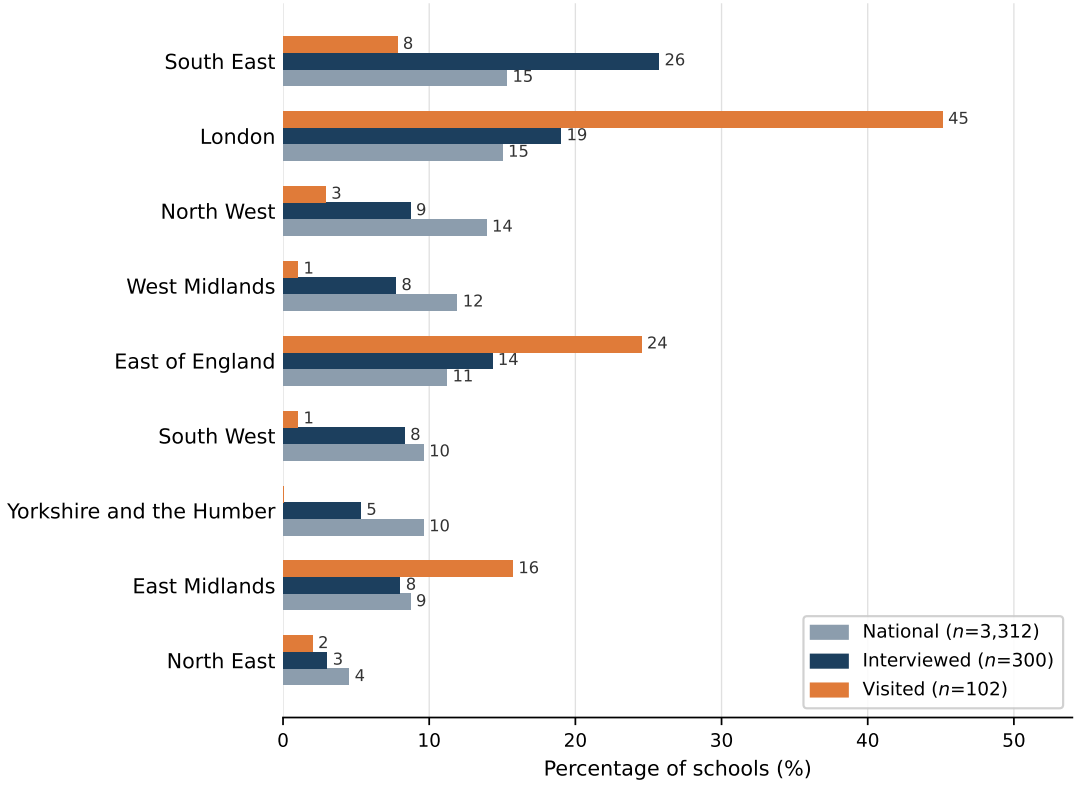
The main exception is regional distribution (Figure 2A.1). Both samples over-represent London, and the interviewed sample also the South East: taken together, London and the South East account for 44.7% of interviewed schools versus 30.3% nationally. The visited sample is particularly concentrated geographically, with London accounting for 45.1% of visits compared to 15.0% of schools nationally, which reflects the practical constraints of carrying out full-day school visits. Northern England is correspondingly under-represented: the North West (13.9% nationally, 8.7% of interviewed, 2.9% of visited), the West Midlands (11.9%, 7.7%,

1.0%), and Yorkshire and the Humber (9.6%, 5.3%, 0.0%) all feature substantially less in the sample than in the population. Visited schools are also more likely to have a sixth form (75.5% versus 62.7% nationally), consistent with the London concentration given that London schools disproportionately offer post-16 provision.

Both samples also lean towards stronger schools: 25 per cent of visited schools held an Outstanding grade as at August 2024 against 15 per cent nationally, and their mean 2018–19 Progress 8 was +0.17 against +0.04 nationally; the interviewed sample shows the same lean more weakly (19 per cent and +0.08).

The consequence for the chapter is the one stated in Section 2.2: the visited sample describes a population that is somewhat more London-based and somewhat stronger than the national average, and the range of the culture measures within it is correspondingly restricted.

Figure 2A.1: Regional distribution of interviewed and visited schools




*Notes:* The comparison is with the national population of open secondary and all-through schools in England. Bars show the percentage of schools in each region for the national population ($n = 3{,}312$), interviewed schools ($n = 300$), and visited schools ($n = 102$), the schools matched to the school panel (the 2023/24 regional characteristics panel holds 3,312 of the 3,332 analysis-panel schools; the remainder lack a region record at that date). Regions are ordered by national share.

Table 2A.5: Interviewed and visited schools against the national population

| *Variable* | *Category* | *Percentage of schools* (%) | | | *p*-value | |
|---|---|---|---|---|---|---|
| | | National ($n$ = 3,312) | Interviewed ($n$ = 300) | Visited ($n$ = 102) | Int. | Vis. |
| *School type* | Academy | 74.7 | 72.3 | 74.5 | *ns* | *ns* |
| | Free school | 9.2 | 10.0 | 11.8 | | |
| | Maintained | 15.8 | 17.7 | 13.7 | | |
| *Phase* | Secondary | 95.1 | 94.3 | 93.1 | *ns* | *ns* |
| | All-through | 4.9 | 5.7 | 6.9 | | |
| *Gender* | Mixed | 89.2 | 87.3 | 85.3 | *ns* | *ns* |
| | Girls | 6.1 | 7.0 | 5.9 | | |
| | Boys | 4.3 | 5.7 | 8.8 | | |
| *Admissions policy* | Non-selective | 85.4 | 87.0 | 88.2 | *ns* | *ns* |
| | Selective | 4.9 | 5.3 | 4.9 | | |
| *Urban/rural* | Urban | 87.4 | 88.3 | 90.2 | *ns* | *ns* |
| | Rural | 12.5 | 11.7 | 9.8 | | |
| *Has sixth form* | Yes | 62.7 | 68.0 | 75.5 | *ns* | p<.01 |
| | No | 37.3 | 32.0 | 24.5 | | |
| *Continuous variables* | | | | | | |
| *Number of pupils* | Mean (*SD*) | 1,079 (396) | 1,109 (384) | 1,133 (378) | *ns* | *ns* |
| *FSM percentage* | Mean (*SD*) | 29.9 (14.7) | 27.9 (14.6) | 29.2 (15.6) | p<.05 | *ns* |
| *Ofsted grade (Aug 2024)* | Outstanding | 14.9 | 18.6 | 25.5 | p<.05 | p<.05 |
| | Good | 69.7 | 71.2 | 64.7 | | |
| | Requires improvement | 12.6 | 7.5 | 7.8 | | |
| | Inadequate | 2.8 | 2.7 | 2.0 | | |
| *Prior P8 (2018–19)* | Mean (SD) | +0.04 (0.45) | +0.08 (0.43) | +0.17 (0.44) | *ns* | p<.01 |

*Notes:* National population is all open secondary and all-through schools in England. *p*-values are approximate: $\chi^2$ goodness-of-fit test (observed vs national proportions) for categorical variables; one-sample $z$-test for continuous variables. Rows with <0.5% share in all columns omitted. The structural rows use the schools matched to the school panel (300 interviewed, 102 visited). The Ofsted-grade and prior-P8 rows are computed on the full groups (visited $n = 103$, interviewed $n = 303$, national $n = 3{,}332$; graded 102, 295 and 3,249; with a published 2018–19 score 95, 259 and 2,731); grade tests are $\chi^2$ against the schools in neither sample and the prior-P8 tests are two-sample $t$-tests. Regional distribution is shown separately in Figure 2A.1.

**Coverage of the text collections.** The three collections of text were each gathered from the register of open state secondary schools at a different date, which is why their raw counts differ. Inspection reports were scored for 3,324 schools, behaviour policies were obtained for 3,363, and websites were crawled for 3,370. Set against the analysis panel of 3,332 schools, the coverage counts and shares are those reported with the data: 3,272 reports, 3,294 policies and 3,311 crawls, of which 3,284 contain scorable prose. The few schools in each collection that are outside the panel are schools that closed, merged or changed identifier between the crawl and the construction of the panel.

## 2A.6 Prediction Model Specification

The text representation is TF-IDF over words and two-word phrases (English stop words removed; terms kept if they appear in at least five documents and at most ninety per cent of them; sublinear term frequency; each document's vector normalised to unit length). The model is a ridge regression with an unpenalised intercept, estimated exactly in kernel form. Every reported correlation is leave-one-out: the model is re-estimated with each school excluded, and the penalty is re-chosen within every fold by closed-form inner leave-one-out, so the held-out school takes no part in estimating the model or choosing its penalty. Predictions are clipped to the range of the training scores. Significance comes from a permutation test: the identical pipeline is run on two hundred permutations of the outcome, and the reported $p$ is the share of permutations matching or beating the real correlation.

Before modelling, documents are cleaned of identifying and clerical material: establishment, trust and local-authority names (from the national register), month names and years, postcodes, tokens containing digits, inspection-report page furniture, letter salutations, and framework boilerplate. The demographic adjustment regresses each outcome on the school's free-school-meals share, share of pupils with English as an additional language, prior attainment and report age, and takes the residual as the target. The mark-scheme bands are not changed. The adjustment is applied to the outcome only. The adjusted figure for a mark scheme is the correlation between the band as awarded and this residual.

Alternative representations were evaluated in the identical harness and none improved on the specification above. In brief, adding three- and four-word phrases moved each correlation by less than its sampling variation. Topic-model features predicted every dimension worse than word and phrase counts did. Models estimated on passage embeddings from two pretrained encoders correlate with enacted warmth and teaching at essentially zero. They do reproduce the espoused-strictness result. So the warmth and teaching signals lie in the particular words

used, not in meaning that survives paraphrase. One result disagrees between the two encoders. The larger encoder predicts enacted strictness at +0.33 with mean pooling. The smaller encoder's correlation is near zero, and so is that of every word-based model. Because it appears under one encoder and not the other, it is recorded here as unresolved rather than used. Models built on sentence tags alone (topic and polarity) predicted every score worse than models built on the words. A second variant multiplied each word count by plus one in a positive sentence and minus one in a negative sentence. This raised the correlation for warmth, strictness and teaching alike. The words responsible were clerical ones from the report format. A feature that helps all three dimensions at once, and does so through format words, is picking up how the report was written rather than what the school is like. The chapter's rule discards such features, so the variant was dropped.

## 2A.7 What the Models Use: Examples

This appendix supports the validity of the warmth and teaching prediction models by showing their evidence in readable units. The models work on weighted word and phrase counts, but a sentence's score is simply the sum of its phrases' weights, so every sentence in the corpus can be ranked by how strongly it pushes a school's score up or down. The listings below give the strongest sentences in each direction for each model, followed by the strongest longer phrases (Table 2A.6). The pattern is consistent: sentences describing adults attending to pupils, and teaching designed and checked, raise the scores; stock assurances and provision boilerplate lower them. One oddity in the teaching listing should be read with care: several of the score-raising sentences are criticisms ("pupils are not remembering knowledge deeply"). A report mentions long-term memory, whether in praise or criticism, only when the school teaches in that way. The model counts words and does not register whether a sentence affirms or denies. For teaching this does little harm, because the vocabulary itself is informative. For strictness it is fatal. The

informative strictness sentences are the ones that say behaviour is not good, and the model cannot tell those from the ones that say it is.

## Warmth

**Sentences that raise the score.**

- "They are taught how to show respect for different opinions and listen to the views of others."
- "Sixth-form students are proud role models and make a highly positive contribution to the school and local community."
- "They listen to staff and take into account the pressures on them."
- "Pupils are encouraged to contribute to the school community."
- "For example, leaders listen carefully to what staff tell them about workloads."
- "Early career teachers are happy here and feel well supported."
- "Staff, including early career teachers, are proud to be part of the school."
- "Sixth-form students act as ambassadors for the school by volunteering in the local community."

**Leader-to-staff warmth.** Sentences about leaders treating staff well also raise the warmth score. These sentences describe relations between leaders and staff, not between staff and pupils.

- "They listen to staff and take into account the pressures on them."
- "For example, leaders listen carefully to what staff tell them about workloads."
- "Early career teachers are happy here and feel well supported."
- "Staff, including early career teachers, are proud to be part of the school."

**Sentences that lower the score.**

- "Many pupils feel confident they have a 'trusted adult' they can go to."
- "Pastoral staff work closely with these pupils and their families."
- "They check closely to ensure that disadvantaged pupils have the support they need."
- "This, along with well-established pastoral support, ensures that pupils are kept safe."
- "This helps to ensure that pupils are kept safe and feel well looked after."
- "Leaders make sure that all pupils develop their talents and interests."
- "Parents and carers speak favourably of the school and appreciate the school's efforts to 'see the whole child'."
- "They know they can go to a trusted adult should they have any concerns."

**Unfiltered listing.** The eight highest- and eight lowest-scoring sentences regardless of topic.

- "They are taught how to show respect for different opinions and listen to the views of others."
- "They hold leaders strictly to account for the quality of the school's provision and are highly effective."
- "They listen to others and express their views in a well-managed environment."
- "Sixth-form students are proud role models and make a highly positive contribution to the school and local community."

- "Those responsible for governance know the school's context and provide support and professional challenge where necessary to ensure that the school's high standards are maintained."
- "In lessons, pupils are diligent, focused, and eager to learn."
- "They listen to staff and take into account the pressures on them."
- "Pupils are encouraged to contribute to the school community."
- "However, some pupils, particularly some disadvantaged pupils, are absent too often."
- "Many pupils feel confident they have a 'trusted adult' they can go to."
- "(Information for the school and appropriate authority) In some subjects, the work given to some pupils does not enable them to develop their ideas and extend their answers."
- "Pupils take part in a range of extra-curricular activities to develop their talents and interests, for example basketball, choir, photography, and cookery."
- " Sometimes, the work given to pupils, including those with SEND, does not enable them to achieve the aims and ambition of the curriculum."
- "However, in some subjects, the work given to some pupils does not enable them to develop their ideas and extend their answers as well as they could."
- "Missing out on important learning means that these pupils do not achieve as well as they might."
- "Pastoral staff work closely with these pupils and their families."

## Teaching

**Sentences that raise the score.**

- "In a few subjects, however, pupils are not remembering knowledge deeply in the long term."
- "In lessons, most pupils focus on their learning and learn well."
- "They carefully choose the core knowledge they want pupils to know and remember in the long term."
- "Teachers help pupils to embed learning in their long-term memory."
- "They teach in a routinised, structured way that helps develop long-term memory."
- "This means that some pupils do not always retain knowledge in their long-term memory."
- "Teachers should ensure that pupils' writing is of a high quality by checking that they are spelling correctly and using correct punctuation."
- "In lessons, some pupils with SEND get support from learning support assistants."

**Sentences that lower the score.**

- "The personal, social and health education (PSHE) curriculum is designed to support pupils' wider development."
- "On other occasions, adults do not support pupils to complete work to a high standard."
- "The school has carefully designed the personal, social and health education curriculum so that it supports pupils' wider development."
- "It does so, not only by having high expectations for pupils, but by supporting their individual aspirations and talents."

- "The curriculum in personal, social and health education is designed to help pupils understand important ideas."
- "Well-planned personal, social, health education ensures that pupils learn about relevant and sensitive issues, including personal safety, relationships, sexual health and consent."
- "The school recognises the importance of ensuring that pupils, particularly disadvantaged pupils, and pupils with SEND, gain wider opportunities beyond the curriculum."
- "This is proving successful in helping these pupils become more confident and fluent readers."

**Unfiltered listing.** The eight highest- and eight lowest-scoring sentences regardless of topic.

- "Those responsible for governance know the school's context and provide support and professional challenge where necessary to ensure that the school's high standards are maintained."
- "Leaders are supported and challenged by dedicated and knowledgeable governors, who share their vision of high standards in a supportive school."
- "In a few subjects, however, pupils are not remembering knowledge deeply in the long term."
- "The school's motto is 'excellence with care.' Pupils testify to how this works in practice."
- "They lead charity initiatives and community-based projects to instil a keen sense of responsibility and empathy for others."
- "In lessons pupils settle to their work quickly and show high levels of respect to their teachers and to other pupils."

- "They are taught how to show respect for different opinions and listen to the views of others."
- "In lessons, most pupils focus on their learning and learn well."
- "The personal, social and health education (PSHE) curriculum is designed to support pupils' wider development."
- "Pupils take part in a range of extra-curricular activities to develop their talents and interests, for example basketball, choir, photography, and cookery."
- "On other occasions, adults do not support pupils to complete work to a high standard."
- "Judgements in this report are based on the current inspection framework and also reflect changes that have happened at any point since the last graded inspection."
- "The school offers pupils a range of ways to develop their wider skills and talents."
- "The school has carefully designed the personal, social and health education curriculum so that it supports pupils' wider development."
- "It does so, not only by having high expectations for pupils, but by supporting their individual aspirations and talents."
- "The curriculum in personal, social and health education is designed to help pupils understand important ideas."

Table 2A.6: The phrases with the largest weights

| Warmth | | Teaching | |
|---|---|---|---|
| Raises | Lowers | Raises | Lowers |
| listen | closely | writing | standard |
| lessons pupils | welcoming | lessons pupils | case |
| small number | standard | listen | staff work |
| order | feel confident | share | health |
| exemplary | given pupils | standards | expectations pupils |
| lesson | pupils particularly | explain | pupils achieve |
| guidance | expectations pupils | highly | curriculum school |
| leaders check | including send | lesson | disadvantaged pupils |
| managed | international | motto | pupils kept |
| contribute | case | students benefit | broad ambitious |

*Notes:* What the prediction models use: the ten phrases with the largest positive and negative weights in the warmth and teaching models (inspection reports, demographics removed from the target). Weights from a model fitted to all schools at once, using the penalty value selected in the leave-one-out runs. They illustrate what the model uses. They are not the held-out results.

## 2A.8 Further Parent-Comparison Checks

This section collects the checks that qualify the parent comparison of Section 2.4.5: what the visited schools show, a stricter form of the grade check, and a test that the report text does not contain the parents' answers in a form the model uses.

The visited schools complicate the picture. There the parent measure can be checked against both what the observers saw and what the head said. Against enacted warmth it correlates at +0.17, positive but too imprecise to rely on ($p = 0.08$, $n = 101$). Against the head's espoused warmth it correlates at +0.43. Parents agree with the head more than with the observers. There are two ways to interpret that and the data cannot tell us which is correct. The first is that parents and heads get a picture of the school across an entire year or longer, while the observers saw just one day. On that basis the parents' views provide evidence for the head's account. The second possibility is that the parents and the head, both invested in the school, may report similar things, whatever the school is actually like in practice. The survey therefore cannot judge between what schools say and what observers see.

The check in the main text uses the grade from an earlier inspection. A stricter version uses the grade awarded by the same inspection whose report the model reads. Comparing schools with the same such grade halves the warmth correlation, to +0.16. The grade and the prose come from one visit, so the grade already reflects some of the warmth the prose describes. Removing it therefore removes some warmth as well as a rival explanation. The national parent correlation also falls by about half when the grade is held constant, from +0.29 to +0.14. For scale, the grade alone says little about warmth: on the visited schools it correlates with enacted warmth at only −0.20 (grades run from 1, Outstanding, to 4, so a negative sign means better-graded schools are warmer). The grade alone is only weakly related to warmth. The text prediction is therefore using information beyond the grade.

One caveat needed testing. Inspectors read Parent View during inspections,

and about 3 per cent of reports quote it. The correlation with parents could therefore arise because the report text contains the parents' own answers. Deleting every sentence that mentions the survey and re-running the whole prediction leaves the correlation unchanged at +0.29. The quoted survey material is not what drives the result.

## 2A.9 The Mark Schemes

The seven schemes below are reproduced word for word. They were finalised before any document was scored and not changed afterwards. Each was applied once per document by gpt-4o-mini (version-pinned), which returned a band and one quoted evidence sentence. Thirty documents per source were re-scored with caching disabled, with the agreement figures reported in the chapter, and the full grid was re-scored once by gpt-4o as a sensitivity check, reported alongside and never used to select results.

The pinned model versions: every scheme's primary scorer is `gpt-4o-mini`, pinned to the dated snapshot `gpt-4o-mini-2024-07-18`; the whole-grid sensitivity re-score uses `gpt-4o`, pinned to `gpt-4o-2024-08-06` (`model_pins.py`, verified against the provider's model list before scoring).

### 1. Inspection reports — Warmth

What is measured: how the report describes the relationships between staff and pupils. Score the relational quality described, never the pastoral provision listed.

1. Relationships described as poor: pupils unknown to staff, uncared for, or unsafe because of how staff relate to them.

2. Any credible negative relational signal — pupils not always heard, bullying unresolved, staff not accessible — even alongside positives.

3. DEFAULT. A safe and generally positive school, described through labels, provision or outcomes ("caring school", "pupils feel safe", "pastoral team is effective") without describing how staff and pupils actually relate.

4. The report describes the relationships themselves: staff know pupils well, warm interactions observed, pupils feel valued by staff as individuals.

5. Relational warmth described as a defining feature of the school, in emphatic terms, across several passages.

Rules: if the report does not discuss relationships, score 3 — silence is not coldness. The 3/4 line: does the sentence describe how staff and pupils relate, or the school's character and what provision exists? In doubt between two bands, take the lower.

## 2. Inspection reports — Strictness

What is measured: what the report says about conduct standards and how consistently the adults enforce them.

1. Disruption is a persistent, documented problem; rules unclear or not applied.

2. Standards exist but enforcement is patchy; behaviour is named as an area for improvement, or inspectors saw low-level disruption.

3. DEFAULT. Clear expectations, generally maintained; behaviour mentioned but unremarkable.

4. The report as a whole presents conduct as a strength, in lessons and around the building.

5. Conduct is exceptional and unmistakably a defining feature of the school.

Rules: if behaviour is simply not discussed, score 3 — a 1 or 2 needs positive evidence of a problem. Pupils' opinions about the rules are warmth evidence, never strictness evidence: strictness is what the adults do. In doubt between two bands, take the lower.

## 3. Inspection reports — Teaching

What is measured: how the report describes classroom teaching — explanations, subject knowledge, and the checking of understanding. Curriculum-design language (intent, sequencing) is background, not evidence.

1. Teaching named as a serious failure and a cause of poor outcomes.
2. Teaching inconsistent ("some teachers... not always") and named as requiring improvement.
3. DEFAULT. Competent and unremarkable, or classroom delivery not discussed at all.
4. Teaching described positively and with specifics across most of the school ("typically", "most teachers").
5. Teaching described as expert and a defining strength ("consistently", "across all subjects").

Rules: score how teachers teach, not what the curriculum intends. The consistency words carry the ladder: "consistently" points to 5, "typically" to 4, "some" to 2. In doubt, or with no delivery evidence, score 3.

## 4. Behaviour policies — Strictness

What is measured: how far the policy removes staff discretion and prescribes pupil conduct. Score the machinery the policy describes, not how severe its language sounds.

1. No consequence language at all — the policy is aspirational or entirely about rewards.
2. Consequence language is present, but no day-to-day system a new member of staff could act on tomorrow.

3. DEFAULT. A working day-to-day system, applied by staff judgement from a menu of options. The standard English secondary policy.

4. A named tracking or tier system: a pupil's position in the system, not staff judgement, decides which consequence follows.

5. As 4, and the policy prescribes a default deportment norm — a SLANT-type routine or a similar required way of sitting, attending or moving.

Rules: the test for 3 — could a new teacher read this policy and know what to do tomorrow when a lesson is disrupted? Harsh penalties at staff discretion are LESS strict on this construct than mild ones applied automatically.

## 5. Behaviour policies — Warmth

What is measured: what the policy tells staff about how to relate to pupils. Rewards and pastoral structures are provision, not warmth.

1. No positive or relational content beyond the sanctions system.

2. Positive values asserted ("every child matters") with no described mechanism and no guidance to staff.

3. DEFAULT. A positive mechanism described — named rewards with the occasions they are given on, or named pastoral routes — but no guidance to staff on how to relate to pupils. Most policies.

4. Explicit operational guidance to staff on relating to pupils as individuals: greet at the door, learn names, how to hold a repair conversation.

5. As 4, and relational quality is the organising principle: consequences framed as relationship-preserving, and the policy says the relationship survives the sanction.

Rules: a scheduled restorative meeting is machinery; only telling staff how to conduct it is guidance. If in doubt whether a sentence is guidance, it is not — a rich reward economy cannot pass 3.

## 6. Websites — Strictness

What is measured: whether the site's high-expectations language is about behaviour at all, and how far the school goes in claiming strictness of itself.

1. No mention of high standards or high expectations anywhere — even a cursory mention moves it up.

2. High standards or expectations claimed, but generic — not tied to behaviour or conduct.

3. DEFAULT. High expectations explicitly about behaviour or conduct — or the school asserting it is demanding, making no apologies.

4. Behaviour treated as central: expected at all times, pupils required to become self-disciplined, rewards-and-sanctions machinery on the site, or behaviour named as the foundation of success.

5. The school states that it is strict: strict routines and protocols, zero tolerance, no excuses, the highest possible standards.

Rules: aspiration language alone ("achieve their full potential") does not reach band 2. It is the explicit statement of strictness that earns the 5, not the severity of the machinery.

## 7. Websites — Warmth

What is measured: how far the site presents care for pupils and staff–pupil relationships as what the school is about, in the school's own words. The axis: absent — courtesy — claimed — stressed — central.

1. No care or support language at all. The school presents itself entirely through results, facilities, opportunities and ambition.

2. The courtesy register only: generic supportive-atmosphere phrases ("a supportive learning environment", "care and support for each individual") appearing in passing, typically once inside the head's welcome, and not returned to.

3. DEFAULT. Care claimed as a feature of the school: pastoral care named, support for individual pupils emphasised, or a caring atmosphere claimed prominently or in more than one place.

4. Care or relationships stressed as important and returned to: pupils described as known and valued as individuals, care presented as part of what the school is for, across more than one page or passage.

5. Staff–pupil relationships explicitly named and presented as central to what the school is — and more than once. Rare by design.

Rules: the 2/3 line is courtesy versus feature — if the care language lives only in the head's welcome paragraph, score 2. Band 5 needs the relationships themselves named (not just "caring" or "supportive"), plus centrality, plus recurrence. Quoted Ofsted praise does not count; the school's own and its trust's words do. In doubt between two bands, take the lower.

# Chapter 3

# School Culture and Pupil Progress: Evidence from English Secondary Schools

*Damian Phelan*[1]

***Abstract.*** *This paper asks whether the culture of a school affects how much progress its pupils make. I measure the warmth and strictness of 103 English secondary schools by direct observation, and relate those scores to Progress 8, the Department for Education's measure of pupil progress. The estimates are associations, with intake, school characteristics and the pre-pandemic inspection grade held constant. There are four main results. First, both dimensions are associated with progress: a one-standard-deviation increase in warmth is associated with* 0.31 *of a standard deviation of Progress 8, and a one-standard-deviation increase in strictness with* 0.23*. Second, the quality of the teaching observed does not account for either association. Third, schools in the top half on both dimensions make* 0.86 *of a standard deviation more progress than the rest, and their advantage is the sum of the two separate associations: the coefficient on the interaction is not significantly different from zero. Also within the third result, the warmth association depends on how warmth is measured: replacing the observers' scores with the headteacher's own ratings of the same schools leaves no warmth association, while strictness remains associated with progress. Fourth, the results extend nationally: culture scores obtained from inspection reports predict progress across effectively all English state secondary schools. A one-standard-deviation increase in the strictness score is associated with 0.13 of a grade, and in the warmth score with 0.08.*

[1]Department of Economics, University College London, and Institute for Fiscal Studies. The school visits and headteacher interviews were approved by the UCL Research Ethics Committee (project 22979.001). I thank seminar participants at the Institute for Fiscal Studies and UCL for comments.

## 3.1 Introduction

This paper asks whether the culture of a school affects how much progress its pupils make. I estimate the association between the warmth and strictness of English secondary schools, measured by direct observation, and the progress their pupils make between the end of primary school at age 11 and their GCSEs, the national examinations taken at 16. Warmth is the quality of the relationships between adults and pupils, in lessons and around the school. Strictness is the orderliness of the school and how consistently its behaviour rules are enforced. These are the two dimensions of culture throughout the chapter. I use the enacted culture scores that were constructed in the previous chapter for the 103 visited schools, enacted meaning built from what the observers saw rather than from what the school says about itself, together with the Progress 8 measure published by the Department for Education, which records each school's average pupil progress relative to pupils nationally with the same prior attainment.

Whether the school a child attends makes a difference beyond the ability and family circumstances the child arrives with, and if so through what mechanism, is a long-standing question. The early answers were negative. Coleman et al. (1966) found that school resources explained little of the variation in attainment once family background was taken into account, and Hanushek (1997), reviewing the evidence three decades later, found no consistent relationship between expenditure and performance. The class-size studies that followed (Krueger, 1999; Krueger and Whitmore, 2001; Angrist and Lavy, 1999) showed that resources can matter, and English evidence finds small positive effects of spending concentrated among disadvantaged pupils (Levačić et al., 2005; Jenkins et al., 2006; Holmlund et al., 2010; Machin et al., 2010), but these effects are modest against the school-to-school variation they leave unexplained.

Two lines of work have looked for the source of those school-to-school differences in something other than resources. Rutter et al. (1979), studying twelve inner-London secondary schools, placed it in what they called ethos, meaning the

expectations and routine practice that a school sustains, and the effective-schools syntheses that followed listed an orderly environment and a supportive climate among its correlates (Edmonds, 1979; Sammons et al., 1995; Teddlie and Reynolds, 2000). The teacher-quality literature placed it in the classroom: Rivkin et al. (2005) and Chetty et al. (2014) show that variation in teacher effectiveness is large and consequential, and English estimates (Slater et al., 2012; Murphy and Machin, 2011) are of the same order. What neither line of work settles is whether a school's culture, as distinct from the quality of its individual teachers, predicts how much progress its pupils make, and that is the question here.

Two strands of the literature are closest to this paper. The first is the authoritative-school literature (Pellerin, 2005; Gregory et al., 2010), which builds on the parenting typology of Maccoby and Martin (1983), in which the combination of responsiveness and demandingness predicts the best child outcomes (Majumder, 2016; Zhang et al., 2017). Warmth and strictness are the school-level counterparts of that pair, and the literatures that follow each have their own names for the same two things: responsiveness, support and warmth on one side; demandingness, structure and discipline on the other. This chapter uses warmth and strictness throughout. The literature finds that schools which combine the two produce better behavioural and engagement outcomes, but it has rarely had an academic-progress outcome or a direct measure of both dimensions.

The second strand is the literature on high-expectation, high-structure charter and academy models, which finds large effects of a bundle of practices. In the United States, effective charter schools are marked by strict discipline and high expectations (Abdulkadiroğlu et al., 2011; Angrist et al., 2010); an index of five such practices explains about half of the variation in charter effectiveness, while class size and spending explain none of it (Dobbie and Fryer, 2013); and the bundle transplanted into ordinary public schools raises mathematics attainment (Fryer, 2014). Chabrier et al. (2016) and Epple et al. (2016) review the wider evidence. Most charter schools do no better than the ordinary public schools their

pupils would otherwise have attended, and the effective ones are again those with strict discipline and high expectations.

In England, academies are state-funded schools that operate outside local-authority control, the closest English counterpart to a charter school. The first wave, which targeted under-performing schools and gave them autonomy, raised attainment (Machin and Vernoit, 2011; Eyles et al., 2016). The later converter academies show much weaker effects (Eyles et al., 2018). This suggests that autonomy itself is not the operative channel. Machin and Sandi (2020) show that the early academies' higher exclusion rates reflected a change in discipline rather than the removal of weak pupils. What the improving academies demonstrably changed, then, was behaviour, which is the strictness dimension this paper measures directly.

In all of this work, warmth and strictness are never observed separately. An effective charter school of this kind is strict and warm at once: tight behaviour rules, and adults who know every pupil individually. The studies compare schools that run the whole package with schools that do not, so they can say that the package works, but not which part of it does. This paper separates the parts. It measures warmth and strictness separately, in ordinary schools rather than in schools designed around that combination, and estimates the association of each with progress while holding the other constant.

The estimates are associations, not causal effects, and every one is presented as such. What the design does is narrow the list of other explanations. The outcome is value-added, so the largest confound in any school comparison, that stronger schools take stronger pupils, is removed at the start. The regressions then hold constant intake composition, prior attainment, school type, size, location and the school's pre-pandemic inspection grade, so a culture coefficient should not simply capture who attends the school or how it was regarded before the outcome cohorts arrived.

The analysis is then extended beyond the visited schools. The previous chap-

ter's text-based measures give culture scores for effectively every state secondary school in England. The associations reported below appear again across that national population, on 3,052 schools and all significant at the 1 per cent level, so they do not depend on which schools agreed to be visited. The one thing the design cannot rule out is a persistent characteristic of a school that produces both its culture and its results.

There are four main results. First, both dimensions are associated with progress. A one-standard-deviation increase in enacted warmth is associated with Progress 8 that is 0.15 of a grade higher; a one-standard-deviation increase in enacted strictness with 0.11 of a grade. In standard deviations of Progress 8 across the schools these estimates use, that is 0.31 and 0.23. The two estimates are close in size, and the difference between them is not statistically significant, so the data do not rank the two dimensions. Both hold whether the inspection grade is left out, or the school's current grade is used, or the grade it held before the pandemic. Leaving the grade out can make the warmth estimate too large, because warm schools may simply be good schools, and good schools do well on everything. Including it can make the estimate too small, because inspectors award the grade partly on the behaviour and relationships the warmth score measures, so controlling for the grade takes away some of the warmth too. The estimate hardly changes whichever way the grade is handled, so neither problem is big enough to matter.

The two associations differ in how they depend on the controls. The strictness association is visible with no controls at all: stricter schools simply make more progress. The warmth association is not. At first sight warmth appears unrelated to progress, and the reason is an intake pattern that hides it. Schools with many pupils who speak English as an additional language tend to make more progress, and the observers scored those schools lower on warmth. A comparison of all schools together therefore combines two things. Warmth is positively associated with progress. The share of pupils with English as an additional language (EAL)

is positively associated with progress and negatively with scored warmth. The two roughly cancel. Compare schools with similar intakes and the warmth association emerges. The sensitivity analysis later in the chapter finds that an omitted variable would need to be more strongly related to both the score and progress to remove the warmth association than to remove the strictness association.

The two dimensions also carry separate information. Entered on its own, each has a coefficient of about 0.2. Entered together, each falls, but neither falls to zero, so each is associated with progress in a way the other does not capture.

Second, conditioning on enacted teaching quality does not account for the association. Adding the visit's teaching score leaves both culture coefficients essentially unchanged, and the teaching coefficient itself is close to zero. This matters because teaching quality is the most obvious rival explanation: a warm and orderly school might simply be a school whose teachers teach well. On this evidence, the culture association is not teaching quality under another name.

Third, schools in the top half on both warmth and strictness make two-fifths of a GCSE grade more progress per subject than the rest. Their advantage is the sum of the two separate associations: the coefficient on the interaction of warmth and strictness is not significantly different from zero (0.039, $p = 0.27$). The third result has a second part: the warmth that predicts progress is the warmth the observers saw, not the warmth the head described. Replacing the observers' scores with the headteacher's own ratings of the same schools leaves the warmth coefficient at 0.051 ($p = 0.29$), while the strictness coefficient is 0.120 ($p = 0.018$), close to its value on the observers' scores.

The fourth result is the national extension. Through the previous chapter's text-derived scores, predicted warmth, predicted teaching and mark-scheme strictness each predict progress across the national population. The associations hold with the inspection grade held constant, and they hold on reports written before either outcome cohort's GCSE results existed, so the scores are not simply the inspection grade in another form, and the associations do not come from inspectors describing

results they already knew.

The next section describes the data. The paper then sets out the specification and what it can and cannot identify, reports the results and their robustness, and concludes.

## 3.2 Data

**Outcomes.** The outcome is Progress 8, which is the Department for Education's school-level value-added measure. For each pupil, attainment across eight GCSE slots, weighted towards English and Maths, is compared with the national average attainment of pupils who left primary school with the same Key Stage 2 result, and the differences are then averaged across the school's cohort. A school at zero is exactly average for its intake. A score of $+0.5$ means that its pupils achieved half a grade more per subject than pupils elsewhere with the same starting point, and $-0.5$ means half a grade less. Nationally the measure has a standard deviation across schools of about 0.5 and an interquartile range of about two-thirds of a grade. This means that the difference between a school at the first quartile and one at the third is roughly two-thirds of a grade in every subject for every pupil.

The outcome is a value-added measure, and value-added measures have their critics. Gorard (2010); Gorard et al. (2013) argue that school-level estimates are dominated by error and unstable from year to year. I use Progress 8 nonetheless, for three reasons. It is the measure that the Department for Education publishes and by which schools are judged; it conditions on the single strongest predictor of GCSE performance, which is prior attainment at Key Stage 2; and it allows the visited schools to be placed against every school in the country on the same footing. I meet the concern about instability by averaging two cohorts and by showing, in the appendix, that a school's Progress 8 is stable across years (Section 3A.2) relative to the differences between schools.

I report overall Progress 8 and its English and Maths components, which are the cleanest measures the tables offer because every pupil sits both of those exam-

inations. The remaining components, EBacc and Open, are partly determined by which subjects a school enters its pupils for, and they are reported in Section 3A.6 together with a direct analysis of subject entry rates; Section 3A.1 sets out how the components are constructed.

The primary outcome is the two-year average of the 2022/23 and 2023/24 cohorts, which are the two post-pandemic years for which a Key Stage 2 baseline exists. Averaging the two reduces the year-to-year noise in a single cohort's figure, and it also dilutes any cohort-specific disruption: the Department advised caution over the 2022/23 measures because of the pandemic's effect on that cohort. The 2024/25 cohort has no baseline, because its Key Stage 2 tests were cancelled in 2020, and Section 3A.6 reports a pseudo-progress measure for it that correlates +0.81 with real Progress 8.

**Culture measures.** The explanatory variables of interest are the enacted warmth and strictness scores of the previous chapter, measured on a 1–5 scale for the 103 visited schools. They are entered per standard deviation so that the two coefficients are comparable. Across the visited schools enacted warmth has a mean of 3.2 and a standard deviation of 0.5, and enacted strictness a mean of 3.5 and a standard deviation of 0.4, so a standard deviation is about half a point on the five-point scale.

Two properties of these scores, which were established in the previous chapter, bear on how the estimates below should be read. The first is that a school's score is an average over lessons that differ from one another more than schools differ from each other. Within-school lesson variation is about three times the between-school variation for in-lesson strictness. This means that "a strict school" here is a school whose lessons are orderly on average, and not one in which every lesson is.

The second is that the scores carry measurement error. Their reliability is about 0.7, meaning that about 70 per cent of the variation in a score across schools reflects real differences between schools and the rest is error. For a score

entered on its own, error of this kind biases the coefficient towards zero, so those estimates are conservative. When the two scores enter together, the effect of measurement error is harder to state. Warmth and strictness are correlated. When two correlated scores enter the same regression, each coefficient measures the part of the association that is not shared with the other. Noise interferes with exactly that decision: when one score is measured with more error than the other, some of its association is credited to the other score instead. So when the two scores enter together, the error makes the division between the two coefficients uncertain: part of the warmth association can appear as strictness, and the reverse. The error cannot produce the overall association. It moves estimates towards zero, so if anything the association between culture and progress is understated. Section 3A.3 makes that argument formally, treating the two raters' scores as two independent measurements of the same school. About a third of the observed lessons, and 32 of the 103 visits as a whole, were rated by one researcher rather than two, and the robustness section checks that the estimates do not depend on those schools.

The visit's teaching-practice score, built from the ten lesson items on questioning, feedback, explanation and structure, enters as the rival explanation to be tested. It is the most obvious rival explanation for any association between culture and progress, since a school whose teachers are warm and orderly may also be a school whose teachers teach well. The espoused scores, from the headteacher interview, enter one robustness specification.

Three text-derived scores enter the national extension, all built and validated in the previous chapter and all using the school's most recent inspection report with its stated grades removed. Warmth and teaching come from prediction models: each learns, from the visited schools, which words and phrases of a report go with higher enacted scores, and each visited school's prediction is made by a model fitted on the other visited schools, so a school's own report is never in the training data for its own prediction. Each enacted score was first regressed

on the intake controls, and the models were trained to predict the residual from that regression. A model that only identified a school's intake from its report would therefore predict nothing. The correlation between the predictions and the enacted scores is +0.33 for warmth and +0.28 for teaching. Strictness is measured differently, by a one-page mark scheme, written and fixed before any report was scored, and applied once per report by a language model. Its bands correlate with enacted strictness at +0.30 before the demographic adjustment of the previous chapter, and +0.25 after it.

One further website-derived measure enters an exploratory section only: an indicator of a traditional pedagogy.

The two dimensions are measured by different methods, for a reason given in the previous chapter. Warmth is predicted from the frequency of ordinary words and phrases in a report. Strictness is not. Reports signal strictness in a few specific statements about behaviour, so strictness is scored by a language model applying a mark scheme that lists those statements. Figure 3.2, reported with the national extension, shows the national distribution of the strictness bands.

**Controls.** All specifications condition on the same set of controls. The first is the mean Key Stage 2 score of the cohort. Progress 8 already compares each pupil with pupils who left primary school at the same level, so intake should not matter to it, but the adjustment may be imperfect. The control protects the estimates if it is. The next are the shares of pupils eligible for free school meals, with English as an additional language (EAL) and with special educational needs, which are the composition measures most strongly associated with progress.

Then comes school size, logged so that the control compares proportional rather than absolute differences in size; academy status, urban location and selective admissions; and the number of years since the school's last inspection, since a school that was inspected long ago is described by an older report.

Finally, the school's overall inspection grade as at August 2019 enters as indicators for grades below Good, pooling Outstanding with Good. Outstanding

schools were exempt from routine inspection before 2020, so an Outstanding grade was usually awarded longer ago than any other grade, and is the most out of date. I use the 2019 grade rather than the current one because a contemporaneous grade is partly an outcome of the culture being measured. I enter it at all because it is the best available summary of how the school was regarded before the outcome cohorts arrived.

Seven of the 103 visited schools carry no 2019 grade under their current identity in the Department's records, mostly because a school that converts to an academy is registered afresh under a new reference number, and the old grade stays with the old registration. In these cases the school itself usually continues, with the same site and pupils, and its earlier registration is recorded as its predecessor. One of the seven is excluded from the outcome regressions anyway as a late-entry school (schools admitting only from age 13; defined below), and one lacks the years-since-inspection control, so five remain in the estimation sample. For four of the five, the predecessor's 2019 grade fills the gap. The fifth has no predecessor recorded in the Department's register, so no 2019 grade exists to inherit, and it is dropped where the grade control is required.

**Sample.** The estimation sample starts from the 103 visited schools, and three exclusions reduce it. The first is schools whose statutory entry age is 13 or above (university technical colleges, studio schools and similar; 98 schools nationally, two of them among those visited), which are excluded from every outcome regression. Progress 8 measures progress from age 11, and these schools enrol their pupils at 13 or 14, so their figure attributes to them progress that was made elsewhere. The second is one visited school that has never been inspected under its current registration, so the years-since-inspection control cannot be computed for it. The third is the school with no recorded predecessor, described above. The primary estimation sample is therefore $n = 99$: 103 schools, less the two late-entry schools, one missing control and one unfillable grade.

Table 3.1 and Table 3.2 report descriptive statistics for the outcomes and

the controls. The visited schools' mean Progress 8 is about a quarter of a grade above zero, with a standard deviation a little below the national one, which is the restricted range noted in the previous chapter.

The visited schools are stronger on average than the national population, so the estimates apply most directly to a population somewhat above the national average. The sample also contains few very weak schools, so both the culture scores and Progress 8 cover a narrower range than they do nationally. An association is estimated from differences between schools, so when the schools differ less, an association is estimated less precisely, and a real association becomes harder to detect. If anything, then, this feature of the sample understates the associations; it cannot have created them. Two checks on the sample follow. The first asks whether the visited schools look like other English secondary schools. The second asks whether the culture scores are simply correlated with the schools' intakes.

On the first, the visited schools are close on the controls to the interviewed schools, the larger pool of 303 from which they were recruited (the descriptive tables report both groups), and to the country as a whole: 29 per cent of pupils eligible for free school meals, 16 per cent with special educational needs, three-quarters academies, nine in ten urban, one in twenty selective, and a 2019 grade distribution of roughly a quarter Outstanding, five-eighths Good and one in nine below Good. The one visible difference is the share of pupils with English as an additional language: 28 per cent, against 21 per cent in the interviewed sample. The difference arises because London schools are over-represented among the visited schools, and London has a high share of such pupils. The share is controlled for throughout.

On the second, the correlations between the culture scores and the controls, reported in Table 3A.17, are all below |0.3|. The culture measures are not close proxies for intake or school type, so the coefficients below cannot simply be reflecting differences in pupil composition.

Table 3.1: Progress 8 outcomes: summary statistics

| Outcome | Sample | *N* | Mean | SD | Min | Max |
|---|---|---|---|---|---|---|
| P8 Overall (2-yr avg) | Visited | 103 | 0.252 | 0.478 | -0.700 | 1.345 |
| | Interviewed | 299 | 0.095 | 0.487 | -1.655 | 1.345 |
| | National | 3,240 | -0.015 | 0.510 | -3.570 | 2.465 |
| P8 English (2-yr avg) | Visited | 103 | 0.256 | 0.507 | -0.750 | 1.545 |
| | Interviewed | 299 | 0.093 | 0.485 | -1.485 | 1.545 |
| | National | 3,240 | -0.017 | 0.519 | -4.035 | 2.360 |
| P8 Maths (2-yr avg) | Visited | 103 | 0.270 | 0.449 | -0.595 | 1.345 |
| | Interviewed | 299 | 0.093 | 0.470 | -1.370 | 1.345 |
| | National | 3,240 | -0.017 | 0.487 | -2.030 | 2.660 |
| Att8 English (2024–25) | Visited | 103 | 10.658 | 1.757 | 7.900 | 17.300 |
| | Interviewed | 300 | 10.277 | 1.696 | 6.700 | 17.300 |
| | National | 3,274 | 10.049 | 1.702 | 0.100 | 17.300 |
| Att8 Maths (2024–25) | Visited | 103 | 9.887 | 1.975 | 6.700 | 17.400 |
| | Interviewed | 300 | 9.514 | 1.891 | 5.400 | 17.400 |
| | National | 3,274 | 9.263 | 1.879 | 0.300 | 17.600 |

*Notes:* The interviewed sample numbers 303 schools and contains the 103 visited schools; *N* falls short of 303 where an outcome is unpublished for a school. P8 outcomes are 2-year averages of 2022–23 and 2023–24 component scores where both years are available. Att8 2024–25 components are the contemporaneous robustness outcome; no Progress 8 is available for this cohort because the cohort were in Year 6 in 2019–20 when Key Stage 2 assessments were cancelled.

## 3.3 Empirical Specification

The primary specification is

$$Y_s = \alpha + \beta_W W_s + \beta_S S_s + \mathbf{X}_s'\gamma + \sum_g \delta_g \mathbb{1}[G_s = g] + \varepsilon_s, \tag{3.1}$$

where $Y_s$ is the two-year average Progress 8 (overall, English or Maths) of school $s$, $W_s$ and $S_s$ are its enacted warmth and strictness scores in standard-deviation units, $\mathbf{X}_s$ is the set of controls listed above and $G_s$ is its 2019 inspection grade. Standard errors are heteroskedasticity-consistent (HC3), which is the small-sample variant that is appropriate when $n$ is near 100.

I estimate three stages: equation (3.1); the same equation with the teaching-practice score $T_s$ in place of $W_s$ and $S_s$; and the equation with all three. Each dimension is also entered on its own: warmth without strictness, and strictness without warmth, in separate regressions.

The national extension uses the same equation on the national sample, with a text-derived score in place of $W_s$ and $S_s$. There are three such scores from the

Table 3.2: Control variables: summary statistics

| | Visited ($N = 103$) | | | Interviewed ($N = 303$) | | |
|---|---|---|---|---|---|---|
| Variable | $N$ | Mean | SD | $N$ | Mean | SD |
| *Pupil composition and prior attainment* | | | | | | |
| Mean KS2 score | 103 | 104.909 | 2.895 | 300 | 104.636 | 2.703 |
| FSM eligible (%) | 103 | 29.2 | 15.6 | 303 | 27.9 | 14.6 |
| EAL share (%) | 103 | 27.8 | 26.1 | 300 | 21.3 | 22.2 |
| SEN share (%) | 103 | 15.6 | 7.2 | 300 | 15.6 | 7.3 |
| Log school size | 103 | 6.964 | 0.366 | 303 | 6.938 | 0.398 |
| Yrs since Ofsted | 102 | 4.834 | 3.890 | 295 | 4.918 | 4.007 |
| *School characteristics (share)* | | | | | | |
| Academy school | 103 | 73.8% | | 303 | 71.6% | |
| Urban location | 103 | 90.3% | | 303 | 88.4% | |
| Selective admissions | 103 | 4.9% | | 303 | 5.3% | |
| *Pre-COVID Ofsted grade (as at 31 August 2019)* | | | | | | |
| Grade 1 (Outstanding) | 25 | 26.0% | | 63 | 23.6% | |
| Grade 2 (Good) | 60 | 62.5% | | 170 | 63.7% | |
| Grade 3 (Requires improvement) | 10 | 10.4% | | 27 | 10.1% | |
| Grade 4 (Inadequate) | 1 | 1.0% | | 7 | 2.6% | |
| No pre-COVID grade | 7 | | | 36 | | |

*Notes:* The interviewed sample contains the visited schools; within a column, $N$ falls below the sample size only where a control is missing for a school. KS2 is the mean scaled score of the 2023–24 KS4 cohort at Key Stage 2. FSM is the share eligible for the pupil premium (FSM6 definition); EAL and SEN are KS4 cohort shares from the 2023–24 performance tables. Selective admissions is the share of schools recorded as selective in the performance tables. Pre-COVID Ofsted grade is the overall effectiveness grade from the Ofsted Management Information snapshot as at 31 August 2019. Grade percentages are shares of the graded schools (96 visited, 267 interviewed), not of the sample.

previous chapter: predicted warmth, predicted teaching, and the mark-scheme strictness bands. Each score is divided by its standard deviation, so its coefficient reads as grades of Progress 8 per standard deviation, like the visited coefficients. Each score then enters a separate regression of Progress 8 on that score and the full control set, run on every school with an outcome. Each of the three regressions is estimated twice: once with the 2019 grade dummies among the controls and once without them.

The estimates are associations rather than causal effects, and what the design does and does not achieve can be stated exactly. Because the outcome is value-added, the comparison is between schools whose pupils started from the same place. This means that the first and largest confound in any school-effects estimate, namely that stronger schools take stronger pupils, is removed by construction,

and the Key Stage 2 control removes any intake difference that the value-added adjustment leaves. Half of the visits took place during 2023/24, the second outcome cohort's own year, and half in the year after it. The timing therefore does not rule out ordinary reverse causation: good results may produce a warm and orderly school.

What the design relies on instead is that culture is persistent, so that what was observed in 2023/24 and 2024/25 is a reasonable measure of what the 2022/23 and 2023/24 cohorts experienced. This is an assumption, and it is not tested directly. No school was visited twice, so the persistence of culture itself is never measured. The evidence that can be offered is indirect, and it comes in three parts.

The first is stability. A school's Progress 8, and the structural characteristics that shape it, are stable across years relative to the differences between schools, which is consistent with stable culture without establishing it.

The second is the contemporaneous cohort. The 2024/25 cohort, the one measured at the same time as the fieldwork, gives coefficients close to the primary ones on its pseudo-progress measure. That check is reported with the robustness results.

The third comes from the national extension: old inspection reports predict progress almost as well as recent ones. Some schools were last inspected many years ago, so their reports describe an older version of the school; if culture drifted quickly, those reports should predict today's progress much worse than fresh ones do. They predict only slightly worse: each additional year since a report was written weakens the association between strictness and progress by about 0.003 ($p = 0.09$). The weakening is treated as real rather than dismissed, but it is small: at that rate, ten years of report age reduce the association by about 0.03.

Two features of the design also limit what the estimates could be picking up besides culture. Because the 2019 grade is held constant, the comparison is between schools that the inspectorate regarded alike before the outcome cohorts

arrived. This matters because reputation can raise results on its own. A school with a good name attracts committed staff and motivated families, and they could raise results whatever the culture is like. The 2019 grade is the most visible summary of reputation. Controlling for it removes the part of reputation that the grade measures. Any part of reputation that the grade does not measure remains. Finally, because the sample excludes schools that enrol at 13 or 14, no school is credited with progress that was made at another school.

What the design cannot do is rule out the possibility that some persistent characteristic of a school produces both its culture and its results. This might be a headteacher who is effective in ways that the visit does not measure, or the part of a school's reputation that the grade does not capture.

The data cannot rule this out, but they can test parts of it. The teaching-practice score is observed alongside the culture scores, so the most obvious such characteristic, that warm and orderly schools are simply schools whose teachers teach well, can be tested directly rather than left as an assumption. A separate doubt, that the associations are specific to the schools that agreed to be visited, is answered by the national extension. The associations appear again across the national population, where culture is scored from inspection reports rather than from visits, and they hold with the inspection grade held constant.

One sorting concern remains. Pupils with social, emotional and mental-health needs tend to make less measured progress. If strict schools retain fewer of them, whether through exclusion or through families leaving, their measured progress would rise without their culture improving anything. The robustness section adds each school's share of such pupils, measured before the outcome years, as a control.

## 3.4 Results

### 3.4.1 Estimates on the visited schools

Table 3.3 reports the primary estimates, which show that culture is associated with progress. In the primary specification enacted warmth has a coefficient of $\hat{\beta}_W = 0.146$ ($p = 0.001$) and enacted strictness a coefficient of $\hat{\beta}_S = 0.106$ ($p = 0.012$) on overall Progress 8. Warmth is the larger of the two coefficients, although the difference between them is not itself significant ($p = 0.60$). The table shows the same equation under four treatments of the grade control, from requiring an original 2019 grade ($n = 95$) to dropping the control altogether ($n = 100$), and the coefficients move by no more than 0.02 across the four. Both dimensions are associated with progress, which is what the authoritative-school literature would expect.

Table 3.3: Four treatments of the inspection-grade control

| Grade-control specification | $N$ | Warmth ($W$) | Strictness ($S$) |
|---|---|---|---|
| Grade as recorded (missing dropped) | 95 | 0.166*** | 0.098** |
| | | (0.041) | (0.041) |
| Predecessor-filled grade (primary) | 99 | 0.146*** | 0.106** |
| | | (0.043) | (0.041) |
| Missing-grade category retained | 100 | 0.146*** | 0.106** |
| | | (0.043) | (0.041) |
| No grade control | 100 | 0.153*** | 0.097** |
| | | (0.041) | (0.039) |

*Notes:* Each row is the same regression of average Progress 8 on enacted warmth ($W$) and enacted strictness ($S$), each per standard deviation, with the full control set; the rows differ only in how schools with a missing 2019 grade are handled. Schools with a statutory admission age of 13 or above are excluded throughout. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Seven visited academies lacked a 2019 grade under their current URN; of the five in the estimation sample, four are filled from the predecessor school's grade (a fifth predecessor grade belongs to a school excluded for a missing control). The filled and category-retained rows print identical estimates because the retained category contains a single school, which its own dummy fits exactly.

These magnitudes can be put on a familiar scale. A standard deviation of enacted warmth is about half a point on the five-point scale, and the associated 0.15 of a grade is 0.31 of a standard deviation of Progress 8 across the schools

these estimates use. Two standard deviations on both dimensions together are associated with about half a grade per subject per pupil, the size of difference that separates a well-above-average school from an average one in the Department for Education's published performance tables.

These associations are large, and it is fair to ask what they should be compared with. The estimates compare whole schools with one another, and a school that is strong in one respect is usually strong in others, so numbers of this kind come out larger than the effect of any single part of a school, such as one teacher. The right comparison is therefore with other whole-school estimates, and the closest is the charter-school literature. Pupils who attend the schools that literature identifies as effective gain 0.2 to 0.4 standard deviations of attainment for each year they spend there (Abdulkadiroğlu et al., 2011; Angrist et al., 2010). An index of five school practices, several of them close to the strictness measured here, explains about half of the variation in how effective those schools are (Dobbie and Fryer, 2013). Set against those figures, a third of a standard deviation of Progress 8 for a full standard deviation of observed culture is large but of the same order.

Each dimension, entered on its own, has a coefficient of about 0.2: Table 3.4 reports 0.219 for warmth and 0.199 for strictness. When the two enter jointly, the warmth coefficient is two-thirds of its value when entered alone and strictness about half. The two scores overlap, as their correlation of 0.62 implies, but each keeps a substantial association of its own. If the two scores measured one underlying quality rather than two, then entered together only one of them would have a non-zero coefficient. Neither falls to zero, so the two behave as separate dimensions. That correlation, though, partly reflects how the scores were made: the same observers scored both dimensions at the same school on the same day, and this raises the correlation between the two scores. The previous chapter estimated what the correlation would be if the two dimensions were scored by different observers on different days, which removes the part that comes from shared scoring. That estimate is about 0.36, so the case for treating them as two

separate things is stronger still.

Table 3.4: Warmth and strictness alone and together

| Specification | Term | Overall | English | Maths |
|---|---|---|---|---|
| $S$ only | Strictness ($S$) | 0.199*** | 0.174*** | 0.162*** |
| | | (0.033) | (0.038) | (0.034) |
| $W$ only | Warmth ($W$) | 0.219*** | 0.211*** | 0.187*** |
| | | (0.033) | (0.035) | (0.039) |
| Joint | Warmth ($W$) | 0.146*** | 0.162*** | 0.135** |
| | | (0.043) | (0.042) | (0.052) |
| Joint | Strictness ($S$) | 0.106** | 0.071 | 0.076* |
| | | (0.041) | (0.044) | (0.045) |
| $N$ | | 99 | 99 | 99 |

*Notes:* Each column is a Progress 8 outcome; the first two rows enter one culture dimension at a time and the final two rows enter both jointly (coefficients per standard deviation), always with the full control set and the predecessor-filled grade control. The drop from the univariate to the joint coefficients shows the two dimensions share some variance, but each keeps an independent association when the other is held fixed. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Visited sample, late-entry schools excluded. EBacc and Open components are reported in the chapter appendix.

The controls change the two coefficients in different ways. With no controls at all, warmth has a coefficient of 0.082 on the same schools ($p = 0.16$) and strictness 0.145 ($p = 0.006$): unconditionally, the strict schools show the clearer progress advantage. Adding the share of pupils with English as an additional language, and nothing else, moves the warmth coefficient to 0.151 and the strictness coefficient to 0.101, which is close to the values the two coefficients take in the full specification (the no-control estimates are tabulated with the sensitivity analysis, Table 3A.18).

The reason is that schools with many EAL pupils make more progress on average and are scored somewhat lower on enacted warmth, so a comparison that ignores intake composition hides the warmth association. The warmth result is therefore a conditional one. It appears when schools are compared with others of similar intake, which is the comparison the specification is built to make.

The fact that the warmth association appears only after the EAL share is controlled for matters when reading the sensitivity analysis below. The lower warmth scores of high-EAL schools could partly reflect how observers read warmth in different contexts rather than the schools themselves. The regression cannot

tell these two cases apart. In the first, the scores are accurate: high-EAL schools really are, on average, less warm as the items measure it, and comparing schools of similar intake then reveals a true association between warmth and progress. In the second, the observers read warmth differently in high-EAL contexts, so those schools' low scores partly reflect the scoring rather than the schools. The adjusted association then mixes a fact about the schools with a correction for how the observers scored.

At the level of the components, warmth is significant for both subjects (English 0.162, $p < 0.001$; Maths 0.135, $p = 0.011$), while strictness falls short of significance in each taken alone (English 0.071, $p = 0.11$; Maths 0.076, $p = 0.09$). Overall Progress 8 is not the sum of these two components: it also counts the EBacc and Open slots, and it is there that the strictness association is clearest (EBacc 0.138, $p = 0.008$; Open 0.119, $p = 0.035$). Those two components mix progress with the school's entry decisions, so part of the overall strictness association could come from stricter schools entering pupils for different subjects rather than from their pupils progressing more. Testing that explanation means first showing that stricter schools do enter pupils differently, and the visited sample is too small to show it: there, strictness does not significantly predict EBacc entry. The national sample introduced below is large enough. There, adding the school's EBacc entry rate to the regression removes only 10 to 14 per cent of the strictness association, so entry decisions account for little of it. The full component estimates and the entry analysis are in Section 3A.6.

Table 3.5 reports the second and third stages, which show that teaching quality does not account for the association. On its own, the visit's teaching-practice score predicts overall Progress 8 at 0.206 ($p < 0.001$), but it explains less of the variation between schools than the two culture scores do ($R^2$ of 0.55 against 0.62). When it is added to the primary specification it enters at 0.015 ($p = 0.82$), and the culture coefficients are essentially unchanged (0.137 and 0.103).

There are two possible interpretations of this result, and the data cannot

distinguish between them. The first is that culture and teaching are genuinely different predictors: what a school is like between and around lessons matters over and above how well its lessons are taught. The second is that the three scores overlap too much for the regression to separate them. Teaching is highly correlated with the two culture scores. When correlated scores enter together, noise in one of them can shift part of its association onto the others. If that has happened here, the near-zero teaching coefficient understates what instruction contributes. The shift would be more likely if the teaching score were noisier than the culture scores. At school level it is not: although the observers agree least on the teaching items lesson by lesson, the averaged school score is about as reliable as the culture scores (0.67, against 0.60 and 0.58 for the in-lesson components).

Two facts favour the first interpretation without settling the matter. Firstly, the teaching score on its own predicts progress well, so it is not noise. Secondly, the sub-score regressions reported below find that the warmth association comes from warmth outside lessons. The teaching score is built from lesson observations only, so it cannot measure warmth outside lessons. Settling the question fully would need a measure of teaching quality independent of the visit, and none exists for these schools. The claim made here is therefore the narrower one: conditioning on enacted teaching quality does not account for the culture association.

Figure 3.1 plots the visited schools by their two culture scores, shaded by progress. Schools in the warm-and-strict quadrant are mostly above-average performers. The size of their advantage can be estimated in the same way as the main results: a new variable marks whether a school is in that quadrant, one if it is and zero if not, and Progress 8 is regressed on it together with the full control set. Quadrant schools make 0.40 of a grade more progress than other schools with similar intakes ($p < 0.001$; the authoritative-versus-rest row of Table 3A.12). The question is what produces the advantage: the two separate associations added together, or something extra that appears only when a school has both dimensions. An interaction term in the regression would measure any extra advantage from

Table 3.5: Stages 2 and 3 under the primary specification

| | *Overall P8* | *English* | *Maths* |
|---|---|---|---|
| *Stage 2: teaching quality only* | | | |
| Teaching $T$ | 0.206*** | 0.188*** | 0.165*** |
| | (0.042) | (0.042) | (0.048) |
| $R^2$ | 0.555 | 0.556 | 0.477 |
| *Stage 3: warmth, strictness and teaching* | | | |
| Warmth $W$ | 0.137** | 0.166** | 0.144** |
| | (0.058) | (0.064) | (0.062) |
| Strictness $S$ | 0.103** | 0.073 | 0.079 |
| | (0.042) | (0.046) | (0.047) |
| Teaching $T$ | 0.015 | −0.007 | −0.014 |
| | (0.064) | (0.073) | (0.069) |
| $R^2$ | 0.624 | 0.615 | 0.540 |

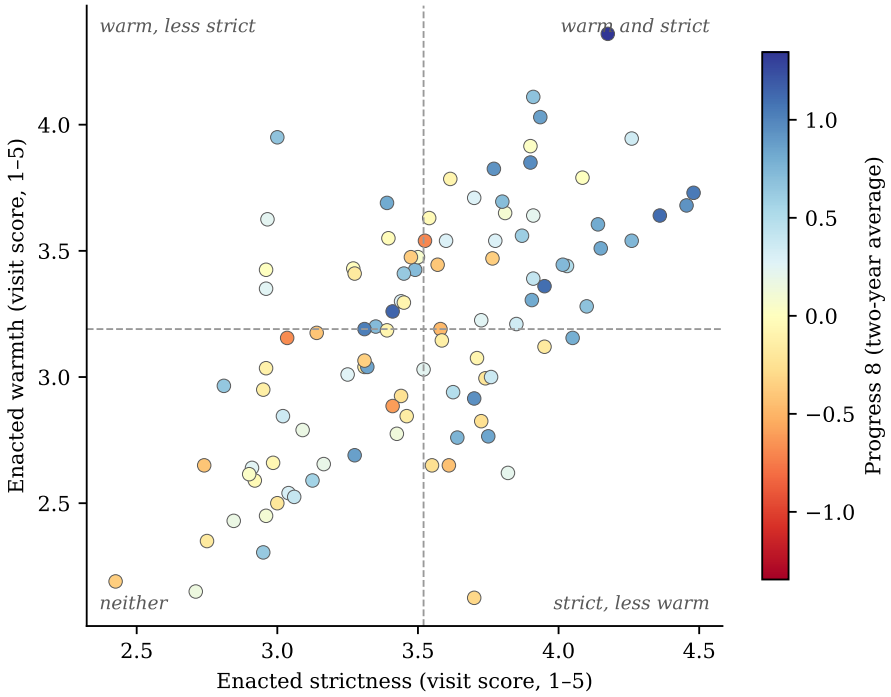
*Notes*: Stages 2 and 3 under the primary specification: teaching quality as the sole culture predictor (Stage 2), and teaching quality added to the warmth-and-strictness model (Stage 3). Coefficients are per standard deviation of the predictor; HC3 robust standard errors; $n = 99$. The full five-outcome versions, estimated on the same specification, are reported in the appendix. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Primary specification throughout: full control set, predecessor-filled pre-COVID grade, late-entry schools excluded.

having both. Its coefficient is small and not significantly different from zero (0.039, $p = 0.27$). The advantage is therefore the sum of the two separate associations, with no bonus from the combination. The national extension below repeats this test at thirty times the sample and reaches the same conclusion.

One caution applies to this test. The interaction term is built by multiplying the two scores, and both scores are measured with error. Multiplying them makes the error worse, and the extra error drags the estimate towards zero. A zero from this test can therefore mean two things: there is no extra benefit from the combination, or there is a small one and the noise is hiding it. The test cannot tell those apart. The comparison reported above, in which the schools in the top half on both warmth and strictness are compared with all other schools, does not have this problem, because it never multiplies the scores. That comparison points the same way: the advantage of those schools is what the two separate associations would produce together, with no sign of anything extra, and it is the stronger evidence on this question.

A further question is which parts of the scores are associated with progress.

Figure 3.1: Warmth, strictness and Progress 8 in the visited schools




*Notes:* Each dot is a school, shaded by its two-year average Progress 8. Dashed lines mark sample medians. The adjusted quadrant comparison is in Table 3A.12.

Table 3A.9 breaks the enacted scores into their five blocks: warmth in lessons, warmth outside them, strictness in lessons, strictness outside them, and teaching. Item-level associations, corrected for multiple testing, are in Table 3A.10. Entered together, two of the five keep their associations: warmth outside lessons (0.111, $p = 0.006$) and strictness inside them (0.115, $p = 0.008$). The other three blocks have coefficients near zero when all five enter together. Read literally, the relational quality that goes with progress is the one visible in corridors and at break, and the order that goes with it is the one visible in lessons. The five blocks are highly correlated with one another, though: a school that scores high on one tends to score high on the others. When the regression divides one association among several highly correlated measures, small amounts of measurement error can change which of them retains a non-zero coefficient, so the ranking of blocks is fragile at this sample size and the literal interpretation should not be relied on.

Espoused warmth does not predict progress: substituting the headteacher's espoused scores for the enacted ones on the same schools reduces the warmth

coefficient to 0.051 ($p = 0.29$), while espoused strictness retains an association of the same size as the enacted one (0.120, $p = 0.018$); Table 3.6 reports both regressions. This is the pattern the previous chapter's measurement predicts, since espoused warmth was shown there to be only weakly related to enacted warmth. The result is also a check on the enacted measures: had the espoused scores predicted progress as well, the visits would have added nothing that an hour with the headteacher could not supply. Section 3A.6 adds three qualifications and related results.

Table 3.6: Espoused culture and Progress 8

| | Overall | English | Maths | EBacc | Open |
|---|---|---|---|---|---|
| Espoused warmth | 0.051 | 0.010 | 0.092* | 0.061 | 0.002 |
| | (0.048) | (0.046) | (0.049) | (0.062) | (0.048) |
| Espoused strictness | 0.120** | 0.113** | 0.040 | 0.130** | 0.167*** |
| | (0.050) | (0.050) | (0.047) | (0.060) | (0.057) |
| $N$ | 99 | 99 | 99 | 96 | 96 |
| $R^2$ | 0.502 | 0.500 | 0.435 | 0.472 | 0.476 |

*Notes:* Each column regresses the named Progress 8 component on espoused warmth and espoused strictness — the standardised headteacher-interview statement-battery scores — with the full control set, on the visited schools. The Overall, English and Maths columns use the primary specification (predecessor-filled pre-COVID inspection grade, late-entry schools excluded; $n = 99$); the EBacc and Open columns are estimated on the same control set before the predecessor-grade fill and before the late-entry exclusion ($n = 96$; with that exclusion applied the unfilled-grade sample has 95 schools), the only form in which the espoused estimates for those two components exist. The enacted estimates for EBacc and Open are available on the primary specification and are reported in Table 3A.3. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

### 3.4.2 The national extension

The estimates so far rest on a hundred schools that agreed to be visited. The national extension asks whether the same associations appear in schools that were not visited. Table 3.7 reports these regressions. For 3,052 schools with an outcome, some thirty times the visited sample,[2] warmth and teaching are predicted

[2]The national sample starts from the 3,332 schools of the national population: 60 schools have no scored inspection report, mostly because they opened too recently to have been inspected; a further 80 have no published Progress 8, mostly because their first cohort has not yet reached

from each school's inspection report by the models of the previous chapter, and strictness is scored by the one-page mark scheme. No school's warmth or teaching prediction comes from a model that was fitted on its own report.

All three coefficients are positive, with small standard errors. +0.078 per standard deviation for warmth, +0.073 for teaching and +0.130 for strictness, each with $p < 0.001$.

The three coefficients cannot be compared with one another, because the three scores are not equally good measures. Each score is aimed at one of the previous chapter's three measures, warmth, teaching or strictness, but captures only a share of what it is aimed at, and each also picks up a different amount of the rest of what a report says about a school. A larger coefficient can therefore reflect a stronger association. It can also reflect a score that picks up more of the other things a report says about a school, such as its general quality, which are themselves correlated with progress. What the scores' imperfection cannot do is create the associations. The part that a score misses acts as noise, and noise pushes a coefficient towards zero rather than away from it, so with perfect scores the coefficients would if anything be larger.

The imperfection raises a different concern. Besides culture, each score picks up other things a report says about a school, and those other things might predict progress on their own. The grade and timing checks below are aimed at exactly that possibility.

Four checks address particular doubts about these estimates. The first concerns the inspection grade. A score obtained from a report might merely restate the inspectorate's grade, in which case its association would be the grade's association. Adding the 2019 grade dummies changes the coefficients very little (warmth +0.074, teaching +0.072, strictness +0.122). The scores therefore predict progress over and above the grade.

GCSEs; 95 are excluded as late-entry schools, the same exclusion the visited sample applies; and 45 lack a control variable. That leaves 3,052. Section 3A.3 checks whether these omissions could bias the estimates.

Table 3.7: The three national estimates

| Dimension | Instrument | Full controls | +Grade | Pre-2022 | 95% CI |
|---|---|---|---|---|---|
| Warmth | prediction model | +0.078*** | +0.074*** | +0.080*** | [+0.041, +0.089] |
| | | (0.007) | (0.007) | (0.008) | |
| | | $n = 3{,}052$ | $n = 3{,}052$ | $n = 1{,}781$ | |
| Teaching | prediction model | +0.073*** | +0.072*** | +0.072*** | [+0.039, +0.086] |
| | | (0.008) | (0.008) | (0.009) | |
| | | $n = 3{,}052$ | $n = 3{,}052$ | $n = 1{,}781$ | |
| Strictness | mark scheme | +0.130*** | +0.122*** | +0.126*** | — |
| | | (0.008) | (0.007) | (0.009) | |
| | | $n = 3{,}052$ | $n = 3{,}052$ | $n = 1{,}781$ | |

*Notes:* Each column regresses Progress 8 on one text-derived culture score, in standard-deviation units, with HC3 standard errors. Every school's warmth and teaching predictions come from models that never saw it; the strictness bands come from the frozen one-page mark scheme applied once per report. The last column is the 95 per cent generated-regressor bootstrap interval for the full-controls coefficient (B=500 over the estimation sample); the mark-scheme bands estimate nothing from the sample, so the strictness column has none. Each column reports its own estimation-sample $n$. Standard errors in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

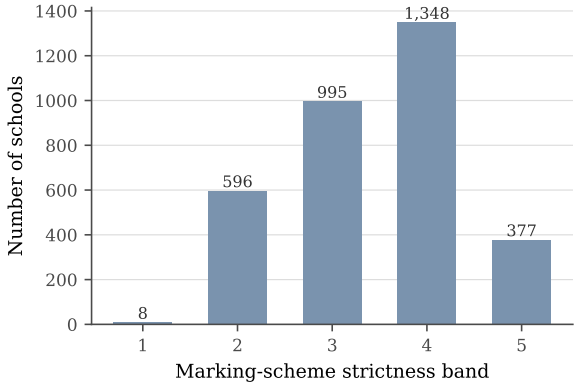
The second concerns timing. On reports published before September 2022, before either outcome cohort's results existed, the coefficients are +0.080, +0.072 and +0.126, essentially their full-sample values, so the associations do not depend on reports that could have been written with knowledge of the outcomes.

The third concerns the models' own estimation error. The warmth and teaching scores are themselves the output of an estimated model, so they carry that model's uncertainty as well as the regression's. Re-estimating the model many times on resampled training schools and rerunning the regression each time gives 95 per cent intervals of [+0.041, +0.089] for warmth and [+0.039, +0.086] for teaching, both away from zero.

The fourth concerns sample dependence: excluding the visited schools from the national sample raises the warmth coefficient (to +0.136) and the teaching coefficient (to +0.122), so neither association depends on the schools that agreed to take part.

The size of each national coefficient can be checked against the visited results. A text score is a noisy version of the enacted score. When a noisy version of a variable replaces the variable itself in a regression, the coefficient falls in proportion to the correlation between the two. The expected national coefficient is therefore

Figure 3.2: The national distribution of strictness bands




*Notes:* Bands are assigned by the one-page mark scheme; 3,272 of the 3,332 schools have a scored report, and the regressions run on the 3,052 with an outcome and full controls. All five bands are used; band 4, conduct presented as a strength, is the national mode.

the visited coefficient multiplied by the correlation between the text score and the enacted score. For warmth and teaching the national coefficients are close to that expected value. The strictness coefficient is about twice that size. The likely reason lies in how the strictness score is produced. The mark scheme assigns its band by reading the whole report, and a report describes more than behaviour: it also gives a general picture of how good the school is. Some of that general picture ends up in the strictness band, and schools whose reports describe them favourably tend to have good results. The strictness coefficient therefore picks up some of that link as well as strictness itself. Because of this, comparing warmth with strictness is best done in the visited sample, where both were measured directly. Section 3A.4 works through the arithmetic for each coefficient.

There is one further concern about how the strictness score is produced. Inspectors know a school's exam results when they write, and good results may lead an inspector to describe the school's behaviour more favourably than they otherwise would. If that happens, the strictness score partly measures results, and part of the strictness coefficient would then reflect earlier results predicting later results, rather than strictness predicting results. The pre-2022 check rules out the direct version of this: reports written before the outcome years still show the association, and those inspectors could not have known the outcomes. It cannot

rule out the indirect version. An inspector who saw the school's earlier results may still have described its behaviour more favourably, and earlier results resemble current ones, so some of the association may still come from results rather than strictness.

The national data can also be used to test whether schools that are both warm and strict do better than the two separate associations imply. Schools in the top half on both predicted warmth and strictness make 0.36 of a grade more progress than other schools with similar intakes, or 0.26 when the inspection grade is also held constant (Table 3A.12). The interaction term is again close to zero (0.004, $p = 0.56$), so here too the advantage is the two separate associations added together rather than anything extra from the combination.

The national comparison of schools in the top half on both dimensions is less exact than the visited one, in two ways. The strictness bands are whole numbers and their median is four, so the higher-strictness half is in practice the schools in the top band. And few national schools are low on warmth while high on strictness, so the group that would show what strictness does without warmth is small.

Progress 8 counts eight subject slots, and a slot that a school does not enter a pupil for scores zero, so a school can raise its score by entering more pupils for the academic subjects of the English Baccalaureate rather than by the progress its pupils make. This is the question left open earlier: whether the strictness association comes from these entry decisions rather than from progress within subjects. Stricter schools do enter more pupils for the academic subjects nationally, but adding the school's EBacc entry rate to the regression removes only 10 to 14 per cent of the strictness association nationally, and about 11 per cent in the visited sample. Most of the association therefore reflects the progress pupils make within the subjects they take. Section 3A.6 reports the entry regressions.

The extension does not make these associations causal. A score built from an inspection report is open to the same doubt as a score built from a visit: some

feature of a school that the control variables do not capture could produce both the score and the results. How strong such a feature would have to be to overturn the visited estimates is the subject of the sensitivity analysis below.

### 3.4.3 Robustness

Table 3.8 reports checks on the warmth and strictness estimates for the visited schools. Each check is aimed at a particular doubt about those estimates.

Two doubts concern how the regression is set up. The first check drops the inspection grade, the strongest control in the specification, in case the culture coefficients depend on its presence. The outcome averages two years of results, which could hide a result that holds in one of those years only, so the second check uses 2023/24 alone.

Three doubts concern which pupils and which schools produce the association. A strict school might raise its measured progress by losing pupils with social, emotional and mental-health needs rather than by teaching them better, so the share of those pupils is added as a control. London schools make more progress than schools elsewhere and differ from them in other ways, so London is controlled for separately. A single unusual school could be producing the association on its own, so each school is dropped in turn, with those results in the appendix.

The remaining doubts concern the measures. The association might come from the value-added adjustment built into Progress 8 rather than from what pupils achieve, so the estimates are repeated on Attainment 8, which has no such adjustment, for the 2024/25 cohort, both in total and for English alone. Attainment 8 is a points total over ten weighted subject slots rather than a grades-per-subject measure, so its coefficients in Table 3.8 look much larger; converted to the same footing, 1.40 points is roughly a seventh of a grade per subject, close to the primary warmth estimate. The culture scores rest on the researchers' ratings, so two checks ask whether the number of raters matters: one adds the share of a school's lessons that were rated by a single researcher, and the other keeps only

the schools where every lesson was rated by at least two.

Every check but two leaves both coefficients significant at the 5 per cent level or better. The exceptions are both for strictness: the Attainment 8 English component, and the subset of schools where every lesson was double-rated. Section 3A.3 walks through the checks one by one.

One further check concerns the standard errors rather than the coefficients. Schools in the same academy trust share policies and often leadership, so their results may not be independent of one another, and treating them as independent would make the standard errors too small. Clustering the standard errors by academy trust in the national sample changes them by at most ten per cent. Each national coefficient is at least nine times its standard error, so a change of that size leaves all three significant at the 1 per cent level (Section 3A.3).

Table 3.8: Robustness of the visited estimates

| Check | $N$ | Warmth ($W$) | Strictness ($S$) | $R^2$ |
|---|---|---|---|---|
| Primary specification | 99 | 0.146*** (0.043) | 0.106** (0.041) | 0.624 |
| No grade control | 100 | 0.153*** (0.041) | 0.097** (0.039) | 0.609 |
| 2023–24 results only | 99 | 0.155*** (0.045) | 0.105** (0.040) | 0.636 |
| SEMH share controlled | 94 | 0.149*** (0.044) | 0.090** (0.038) | 0.618 |
| Warmth × strictness added | 99 | 0.149*** (0.043) | 0.108** (0.041) | 0.632 |
| Attainment 8 total, 2024/25 | 99 | 1.400** (0.535) | 1.317*** (0.471) | 0.877 |
| Attainment 8 English, 2024/25 | 99 | 0.347*** (0.107) | 0.157 (0.100) | 0.830 |
| London controlled | 99 | 0.152*** (0.042) | 0.096** (0.040) | 0.630 |
| Single-rater share controlled | 99 | 0.150*** (0.042) | 0.105** (0.041) | 0.625 |
| Double-rated schools only | 64 | 0.189*** (0.054) | 0.071 (0.048) | 0.706 |

*Notes*: Each row is one check, and reports the warmth and strictness coefficients from a single regression. The outcome is the two-year average of overall Progress 8 unless the row says otherwise, and every row uses the primary specification: full controls, predecessor-filled 2019 grade, late-entry schools excluded. The two Attainment 8 rows use the 2024/25 cohort, which has no Progress 8; the first takes the sum of the four Attainment 8 buckets and the second the English bucket alone, so their coefficients are in points rather than grades. HC3 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The warmth × strictness row adds the interaction of the two standardised scores; the two coefficients shown are then main effects, which are not interpretable on their own. The single-rater row adds the share of a school's observed lessons that were rated by one researcher rather than two; the double-rated row keeps only schools in which every lesson was rated by at least two.

The enacted warmth and strictness scores come from the visits, while the outcomes average the results of two groups of pupils: those who sat their GCSEs in the summer of 2023, and those who sat them in the summer of 2024. Half the visits took place during the 2023/24 school year, while the second group was still in the school. The other half took place the following year, after both groups had

left. In the schools visited during 2023/24, the observers saw the school while the second group was still attending it, though the first group had already left. In the schools visited in 2024/25, both groups had gone: the second up to a year before the visit, and the first up to two years before. The four checks below ask whether the culture the observers scored can stand for the culture those pupils experienced.

The first check looks at how much a school's results move from year to year. The average within-school standard deviation of Progress 8 is 0.20, against a standard deviation across schools of about 0.5 (Table 3A.15). Results are therefore a slow-moving feature of a school rather than something that swings from cohort to cohort. Whether culture changes as slowly is a separate question, and it is the one that matters here. The three checks that follow are about culture itself.

The second check uses the one cohort measured at the same time as the fieldwork. That cohort, 2024/25, has no Progress 8, because the Key Stage 2 tests it would be measured from were cancelled in the first year of the pandemic. A stand-in is built instead by adjusting the cohort's Attainment 8 for the school's Key Stage 2 intake, averaged over nearby years. Built the same way for an earlier year, where Progress 8 does exist, this measure correlates +0.81 with it. The primary regression is then run again with this stand-in as the outcome in place of Progress 8. The coefficients that come back are close to the primary ones: for warmth and for strictness alike, the stand-in coefficient is within 0.04 of a grade of its Progress 8 counterpart (Table 3A.14). The strictness coefficient is, if anything, larger on the stand-in than on Progress 8.

The value of this check is that it removes the timing gap, because the culture and the results now describe the same year. Its weakness is that closing that gap does not show whether culture produces results or results produce culture. The 2024/25 results were published after every visit, so the culture the observers saw cannot have been a response to them. But a school's results are similar from year to year, so a school that did well in 2024/25 was in most cases already doing well

when it was visited, and those earlier results were public. What the check shows is that the association is still there when culture and results belong to the same year. What it cannot show is whether the culture came before the results or followed from them, and that is why it is not treated as evidence of persistence on its own.

The third check uses results from long before the fieldwork. The question is still the same one: does culture change slowly enough that the 2023–25 scores also describe the school in earlier years? If culture is a lasting feature of a school, the culture observed in 2023–25 should relate to results from earlier years as well; if the score mostly reflects the school's recent results, it should not. Regressing pre-pandemic Progress 8, from 2018/19, on the culture scores observed in 2023–25, with that era's control variables where they exist, gives +0.09 for warmth and +0.08 for strictness on the 93 schools with both outcomes. Neither is significant, and both are smaller than the +0.16 and +0.10 that the same schools give for current results.

The result shows us two things. Firstly, the culture observed in 2023–25 is positively related to results from six years earlier, though not significantly so. That is what would be expected if culture lasts, and it is not what would be expected if the observers were simply scoring schools by their recent results. Secondly, the older association is weaker than the current one. That is also what would be expected if culture changes slowly: the further back the results, the less that school resembles the school the observers saw. The second point is only suggestive, because a formal test cannot reject that the older and current associations are the same size ($p = 0.17$ for warmth and 0.62 for strictness with the era's control variables, 0.09 and 0.57 with current ones).

The fourth check uses a change of headteacher as a sign that a school's culture may have changed. The concern throughout this subsection is that the culture observed at the visit may not be the culture of the outcome years, and a new headteacher is the clearest recorded event that could break the link between the two, since the head shapes a school's routines and expectations. The schools are

therefore split by whether the headteacher changed between the first outcome cohort's final year and the end of the visit window, which can be determined for all 99 schools: 58 kept the same headteacher and 41 changed.

In the 58 schools with the same head, a standard deviation of warmth is associated with +0.16 of a grade and a standard deviation of strictness with +0.13, both significant and close to the full-sample estimates of 0.15 and 0.11. In the 41 schools whose head changed, the warmth association is close to zero and neither association is significant.

What the results show is that the main estimates do not depend on the schools where culture is most likely to have changed. The 58 same-head schools are the ones where the culture the observers scored is most likely to be the culture the outcome cohorts experienced, and the associations there are as strong as in the full sample. The weaker associations in the changed-head schools are also consistent with the reasoning behind the check: if a new head changed a school's culture, the scores describe a culture that the outcome pupils did not experience, and the association should weaken. This second observation is weak evidence on its own, because the changed-head group is small, its estimates are noisy, and a formal test cannot reject that the two groups' associations are the same size ($p = 0.27$ for warmth).

The number of schools that changed headteacher also has an implication for the main estimates. If the scores in the 41 changed-head schools are noisier measures of the culture the outcome cohorts experienced, that noise pushes coefficients towards zero, so including those schools would if anything make the full-sample estimates smaller, not larger. Table 3.9 reports the estimates from the third and fourth checks with their standard errors.

A related check asks whose progress the associations reflect. The Department publishes Progress 8 separately for disadvantaged pupils, and the primary regressions can be re-run on that outcome. The associations are, if anything, stronger for disadvantaged pupils than for their peers: nationally, a standard deviation

Table 3.9: Timing checks on the visited-school estimates

| | Warmth | (se) | Strictness | (se) | $n$ |
|---|---|---|---|---|---|
| *Panel A: results from before the fieldwork (outcome is 2018/19 Progress 8 unless stated)* | | | | | |
| 2018/19 P8, era intake controls | +0.093 | (0.058) | +0.078 | (0.049) | 93 |
| 2018/19 P8, all-current controls | +0.086 | (0.052) | +0.078* | (0.046) | 93 |
| Contemporaneous P8, same schools | +0.159*** | (0.044) | +0.099** | (0.041) | 93 |
| *Panel B: same or changed headteacher (contemporaneous P8, primary specification)* | | | | | |
| Same head, Sept 2022 and July 2025 | +0.160*** | (0.044) | +0.130** | (0.052) | 58 |
| Head changed between those dates | +0.028 | (0.111) | +0.121 | (0.100) | 41 |

*Notes:* Every row regresses the stated outcome on standardised enacted warmth and strictness with the full control set plus pre-COVID inspection-grade indicators (grades 1–2 pooled base; schools with a missing grade drop), HC3 standard errors, late-entry schools excluded; each row reports its own estimation-sample $n$. Panel A relates the culture scores to results from a cohort the visits never observed: the 2018/19 school-level Progress 8, with the intake controls either swapped to that era's values (KS2 APS of the GCSE cohort and FSM share) or held at current values, beside the contemporaneous outcome on the same 93 schools. Panel B splits the primary specification by whether the school had the same headteacher in the 1 September 2022 and 1 July 2025 GIAS extracts, that is, from the first outcome cohort's final year to the end of the visit window. Standard errors in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. The headteacher-change flag covers every estimation-sample school.

of the strictness bands is associated with 0.15 of a grade of disadvantaged-pupil progress against 0.12 for other pupils, and on the visited schools both culture dimensions hold on the disadvantaged-pupil outcome (warmth 0.14, strictness 0.13, both $p \approx 0.02$). Whatever the associations reflect, they are not gains concentrated among the already advantaged.

The last check concerns the possibility that some characteristic of a school that the control variables do not capture produces both its culture scores and its results. This is the doubt the design cannot remove, and no test can rule it out. What two standard sensitivity measures can do is calculate how strong such a characteristic would have to be to explain the estimates away (Table 3A.18). Adding the measured controls moves each coefficient by a known amount. The first measure extrapolates from that movement. It asks how strongly an unmeasured variable would have to be related to both the culture score and progress, compared with the measured controls, to move the coefficient the rest of the way to zero. For strictness, an unmeasured variable related to both about as strongly as the measured controls would be enough. For warmth, the same amount of confounding would leave about three-quarters of the association in place. The second measure

asks a related question without reference to the controls. It gives the smallest association an unmeasured characteristic would need with both the culture score and progress to account for the whole estimate. Both are reported in Table 3A.18. Two limits apply. Firstly, both calculations assume that any bias comes from an omitted variable. They give no information about whether results cause culture rather than the reverse. Secondly, the measures start from the estimated coefficients. The culture scores contain measurement error, and error of that kind pulls an estimated coefficient towards zero, so the coefficients the measures start from are likely smaller than the true associations. A larger association takes more confounding to remove, so the measures likelier understate than overstate how much confounding the associations could withstand. Section 3A.3 sets out both measures in full.

### 3.4.4 Exploratory results on traditional pedagogy

The last results concern what schools say about how they teach. In the public debate about school improvement, the schools known for combining warmth with strictness often also describe their teaching as traditional: knowledge is taught directly by subject-expert teachers and practised until it is remembered, and creative work is treated as something that builds on knowledge rather than something that develops alongside it. Whether a school takes this approach cannot be read from its inspection report, because reports assess the quality of a school's teaching but rarely describe its approach. The one public source in which schools state their approach is their own website, so traditional pedagogy is measured from each school's site: an indicator that marks a school when its website presents knowledge as the organising principle of the curriculum, or describes teaching as instruction by subject experts. Every school whose website contains any of the language associated with this approach was read in full against a written rule, twice and independently, and the readings agreed on 95 per cent of them. Section 3A.5 sets out how the indicator was built and how accurate it is. The

indicator marks 60 schools nationally, 50 of them in the estimation sample.

Schools whose websites espouse traditional pedagogy make +0.177 of a grade per subject more progress than otherwise similar schools ($p = 0.02$, $n = 3{,}075$). The association changes little when the school's pre-pandemic inspection grade is added to the control variables (+0.179).

The indicator can also be put together with the two national culture measures. Of the 299 schools in the top half on both predicted warmth and the strictness bands, 8 also espouse traditional pedagogy, so the full combination is rare, and the warm-and-strict schools are only a little more likely to espouse it than the rest (3 per cent against 2).

In raw terms, the 8 schools average +1.28 on Progress 8, so their pupils make about a grade and a quarter more progress per subject than the national average. Part of that gap may reflect who attends these schools rather than what the schools do. Parents choose these schools partly for their stated approach. Parents who want a strict, knowledge-led education for their children may differ from other parents in how much they support their children's schooling, and pupils with such parents might make more progress in any school. The controls can address the measurable part of this, comparing the 8 with otherwise similar schools on intake, size, type and location. Entering an indicator for the 8 into the same regression as the national estimates, with the full control set and the pre-pandemic inspection grade held constant, gives 0.98 of a grade ($p < 0.001$).

The 8 can also be compared only with the other warm-and-strict schools, to ask whether traditional pedagogy adds anything beyond the two culture dimensions. Among the 299 schools in that group, the ones that also espouse traditional pedagogy make 0.62 of a grade more progress than the ones that do not ($p = 0.02$).

With only 8 schools in the group, both comparisons depend on which schools it contains. Dropping each of the 8 in turn moves the first comparison between 0.86 and 1.11 of a grade, so that comparison holds whichever school is removed. The second is the weaker one: dropping the one school whose removal changes the

estimate most reduces the within-group difference from 0.62 to 0.48 of a grade, which is at the boundary of significance ($p = 0.05$).

These results are exploratory, meaning they were produced to see whether a pattern exists rather than to test a hypothesis fixed in advance, and they carry four limits because of it. The measures come from what schools choose to say about themselves. The traditional-pedagogy indicator misses schools that follow the approach without writing about it. The 8-school group is small enough that a few unusual schools can move its estimates substantially, as the leave-one-out figures above show. And a significance level assumes that the comparison was chosen before the data were seen; these comparisons were chosen partly because the data looked interesting, and comparisons chosen that way will sometimes look strong by chance alone, so the significance levels here overstate the evidence in a way that those elsewhere in the chapter do not. The associations are therefore reported as descriptions rather than tests.

## 3.5 Conclusion

Observed school culture is associated with pupil progress. Across the 103 English secondary schools that were visited, a standard deviation of enacted warmth is associated with Progress 8 that is 0.15 of a grade higher, and a standard deviation of enacted strictness with 0.11 of a grade. The coefficients barely move whether the pre-pandemic inspection grade is included among the control variables, excluded, or filled in from a predecessor school where it is missing. The two coefficients are close in size, and a formal test cannot distinguish them.

The two associations simply add together: a school that is high on both dimensions gains no more and no less than the sum of the two. Schools in the top half on both outperform the rest by two-fifths of a grade. Conditioning on enacted teaching quality does not account for the association. Substituting the headteacher's own description of the school for the observed scores removes the warmth association, so the association belongs to what the observers saw rather

than to what the school says about itself. And the associations do not depend on which schools agreed to be visited: all three scores predict progress across the national population, with warmth and teaching predicted from inspection reports by the prediction models and strictness scored by the one-page mark scheme. Each of the three national associations holds under two checks. It holds when the pre-pandemic inspection grade is added to the control variables, so the scores carry information beyond the report's overall verdict. And it holds on the reports written before the outcome cohorts sat their GCSEs, whose inspectors could not have known the results the scores now predict.

There are two implications for the current policy debate. Both depend on the association being at least partly causal, which the design cannot prove, though the checks above have removed many of the other explanations.

The first implication concerns the behaviour debate, which treats discipline as the thing that improves schools and relational culture as secondary. The estimates here do not rank the two that way: warmth matters about as much as strictness, and a school high on both gains the sum of the two associations and nothing more. On these estimates, a policy that strengthened behaviour while relationships weakened would gain about as much on one dimension as it lost on the other. Schools in the top half on both dimensions make two-fifths of a grade more progress than the rest, with intake held constant. The four-way split by warmth and strictness is reported in Table 3A.12 but not used here, because two of the four groups contain too few schools.

The second implication concerns where in the school the warmth that matters is found. The sub-score results place the association in warmth outside lessons and in orderliness inside them. The warmth that goes with progress is therefore not a way of teaching; it is how adults and pupils treat each other in corridors, at break and at the gate.

The estimates are associations, not causal effects. Several features of the design make it less likely that the associations are spurious. The outcome is value-added,

the controls are extensive, the coefficients barely move across specifications, and school characteristics persist across years. What the design cannot rule out is that something about a school which the controls do not capture, present throughout the period, produces both its culture and its results, because every school is observed only once. There are three other limitations. The visited sample is small, and its schools are stronger than the national average. Culture was measured during or after the final outcome year rather than before it. And a single visit measures a school with considerable error.

Two designs would go further than this one. Observing culture before an outcome cohort arrived and again after it left would show which comes first, the culture or the results. And following schools whose culture changed, because the headteacher changed or a new behaviour policy was introduced, would show whether results change with it.

Within those limits, the results pick out the relational and disciplinary features of a school that go together with its pupils' progress. They also show that these features, and teaching besides, can be measured for every school in the country from the public record: warmth and teaching by a model that counts words in inspection reports, and strictness by a language model applying a one-page mark scheme to the report.

# Appendix to Chapter 3

This appendix collects the supporting material for the analyses reported in the chapter. It explains how the Progress 8 measures and the subject entry rates are calculated. It sets out the evidence on the stability of school characteristics that underpins the choice of outcome years, and gives the full five-outcome versions of the main results tables (the chapter body reports overall, English and Maths Progress 8, and the EBacc and Open components appear here) together with the secondary and exploratory analyses.

## 3A.1 How the Progress 8 Measures Are Calculated

The eight slots of Progress 8 (P8 below) are structured, and this structure is why the thesis treats the components differently. The measure itself is as the body defines it: attainment across eight GCSE slots, compared with the national average for pupils with the same Key Stage 2 starting point, and averaged across the school. English and Maths each fill two slots and are double-weighted. Every pupil sits both, so these components measure progress and nothing else. The EBacc component covers three slots that must be filled from a restricted academic list: sciences, computer science, history, geography, and modern foreign languages. The Open component covers the final three slots, which may be filled by any approved qualification, including arts and vocational subjects. For these last six slots the school makes entry decisions, and an unfilled slot contributes zero progress to the school's average. Consequently, a school's EBacc and Open components mix two different things: how much progress pupils make in the subjects they enter, and which subjects the school enters them for. A school that enters few pupils for languages will carry empty EBacc slots and a depressed EBacc component even if the pupils it does enter progress well.

This structure generates a theoretical prediction about warmth and strictness. Schools higher on strictness, characterised by structured behaviour management, high academic expectations, and intensive pupil steering, are more likely to direct pupils into the EBacc pathway, and could therefore raise their EBacc component through entry as well as through progress itself. Schools higher on warmth might generate stronger progress in the more elective Open component. The five-outcome versions of the main results tables in this appendix provide evidence on that prediction, and the entry-rate analysis below addresses the pathway channel directly.

The entry rates themselves are calculated from the pupil count columns of the 2023/24 performance tables, because the Department for Education stopped publishing entry rates in percentage form in recent releases. The EBacc entry rate is the number of pupils entered for the full EBacc suite divided by the number of pupils at the end of Key Stage 4, and the humanities and languages rates are constructed in the same way from their respective counts. Entry rates are available only at the subject-group level. Individual subjects within a group cannot be distinguished in the data collected for this project.

## 3A.2 Structural Stability of School Characteristics

The intra-class correlation (ICC) decomposes total variance in a panel variable into a between-school component and a within-school over-time component. An ICC close to one indicates that schools are much more different from each other than they are from their own past selves. In other words, year-on-year fluctuation within a school is small relative to the persistent differences between schools. ICC is estimated by one-way random-effects ANOVA on the school-year panel, retaining only schools with at least three years of non-missing data for the variable in question.

Table 3A.1 reports ICC estimates and mean absolute year-on-year changes for nine panel variables covering pupil intake composition, special educational

needs prevalence, prior attainment, pupil progress, and per-pupil expenditure. Six of nine variables exceed ICC = 0.69, and all exceed 0.49. Intake characteristics (the share of pupils with English as an Additional Language, school roll, and the mean KS2 prior attainment of the entering cohort) are especially stable, with ICC values between 0.81 and 0.92 and small mean absolute year-on-year changes relative to their between-school spread. Special educational needs prevalence (EHC plan share, ICC = 0.67; SEN support share, ICC = 0.65) and school-level Progress 8 (ICC = 0.73) show moderate to high stability. Per-pupil expenditure, at ICC = 0.52, is the most volatile, which reflects its sensitivity to changes in national funding policy. Even so, the majority of the variation in expenditure is between schools rather than within schools over time.

The ICC evidence quantifies the stability that the design assumes. A school's intake, size and results change little from year to year compared with the differences between schools. This is consistent with the assumption that its culture also changed little between the outcome years and the visit, though it does not test that assumption.

The same analysis at the level of the thirteen primary-need categories within the SEN population shows moderate stability for the Social, Emotional and Mental Health (SEMH) category (ICC = 0.545), which supports the baseline SEMH share entering as an additional control in the robustness specification of Table 3.8. The lower stability of SEMH relative to the aggregate SEN support share (ICC = 0.65) is consistent with SEMH composition being partly responsive to school behaviour and exclusion practices, rather than being determined solely by fixed intake characteristics.

Table 3A.2 reports ICC estimates separately for the four Progress 8 components reported in this chapter. Among the Progress 8 components, stability varies from ICC = 0.587 for the Open component to ICC = 0.748 for EBacc, with English and Maths sitting between these at ICC $\approx$ 0.67. EBacc is more stable than Open because curriculum-pathway decisions, meaning which subjects pupils are entered

Table 3A.1: Structural stability of school characteristics

| Variable | Schools | Mean yrs | ICC | Mean YoY |
|---|---|---|---|---|
| EAL share (%) | 3,286 | 7.7 | 0.922 | 3.7 pp |
| Pupils on roll | 3,363 | 9.8 | 0.920 | 37 pupils |
| KS2 scaled score (2021–24) | 3,215 | 3.0 | 0.910 | 0.86 pts |
| KS2 APS (2015–19) | 3,100 | 4.0 | 0.812 | 0.65 pts |
| FSM share (%) | 3,363 | 8.8 | 0.742 | 2.3 pp |
| Progress 8 | 3,256 | 6.8 | 0.727 | 0.21 pts |
| EHC plan share (%) | 3,355 | 9.6 | 0.669 | 0.4 pp |
| SEN support share (%) | 3,361 | 9.8 | 0.653 | 1.9 pp |
| Per-pupil expenditure | 3,331 | 6.7 | 0.516 | £599 |

*Notes:* The panel covers 2015–16 to 2024–25. ICC = intra-class correlation = between-school variance / (between-school + within-school variance), estimated via one-way random-effects ANOVA on the school-year panel; schools with fewer than three years of non-missing data are excluded. *Mean yrs*: mean number of academic years of non-missing data per school. *Mean YoY*: mean absolute year-on-year change per school. EAL = English as an Additional Language; FSM = free school meals (school census); EHC = Education, Health and Care plan; SEN = special educational needs. The two KS2 measures use different scales from successive assessment frameworks and are therefore reported separately; no KS4 performance data were published for 2019–20 or 2020–21 owing to the suspension of national examinations. Progress 8 is nationally standardised with a mean near zero; the mean YoY figure is an absolute change, not a percentage. Per-pupil expenditure combines Consistent Financial Reporting (maintained schools) and Academies Accounts Return (academies); the 2015–16 and 2024–25 expenditure data are incomplete.

for, are governed by stable school policy and change little from year to year, while the more varied Open component (arts, vocational and additional GCSEs) is subject to greater annual variation in subject mix. All four P8 components exceed the per-pupil expenditure ICC of 0.52, which is the weakest result in the main stability table. Each component is therefore at least as stable across years as the least stable structural characteristic. The case for treating a two-year average as representative of the school holds for each component as well as for the overall measure.

The Open component's lower stability means that the warmth coefficient is estimated against a noisier outcome. The chapter's differential prediction, set out in Section 3A.1, expects warmth to show most strongly in the Open component and strictness in the EBacc component. With Open at ICC = 0.587 against 0.748 for EBacc, the warmth side of that prediction is tested against more noise and may be understated. Any differential finding between components must be read against this difference in outcome noise.

Table 3A.2: Structural stability of the Progress 8 components

| Variable | Schools | Mean yrs | ICC | Mean YoY |
|---|---|---|---|---|
| English | 3,256 | 6.8 | 0.666 | 0.250 pts |
| Maths | 3,256 | 6.8 | 0.667 | 0.225 pts |
| EBacc | 3,256 | 6.8 | 0.748 | 0.243 pts |
| Open | 3,256 | 6.8 | 0.587 | 0.304 pts |
| Overall | 3,256 | 6.8 | 0.727 | 0.210 pts |

*Notes:* ICC and mean years are defined as in Table 3A.1. The panel covers 2015–16 to 2023–24, excluding 2019–20 and 2020–21, for which no KS4 performance data were published owing to the suspension of national examinations. The components are English (two slots, double-weighted), Maths (two slots, double-weighted), EBacc (sciences, computer science, history, geography, and modern foreign languages), and Open (any additional approved qualification).

## 3A.3 Robustness in Detail

This section walks through the checks summarised with the results: the alternative specifications of Table 3.8, the headteacher-change split, the two sensitivity measures of Table 3A.18, the clustering of the national standard errors, and a check on the schools the national sample omits.

Table 3.8 reports the primary estimate under alternative specifications, all on the primary sample, and each of them addresses a particular doubt. The 2019 grade control might create an association that is not there, or remove part of one that is. Table 3.3 has already shown the estimates with the control entered in different ways. Dropping the control altogether gives 0.153 for warmth and 0.097 for strictness ($n = 100$). The pairs of figures below are warmth and strictness in that order.

Averaging two cohorts might smooth over a result that holds in only one of them. The single 2023/24 outcome year gives 0.155 and 0.105.

Strict schools might have fewer pupils with social, emotional and mental-health needs, and gain in measured progress by that route. Adding the 2015/16 baseline share of such pupils, which was measured before any of the culture was observed, gives 0.149 and 0.090 ($n = 94$), and the sorting channel accounts for under a tenth of the strictness association (Table 3A.16).

The association might be specific to the Progress 8 measure. Attainment 8

is a plain grade total with no adjustment for pupils' prior attainment. On the 2024/25 cohort's Attainment 8, both dimensions remain significant at the 5 per cent level on the total, at 1.40 and 1.32 points per standard deviation of the score. Attainment 8 sums grades over ten weighted subject slots, so these coefficients are roughly a seventh of a grade per subject. On the English component alone warmth remains significant (0.35) while strictness does not (0.16, $p = 0.12$). The Key Stage 2 control in these regressions is the school-level mean for the Progress 8 cohorts, since the 2024/25 cohort has no Key Stage 2 results of its own.

The enacted scores might reflect something about the visits themselves. About a third of the observed lessons were rated by one researcher rather than two, those lessons were scored a little lower, and the share of such lessons in a school's visit correlates $-0.25$ with its warmth score. Adding that share as a control leaves the estimates unchanged at 0.150 and 0.105 (the share itself has a coefficient of 0.025, $p = 0.76$).

The same doubt can be met by restricting the sample to the 64 schools in which every lesson was rated by at least two researchers. That gives 0.189 for warmth ($p = 0.001$) and 0.071 for strictness ($p = 0.15$). The warmth estimate is larger on that subset and the strictness estimate smaller, with a standard error about a sixth larger on two-thirds of the sample, so the strictness result is the one that depends on the full sample.

The warmth result might depend on London, since the sample over-represents London, the warmth association appears only once the EAL share is controlled for, and high-EAL schools are concentrated there. Adding a London indicator built from the schools register moves warmth only to 0.152 ($p = 0.001$) and strictness to 0.096, and the indicator itself is not significant (0.113, $p = 0.33$), so the association does not depend on London.

Rater disagreement adds error to the culture scores. Error of that kind pulls the coefficients towards zero. The size of that bias can be estimated directly. Where a school was scored by more than one researcher, building each observer's

warmth and strictness scores separately gives two replicate measurements of the same school whose rater-specific errors are plausibly independent, so one observer's score can be used to predict the other's, and the prediction used in the regression in place of the raw score (Gillen et al., 2019). This removes the error that differs between the two observers. On the 66 schools where a second observer's full set of forms exists, the two observers' school-level scores correlate at 0.77 for warmth and 0.85 for strictness, and the instrumented estimates move the warmth coefficient from 0.17 to 0.20 while leaving strictness essentially unchanged. The correction is modest because it removes only disagreement between raters: both observers saw the same lessons on the same day, so the error that comes from which lessons a visit happened to sample is common to the two replicates and survives. The corrected estimates therefore remain conservative.

The visited estimates do not depend on any one school. Dropping each school in turn moves the warmth coefficient by at most 0.019 and the strictness coefficient by at most 0.017, and no omission moves either across conventional significance.

In the headteacher-change split reported with the timing checks, the 58 schools with the same head in the September 2022 and July 2025 registers give estimates close to the full-sample values for both dimensions, while the 41 schools whose head changed give a near-zero warmth estimate with standard errors more than twice as large; a formal test cannot reject that the two groups' associations are equal.

Two standard measures give the strength an unmeasured confounder would need to overturn the visited estimates (Table 3A.18). Both concern the same doubt: some characteristic of a school that the control variables do not capture might produce both its culture scores and its results. Neither can rule such a characteristic out; each calculates how strong it would have to be to explain the estimates away.

The first is Oster's $\delta$. Adding the full control set moves each coefficient by a known amount, and the measure extrapolates from that movement. It asks

how strongly unmeasured variables would have to be related to both the culture score and progress, compared with the measured controls, to move the coefficient the rest of the way to zero. For strictness the answer is 1.09, just above the conventional benchmark of one, so unmeasured variables related to both about as strongly as the full control set would be enough to remove the association. A $\delta$ above one is usually read as reassurance. That reading does not apply to warmth. Adding the controls raises the warmth coefficient, so the extrapolation moves it away from zero, not towards it. Warmth's large $\delta$ of 1.71 is therefore not by itself evidence of robustness. The informative number for warmth is the coefficient that remains if the unmeasured variables are assumed to be related to both the score and progress as strongly as the controls are, which is 0.112, about three-quarters of the estimate. No measure of this kind can rule out an unmeasured variable related to the score with the opposite sign to the measured controls. Such a variable would lower the coefficient where the controls raised it.

The second is the E-value. It does not use the controls. It gives the minimum association a single unmeasured characteristic would need with both the culture score and progress to account for the whole estimate. For warmth the answer is an association of at least 1.99 in risk-ratio terms with each, and for strictness 1.76. A confounder of that strength would be more strongly related to the score, and to progress, than the score and progress are related to each other.

Both measures assume that any bias comes from an omitted variable. Neither gives information about reverse causation. Both also start from the estimated coefficients. The culture scores contain measurement error, and error of that kind pulls an estimated coefficient towards zero, so the coefficients the measures start from are likely smaller than the true associations. A larger association takes more confounding to remove, so the measures likelier understate than overstate how much confounding the associations could withstand.

Schools in the same academy trust share practices, so the standard errors were also computed with clustering on trust membership from the schools register

(1,595 clusters; three-fifths of schools in multi-school trusts). Clustering moves the standard errors by at most ten per cent and changes no inference. The visited sample has too few multi-school trusts (25 schools across nine shared trusts) for cluster-robust inference to be informative. Clustered standard errors for the visited sample are therefore not reported.

The strictness result does not depend on treating the five bands as equally spaced steps. Comparing each band directly with band 3, progress rises band by band ($-0.27$, $-0.05$, $+0.15$, $+0.43$); Table 3A.11 reports the four estimates.

One further doubt concerns the national sample itself. The national regressions run on 3,052 of the 3,332 panel schools, and the footnote with the results gives the arithmetic: 60 schools have no scored inspection report, 80 no published Progress 8, 95 are excluded as late-entry schools, and 45 lack a control variable. The doubt is whether the missing 280 differ from the rest in a way that could bias the estimated associations.

They differ in how well they are doing. The omitted schools are more disadvantaged and weaker than the estimation sample: the uninspected group averages a free-school-meals share of 37 per cent against the sample's 30, and where an outcome is published the omitted groups' mean Progress 8 sits well below zero ($-0.45$ for the uninspected schools, $-0.44$ for the late-entry schools and $-0.27$ for those missing a control, against $+0.01$ in the sample). The omitted schools are mostly newly opened schools, schools that changed status during the panel, and university technical colleges. The national estimates therefore describe established mainstream schools.

Being weaker is not, by itself, a source of bias. The regression estimates how progress moves with strictness among the schools it includes, so leaving out a group of weak schools changes who is described, not the answer for those included. The estimates would only mislead if the omitted schools' own strictness–progress relationship were different, so that putting them back would change it. For one group that can be tested directly. The 45 schools dropped only for a missing

control have both a strictness band and an outcome. Among them, a standard deviation of the band score goes with 0.29 of a grade (standard error 0.07); in the estimation sample the same figure, again without controls, is 0.23 (0.01). The two are indistinguishable, and the omitted group's is if anything the steeper. For the omissions to inflate the national strictness association, the missing schools would need a weaker relationship than the sample. The one direct test finds the same or a slightly steeper relationship among the omitted schools. Leaving them out therefore cannot have inflated the national estimate. If anything it lowered it.

## 3A.4 Benchmarks for the National Coefficients

Each national coefficient has a benchmark. The national score is a noisy version of the enacted one, so its coefficient should be the visited coefficient multiplied by the correlation between the two. For warmth that benchmark is a third of the visited coefficient of 0.219, roughly 0.07, and the national estimate of +0.078 sits on it. For teaching it is $0.28 \times 0.206 \approx 0.06$, and the estimate of +0.073 sits about a quarter above. For strictness it is $0.30 \times 0.199 \approx 0.06$, and the estimate of +0.130 is roughly double. The benchmarks are rough. Each correlation was estimated on about a hundred visited schools, which are less varied than schools nationally, so each has a sampling error of about 0.1. The strictness score has only five bands, and a coarse score correlates less well with the enacted score than a fine one would. The strictness benchmark is therefore set too low.

The national strictness estimate is about twice its benchmark. The way the two scores were built explains this. The warmth and teaching models were trained only on the part of the enacted score that demographics cannot explain. School names and other words that identify a school were removed from the reports before training. The models therefore predict from the description of the school and not from its identity. The mark scheme is applied to the report as it stands. Its bands pick up conduct, and with it some of the general character of an orderly, effective school, and that general character is itself correlated with results. Part of

the national strictness coefficient reflects that general character and not strictness alone. For the ranking of the two dimensions, the visited sample is the one to trust, because there the dimensions were measured directly, and the warmth coefficient is at least as large as the strictness coefficient.

## 3A.5 The Traditional-Pedagogy Indicator

This section explains how the traditional-pedagogy indicator was built and how accurate it is.

The indicator records what a school's website says about its teaching. Deciding that means reading sentences and judging what they are about. A rule that searches for words cannot do it. Words like "knowledge-rich", "mastery" and "traditional" appear on the websites of many schools that do not teach this way, so finding the word settles nothing.

The reading was done by a large language model, Claude Opus, pinned to the version `claude-opus-5`. There are 3,370 school websites in the collection, and reading all of them was not practical, so the schools were narrowed down first. A school became a candidate if its website used any one of about thirty terms linked to this approach, or two terms from a second, weaker list. This gave 436 candidates. The term lists were kept long, so that few schools which describe this approach would be missed at this stage.

Each of the 436 websites was then read in full against a written rule, which is reproduced in Section 3A.5.1. The rule sets out the question, says that using a word is not the same as giving an account of teaching, lists four common mistakes, and says that doubtful cases are not marked. Every website was read twice, by two separate runs of the model. Each run saw only the rule and the websites, and read them in a different order. Where the two runs disagreed, a third run decided, without seeing either earlier answer. To mark a school, the model had to quote a sentence from that school's website. Every quoted sentence was then checked against the website text by a script.

The two runs agreed on 95 per cent of the 436 websites ($\kappa = 0.81$) and disagreed on 21. Every sentence the model quoted was found in the website it came from. The indicator marks 60 of the 3,370 schools. Because both runs are the same model working from the same rule, that figure is a measure of how repeatable the reading is, and not of how far two independent judges agree.

Accuracy is measured against 50 websites set aside for this purpose. Each of those was read three times by a language model, working from the same rule, and the answer is what two of the three runs said. The version of the model used for those readings was not recorded. The indicator agrees with that answer on 49 of the 50 ($\kappa = 0.94$). Of the ten websites the three runs marked, it marks all ten. Of the 40 they did not mark, it marks one.

That comparison is between one set of model readings and another. I also read those 50 websites myself. My labels and the indicator agree on 86 per cent of them ($\kappa = 0.55$). I then looked at the seven websites we disagreed about and judged that all seven do describe this approach. Those judgements were made after seeing the model's reading, so they are not a test of it, but they are why two further schools are marked.

Three limits remain. The reading is a model's reading, and the accuracy figures above mostly compare it with other readings by the same model. The figures measure agreement about what a website says, which is not the same as evidence about how the school teaches; the measure is of what a school says, throughout. And the collection holds three pages per school, so a school that teaches this way without describing it on those pages cannot be marked.

### 3A.5.1 The rule the model was given

> **The question.** You are reading a crawl of a school's own public website: typically the homepage, an about or welcome page, and a values or ethos page. Three pages is all there is. *Does this school's public self-presentation signal a traditional pedagogical identity?* Traditional pedagogy here means teaching understood as the transmission of knowledge by a subject expert to a class: the teacher instructs

from the front, the curriculum is a body of knowledge to be learned, and the school says so as part of what it is. Answer YES or NO. There is no middle band.

**What this is not.** It is not a judgement about the school; traditional is neither better nor worse than the alternative. It is not about how the school actually teaches, only what it says about itself. It is not about strictness, behaviour, uniform or discipline. It is not about academic ambition: "high expectations", "excellence" and "aspiration" appear on most school websites and say nothing about pedagogy. It is not about a school being old or having a long history.

**The test that does the work.** A naming is not an account. The question is whether the school has said something about how teaching works, not whether it has used a word from a vocabulary list. If the adjectives are stripped out, has this school said anything about what happens in its classrooms?

YES looks like a school that describes teaching as instruction by subject experts; that says knowledge comes first and skills follow from it; that describes whole-class teaching, direct instruction, practice and retrieval, or a knowledge-rich curriculum *and* says what it means by that; that names a pedagogical authority or doctrine and stands by it; or that calls its own teaching traditional and elaborates.

NO looks like a school that uses one of those words in passing and never returns to it; that says it has "subject specialists" or "expert teachers", which nearly every secondary school does; that says it is "knowledge-rich" once, in a list of other things; or that uses "traditional" about something that is not teaching.

**The four traps.** Each is a reason to answer NO. (1) "Subject specialists" or "expert teachers": a description of who is employed, not of how they teach. (2) A doctrine word used once, in passing: the word is the start of the evidence, not the evidence. (3) "Traditional" applied to something that is not pedagogy: traditional values, uniform, expectations of behaviour, or a traditional Key Stage 4 curriculum, which is a fact about curriculum structure. (4) Generic high-standards language: rigour, challenge, stretch, aiming high. This is ambition, not pedagogy.

**The other direction.** Some schools signal the opposite: child-led enquiry, personalised learning journeys, learning styles, the teacher as facilitator, skills before knowledge, or project-based learning as the organising idea. If a progressive framework is the school's primary account of teaching, the answer is NO, even if a traditional-sounding word appears somewhere. A school with no pedagogical claims at all is also NO.

> **How to answer.** Borderline goes NO. If it could be argued either way on what is in front of you, the answer is NO; most schools are NO. Record a verdict, a confidence, and for a YES the exact sentence from the document that carries it. If an exact sentence cannot be copied, the answer is NO. Judge each document only on what is in it, using nothing known about the school from elsewhere, and do not compare documents with each other.

## 3A.6 Additional Tables

Tables 3A.3 to 3A.5 report the full five-outcome versions of the three main-body stages, including the EBacc and Open components. These tables use the primary specification ($n = 99$, predecessor-filled 2019 grades, late-entry schools excluded), which is the same one as the main-body tables, so the Overall, English and Maths columns reproduce the body exactly and the two extra columns can be read on the same footing. In the EBacc and Open columns, warmth and strictness are both significant at the 5 per cent level for EBacc with near-equal magnitudes, and both are significant for Open as well, again with near-equal magnitudes.

The differential prediction set out in Section 3A.1 expected strictness to show most strongly in the EBacc component and warmth in the Open component. That prediction is not supported: both dimensions are associated with both components to a similar degree. As Section 3A.1 explains, both components mix progress with the school's entry decisions, so the entry-rate analysis in Table 3A.6 is the clearer way to answer the pathway question.

The strictness score predicts EBacc, humanities and languages entry nationally, by between 1.4 and 3.3 percentage points per band standard deviation, with or without the grade control. The attenuation figure quoted in the body of the chapter comes from these regressions: adding the EBacc entry rate to the EBacc specification reduces the strictness coefficient by 10 to 14 per cent, so entry decisions explain little of the association.

Table 3A.7 reports descriptive statistics for all five Progress 8 components and

Table 3A.3: Stage 1: total culture association, five outcomes

| | Overall | English | Maths | EBacc | Open |
|---|---|---|---|---|---|
| Warmth ($W$) | 0.146*** | 0.162*** | 0.135** | 0.164*** | 0.131** |
| | (0.043) | (0.042) | (0.052) | (0.052) | (0.057) |
| Strictness ($S$) | 0.106** | 0.071 | 0.076* | 0.138*** | 0.119** |
| | (0.041) | (0.044) | (0.045) | (0.051) | (0.055) |
| $N$ | 99 | 99 | 99 | 99 | 99 |
| $R^2$ | 0.624 | 0.615 | 0.540 | 0.597 | 0.514 |

*Notes:* Primary specification: full control set, predecessor-filled pre-COVID inspection grade, late-entry schools excluded — identical to the trio reported in the chapter body, extended to all five Progress 8 components. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3A.4: Stage 2: teaching quality benchmark, five outcomes

| | Overall | English | Maths | EBacc | Open |
|---|---|---|---|---|---|
| Teaching quality ($T$) | 0.206*** | 0.188*** | 0.165*** | 0.229*** | 0.229*** |
| | (0.042) | (0.042) | (0.048) | (0.053) | (0.047) |
| $N$ | 99 | 99 | 99 | 99 | 99 |
| $R^2$ | 0.555 | 0.556 | 0.477 | 0.502 | 0.488 |

*Notes:* Primary specification: full control set, predecessor-filled pre-COVID inspection grade, late-entry schools excluded — identical to the trio reported in the chapter body, extended to all five Progress 8 components. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3A.5: Stage 3: culture net of teaching, five outcomes

| | Overall | English | Maths | EBacc | Open |
|---|---|---|---|---|---|
| Warmth ($W$) | 0.137** | 0.166** | 0.144** | 0.182** | 0.072 |
| | (0.058) | (0.064) | (0.062) | (0.070) | (0.072) |
| Strictness ($S$) | 0.103** | 0.073 | 0.079 | 0.143*** | 0.102* |
| | (0.042) | (0.046) | (0.047) | (0.053) | (0.057) |
| Teaching quality ($T$) | 0.015 | -0.007 | -0.014 | -0.029 | 0.094 |
| | (0.064) | (0.073) | (0.069) | (0.077) | (0.086) |
| $N$ | 99 | 99 | 99 | 99 | 99 |
| $R^2$ | 0.624 | 0.615 | 0.540 | 0.597 | 0.523 |

*Notes:* Primary specification: full control set, predecessor-filled pre-COVID inspection grade, late-entry schools excluded — identical to the trio reported in the chapter body, extended to all five Progress 8 components. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3A.6: Culture and curriculum entry

| *Panel A: entry-rate regressions* | | | | | |
|---|---|---|---|---|---|
| Grade | Predictor | $N$ | EBacc entry | Humanities entry | Languages entry |
| No | Visited warmth ($W$) | 100 | 0.001 (0.033) | −0.008 (0.014) | 0.017 (0.030) |
| No | Visited strictness ($S$) | 100 | 0.036 (0.032) | 0.026** (0.012) | 0.014 (0.030) |
| No | Marking-scheme bands | 3,076 | 0.033*** (0.004) | 0.016*** (0.002) | 0.029*** (0.004) |
| Yes | Visited warmth ($W$) | 99 | −0.006 (0.033) | −0.009 (0.015) | 0.010 (0.030) |
| Yes | Visited strictness ($S$) | 99 | 0.045 (0.033) | 0.027** (0.013) | 0.021 (0.031) |
| Yes | Marking-scheme bands | 2,753 | 0.022*** (0.004) | 0.014*** (0.003) | 0.018*** (0.004) |
| *Panel B: channel decomposition, strictness → EBacc Progress 8 component* | | | | | |
| Grade | Model | $N$ | Without entry | With entry | Attenuation |
| No | National (marking-scheme bands) | 3,052 | 0.134*** | 0.115*** | 14.4% |
| No | Visited ($S$) | 100 | 0.232*** | 0.205*** | 11.3% |
| Yes | National (marking-scheme bands) | 2,746 | 0.112*** | 0.100*** | 10.2% |
| Yes | Visited ($S$) | 99 | 0.244*** | 0.218*** | 10.7% |

*Notes:* Panel A regresses each entry rate (EBacc, humanities, languages) on the standardised culture score plus controls, for the visit scores and the national marking-scheme strictness bands. The Grade column records whether the predecessor-filled 2019 inspection-grade control is absent (No) or present (Yes); late-entry schools excluded. Panel B reports the strictness coefficient on the EBacc Progress 8 component with and without the EBacc entry rate as a control; the difference shows how much of the association is explained by entry decisions. Standard errors (HC3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The national strictness measure is the marking-scheme band of Chapter 2 (1–5, standardised over each estimation sample). Entry rates are derived from pupil counts.

the four Attainment 8 buckets, and Table 3A.8 the national strictness estimate on all five outcomes. The coefficients run from 0.11 (Maths) to 0.15 (Open). All five have small standard errors, so the overall result does not rest on any one component.

Table 3A.9 enters the five observational blocks of the enacted scores jointly, and Table 3A.10 takes the decomposition to the level of the 33 individual observation items, each entered in its own small model with a false-discovery correction across the set and each item's inter-rater reliability shown beside it. Every one of the 33 items has a positive coefficient and 28 survive the false-discovery correction. The sub-score result reported in the body, that only outside-lesson warmth and in-lesson strictness keep their associations when entered together, therefore says which blocks are associated with progress once the other blocks are held constant. It does not say which items relate to progress at all. Nearly all of them do.

Table 3A.12 reports the full combination analysis for both the visited sample and the national panel. It is the basis for the body's finding that the advantage

Table 3A.7: Progress 8 and Attainment 8 outcomes: summary statistics

| Outcome | Sample | *N* | Mean | SD | Min | Max |
|---|---|---|---|---|---|---|
| P8 Overall (2-yr avg) | Visited | 103 | 0.252 | 0.478 | -0.700 | 1.345 |
| | Interviewed | 299 | 0.095 | 0.487 | -1.655 | 1.345 |
| | National | 3240 | -0.015 | 0.510 | -3.570 | 2.465 |
| P8 English (2-yr avg) | Visited | 103 | 0.256 | 0.507 | -0.750 | 1.545 |
| | Interviewed | 299 | 0.093 | 0.485 | -1.485 | 1.545 |
| | National | 3240 | -0.017 | 0.519 | -4.035 | 2.360 |
| P8 Maths (2-yr avg) | Visited | 103 | 0.270 | 0.449 | -0.595 | 1.345 |
| | Interviewed | 299 | 0.093 | 0.470 | -1.370 | 1.345 |
| | National | 3240 | -0.017 | 0.487 | -2.030 | 2.660 |
| P8 EBacc (2-yr avg) | Visited | 103 | 0.297 | 0.549 | -0.790 | 1.490 |
| | Interviewed | 299 | 0.116 | 0.570 | -1.960 | 1.490 |
| | National | 3240 | -0.020 | 0.587 | -4.950 | 2.655 |
| P8 Open (2-yr avg) | Visited | 103 | 0.172 | 0.528 | -1.095 | 1.435 |
| | Interviewed | 299 | 0.055 | 0.554 | -3.095 | 1.500 |
| | National | 3240 | -0.034 | 0.573 | -5.140 | 2.220 |
| Att8 English (2024–25) | Visited | 103 | 10.658 | 1.757 | 7.900 | 17.300 |
| | Interviewed | 300 | 10.277 | 1.696 | 6.700 | 17.300 |
| | National | 3274 | 10.049 | 1.702 | 0.100 | 17.300 |
| Att8 Maths (2024–25) | Visited | 103 | 9.887 | 1.975 | 6.700 | 17.400 |
| | Interviewed | 300 | 9.514 | 1.891 | 5.400 | 17.400 |
| | National | 3274 | 9.263 | 1.879 | 0.300 | 17.600 |
| Att8 EBacc (2024–25) | Visited | 103 | 14.783 | 3.235 | 9.400 | 26.500 |
| | Interviewed | 300 | 14.156 | 3.118 | 6.200 | 26.500 |
| | National | 3274 | 13.696 | 3.103 | 0.500 | 26.600 |
| Att8 Open (2024–25) | Visited | 103 | 14.676 | 2.936 | 10.000 | 26.300 |
| | Interviewed | 300 | 14.184 | 2.868 | 4.900 | 26.300 |
| | National | 3274 | 13.869 | 2.830 | 0.000 | 26.300 |

*Notes:* The interviewed sample numbers 303 schools and contains the 103 visited schools; $N$ falls short of 303 where an outcome is unpublished for a school. P8 outcomes are 2-year averages of 2022–23 and 2023–24 component scores where both years are available. Att8 2024–25 components are the contemporaneous robustness outcome; no Progress 8 is available for this cohort because the cohort were in Year 6 in 2019–20 when Key Stage 2 assessments were cancelled.

Table 3A.8: The national strictness bands on all five Progress 8 outcomes

| Outcome | Coefficient | (se) | $n$ |
|---|---|---|---|
| Overall | $+0.130^{***}$ | (0.008) | 3,052 |
| English | $+0.114^{***}$ | (0.008) | 3,052 |
| Maths | $+0.110^{***}$ | (0.008) | 3,052 |
| EBacc | $+0.135^{***}$ | (0.009) | 3,052 |
| Open | $+0.153^{***}$ | (0.009) | 3,052 |

*Notes:* Full controls without the inspection-grade control. Standard errors in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

Table 3A.9: The five visit sub-scores entered jointly

| Sub-score | Overall | English | Maths |
|---|---|---|---|
| Classroom warmth (W1) | 0.067 | 0.114 | 0.115 |
| | (0.083) | (0.082) | (0.084) |
| Outside warmth (W2) | 0.111*** | 0.136*** | 0.108** |
| | (0.040) | (0.042) | (0.043) |
| Classroom strictness (S1) | 0.115*** | 0.130*** | 0.106** |
| | (0.042) | (0.038) | (0.052) |
| Outside strictness (S2) | −0.000 | −0.054 | −0.021 |
| | (0.046) | (0.048) | (0.049) |
| Teaching quality (T1) | 0.000 | −0.053 | −0.063 |
| | (0.087) | (0.096) | (0.087) |
| *N* | 99 | 99 | 99 |

*Notes:* Classroom warmth (W1), outside-of-classroom warmth (W2), classroom strictness (S1), outside-of-classroom strictness (S2) and teaching quality (T1) are entered together, each standardised within the estimation sample, with the primary control set. Only outside-of-lesson warmth and in-lesson strictness remain significant when the five are entered together. The other three add little once those two are in the regression. Standard errors (HC3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Primary specification: visited schools, late-entry schools excluded, full control set with the predecessor-filled 2019 grade. The sub-scores are correlated, so the individual coefficients should be read with care: variance inflation factors are 5.77 (W1), 2.02 (W2), 1.93 (S1), 2.15 (S2) and 6.63 (T1) — W1 and T1 overlap heavily, which widens their standard errors.

of warm-and-strict schools equals the sum of the two separate associations.

Entering predicted warmth and the bands together (Table 3A.12) moves warmth from +0.078 to +0.054 and the bands from +0.130 to +0.117, consistent with the two scores picking up some of the same features of a report. The univariate estimates remain the primary numbers because, as the body explains, the two scores pick up different amounts of what a report says besides culture, so their joint coefficients are not comparable.

Two results qualify and extend the body's espoused comparison. First, espoused warmth does not predict progress, and part of the reason may be the scores themselves rather than what heads know. Heads overstate warmth against the enacted scores by 1.0 points on average, against 0.4 for strictness, so the espoused warmth scores cluster near the top of their scale. Scores that differ little between schools cannot predict differences between schools. Second, the size of a head's overstatement does not matter in itself: once the enacted score is in the regression,

Table 3A.10: Item-level associations with Progress 8

| Item | Setting | Sub-score | $\beta$ | SE | $q$ | IRR $r$ | Pairs |
|---|---|---|---|---|---|---|---|
| Outcomes | Classroom | T1 | 0.222† | 0.040 | <0.001 | 0.620 | 432 |
| Concentration | Classroom | S1 | 0.204† | 0.037 | <0.001 | 0.647 | 467 |
| Motivation | Classroom | W1 | 0.195† | 0.044 | <0.001 | 0.610 | 467 |
| Respectfully | Classroom | S1 | 0.182† | 0.047 | <0.001 | 0.654 | 467 |
| Explanation | Classroom | T1 | 0.177† | 0.039 | <0.001 | 0.578 | 454 |
| Students | Outside | W2 | 0.176† | 0.039 | <0.001 | 0.705 | 93 |
| Misbehaviour | Classroom | S1 | 0.170† | 0.035 | <0.001 | 0.702 | 439 |
| Methods | Classroom | T1 | 0.166† | 0.042 | <0.001 | 0.552 | 467 |
| Response | Classroom | S1 | 0.166† | 0.039 | <0.001 | 0.669 | 277 |
| Questioning | Classroom | T1 | 0.165† | 0.048 | 0.002 | 0.585 | 468 |
| Praise | Classroom | W1 | 0.164† | 0.045 | 0.001 | 0.688 | 456 |
| Verbal | Classroom | T1 | 0.153† | 0.053 | 0.008 | 0.562 | 454 |
| Relationships | Outside | W2 | 0.153† | 0.037 | <0.001 | 0.724 | 89 |
| Student | Classroom | W1 | 0.153† | 0.050 | 0.005 | 0.637 | 457 |
| Names | Classroom | W1 | 0.152† | 0.037 | <0.001 | 0.698 | 463 |
| Structure | Classroom | T1 | 0.148† | 0.037 | <0.001 | 0.560 | 461 |
| Interactions | Classroom | W1 | 0.146† | 0.044 | 0.003 | 0.616 | 463 |
| Disruption | Classroom | S1 | 0.146† | 0.041 | 0.002 | 0.788 | 466 |
| Resource | Classroom | T1 | 0.143† | 0.042 | 0.002 | 0.528 | 463 |
| Differentiation | Classroom | T1 | 0.140† | 0.053 | 0.013 | 0.637 | 437 |
| Sanction | Outside | S2 | 0.136† | 0.044 | 0.005 | 0.714 | 87 |
| Corridors | Outside | S2 | 0.134† | 0.044 | 0.005 | 0.657 | 93 |
| Teacher | Classroom | T1 | 0.134† | 0.045 | 0.006 | 0.636 | 465 |
| Interactions | Outside | W2 | 0.128† | 0.049 | 0.014 | 0.684 | 86 |
| Arrival | Outside | S2 | 0.125† | 0.046 | 0.011 | 0.474 | 91 |
| Reward | Outside | – | 0.120† | 0.044 | 0.010 | 0.805 | 83 |
| Verbal | Outside | – | 0.098 | 0.054 | 0.086 | 0.485 | 76 |
| Displays | Outside | – | 0.098† | 0.040 | 0.022 | 0.548 | 91 |
| Recreational | Outside | S2 | 0.094† | 0.040 | 0.025 | 0.532 | 91 |
| Discussion | Classroom | T1 | 0.065 | 0.046 | 0.175 | 0.722 | 217 |
| Differentiation | Outside | – | 0.062 | 0.062 | 0.321 | 0.552 | 70 |
| Canteen | Outside | S2 | 0.058 | 0.038 | 0.146 | 0.727 | 91 |
| Alignment | Outside | – | 0.050 | 0.045 | 0.278 | 0.578 | 84 |

*Notes:* Every row regresses average Progress 8 on that item alone (standardised) plus the primary control set with the predecessor-filled grade; late-entry schools excluded. $N = 98$ for the in-lesson items and as low as 88 for the outside items, which were recorded at fewer schools. Rows are sorted by coefficient size. The $q$ column is the Benjamini–Hochberg false-discovery-rate adjusted $p$-value across all 33 items; a dagger marks items that survive at $q < 0.05$ (28 of 33; 33 carry a positive coefficient). The final columns give each item's inter-rater reliability: the pooled pairwise Pearson correlation between independent observers, and the number of observer pairs it rests on.

the gap between the espoused and enacted scores predicts nothing (Table 3A.13).

The pseudo-progress measure of Table 3A.14 is validated by running the identical construction one year earlier, where it correlates with the published Progress 8 at +0.81. The measure is an imputed value-added outcome for the one cohort measured contemporaneously with the fieldwork. As the body reports, the culture coefficients on the pseudo-outcome sit within 0.04 of their overall real-P8

Table 3A.11: Linearity of the national strictness gradient

| Strictness band | Coefficient | (se) | Schools in band |
|---|---|---|---|
| Band 1 | -0.267*** | (0.089) | 5 |
| Band 2 | -0.048*** | (0.018) | 544 |
| Band 3 | 0 (base) | — | 907 |
| Band 4 | +0.145*** | (0.016) | 1,251 |
| Band 5 | +0.434*** | (0.028) | 345 |

*Notes:* Linearity of the national strictness gradient: average Progress 8 on marking-scheme band dummies (band 3 omitted as the base), replacing the standardised linear band score of the national strictness estimate. Same specification as that estimate: full controls (FSM, EAL, KS2 cohort mean, SEN, log size, years since inspection, academy, urban, selective), no inspection-grade control, HC3 standard errors, late-entry schools excluded, $n = 3{,}052$. Standard errors in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Each coefficient is the adjusted P8 gap between that band and band 3. Band 1 contains very few schools, so its estimate is imprecise.

Table 3A.12: The authoritative-school typology

| | Visited schools | National (text instruments) | |
|---|---|---|---|
| | | Panel A | Panel B |
| *Adjusted quadrant means (mean, SE, n)* | | | |
| Authoritative (high $W$, high $S$) | +0.547 (0.066) | +0.347 (0.025) | +0.275 (0.026) |
| | $n = 33$ | $n = 299$ | $n = 277$ |
| Authoritarian (low $W$, high $S$) | +0.142 (0.075) | +0.240 (0.047) | +0.213 (0.049) |
| | $n = 16$ | $n = 46$ | $n = 44$ |
| Permissive (high $W$, low $S$) | +0.380 (0.073) | +0.022 (0.010) | +0.053 (0.011) |
| | $n = 15$ | $n = 1{,}227$ | $n = 1{,}095$ |
| Neglectful (low $W$, low $S$) | +0.035 (0.058) | −0.078 (0.009) | −0.051 (0.009) |
| | $n = 35$ | $n = 1{,}480$ | $n = 1{,}330$ |
| *Regression tests* | | | |
| Warmth ($W$) | 0.150*** | 0.054*** | 0.055*** |
| | (0.043) | (0.007) | (0.008) |
| Strictness ($S$) | 0.109** | 0.117*** | 0.096*** |
| | (0.042) | (0.008) | (0.008) |
| $W \times S$ interaction | 0.039 | 0.004 | −0.005 |
| | (0.035) | (0.008) | (0.008) |
| Authoritative vs rest | 0.403*** | 0.363*** | 0.262*** |
| | (0.088) | (0.027) | (0.027) |
| $N$ | 99 | 3,052 | 2,746 |

*Notes:* The upper panel gives covariate-adjusted mean Progress 8 for the four quadrants formed by splits on warmth and strictness; the lower panel gives the continuous interaction test and the contrast of the authoritative (high-warmth, high-strictness) quadrant against all others. The first column uses the visit scores on the primary specification; the national columns use the prediction-model warmth and the marking-scheme strictness bands of Chapter 2, without (A) and with (B) the predecessor-filled 2019 inspection-grade control. In the national columns warmth is split at the national median of the predicted score, and "high" strictness means band 5, above the in-sample median of 4, so the high-strictness cells are small. Scores are standardised within each estimation sample; late-entry schools excluded throughout. Standard errors (HC3) in parentheses; the visited-sample quadrant means use HC1, which the single-school cells require. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$. Visited-school quadrants are median splits within the visited sample.

Table 3A.13: The espoused–enacted gap and Progress 8

| | Signed gap | Absolute gap | Absolute gap (+ component quadratics) |
|---|---|---|---|
| *Visited sample (visit vs interview, $n = 99$)* | | | |
| Warmth | 0.047 | 0.032 | 0.011 |
| | (0.040) | (0.046) | (0.058) |
| Strictness | 0.065 | 0.002 | −0.015 |
| | (0.050) | (0.061) | (0.061) |

*Notes:* Each row regresses average Progress 8 on the enacted (observed) score, the espoused-minus-enacted gap, and the primary control set, on the visited schools. The first column enters the signed gap, the second the absolute gap, and the third re-enters the absolute gap alongside squared terms in both the espoused and the enacted score. An absolute gap could predict progress simply because progress is a non-linear function of one of the scores. The squared terms remove that explanation. Standard errors in parentheses (HC3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Scores are z-standardised before the gap is formed; late-entry schools excluded.

counterparts.

Table 3A.15 reports how much a school's Progress 8 moves from year to year. The movement is small. The table also reports Progress 8 in the years before and after a school became an academy, separately for sponsor-led and converter academies, because conversion is the main recorded event that could change a school's culture. Sponsor-led schools decline before converting and recover afterwards. Schools are given a sponsor after a run of poor results. Poor results tend to be followed by better ones whether or not anything changes. The recovery is therefore what one would expect without any effect of conversion. Converters are flat throughout.

Table 3A.16 tests the sorting concern directly, regressing the 2023/24 share of pupils with social, emotional and mental-health needs on strictness, conditional on the school's 2015/16 share and the standard controls. Nationally, stricter schools have a lower current share conditional on their baseline (−0.114 of a percentage point per standard deviation of the score, $p = 0.003$, $n = 2{,}594$); the direction is the same in the visited sample but imprecise (−0.257, $p = 0.36$, $n = 94$), and warmth, which is available only for the visited sample, is unrelated to the change there. The pattern is consistent with sorting, though it cannot say whether stricter schools admit fewer such pupils or whether such pupils leave those schools after admission. Adding the baseline share to the primary specification (Table 3.8)

Table 3A.14: Culture coefficients on the 2024/25 stand-in outcome

| | Predictor | $N$ | Pseudo overall | Pseudo English | Pseudo Maths | Real P8 overall |
|---|---|---|---|---|---|---|
| A | Visited warmth ($W$) | 100 | 0.162*** | 0.182*** | 0.114* | 0.154*** |
| | | | (0.050) | (0.051) | (0.061) | (0.041) |
| A | Visited strictness ($S$) | 100 | 0.135*** | 0.092* | 0.156*** | 0.098** |
| | | | (0.050) | (0.050) | (0.056) | (0.040) |
| A | Marking-scheme bands | 3,076 | 0.152*** | 0.131*** | 0.132*** | 0.129*** |
| | | | (0.009) | (0.010) | (0.010) | (0.008) |
| B | Visited warmth ($W$) | 99 | 0.165*** | 0.195*** | 0.116* | 0.147*** |
| | | | (0.057) | (0.057) | (0.067) | (0.043) |
| B | Visited strictness ($S$) | 99 | 0.135** | 0.081 | 0.157** | 0.107** |
| | | | (0.055) | (0.055) | (0.061) | (0.041) |
| B | Marking-scheme bands | 2,753 | 0.152*** | 0.133*** | 0.133*** | 0.127*** |
| | | | (0.010) | (0.010) | (0.010) | (0.008) |

*Notes:* The pseudo measure is the part of a school's 2024/25 Attainment 8 that is not predicted by its Key Stage 2 intake, averaged over several years and standardised. It is a progress-like score for the cohort measured at the time of the fieldwork. Each cell is the (standardised) culture coefficient from the same specification run on the pseudo outcome and on real (2022–24 average) Progress 8 for the same schools, without (A) and with (B) the predecessor-filled 2019 inspection-grade control; late-entry schools excluded. Standard errors (HC3) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The same construction, rebuilt one year earlier from the 2021–22 and 2022–23 baselines, correlates with the real 2023–24 Progress 8 at +0.81 ($n = 3{,}242$). The marking-scheme rows use the strictness bands of Chapter 2 (1–5, standardised over each estimation sample). The pseudo measure is school-level, not pupil-matched, so it is a robustness check, never a primary outcome.

moves the strictness coefficient from 0.106 to 0.090; re-estimating on the same 94 schools without the control gives 0.097, so about seven per cent of the strictness association is attributable to the control itself and the rest of the movement to the change of sample.

Table 3A.17 reports pairwise Pearson correlations between the three enacted culture scores ($W_{\text{enac}}$, $S_{\text{enac}}$, $T_{\text{enac}}$) and the two espoused scores ($W_{\text{esp}}$, $S_{\text{esp}}$) against each control variable, estimated on the 103 visited schools. No pairwise correlation exceeds $|r| = 0.29$. The largest entries in the table are between espoused strictness and KS2 prior attainment ($r = +0.29$), between $T_{\text{enac}}$ and selective status ($r = +0.27$) and EAL share ($r = -0.26$), and between $W_{\text{enac}}$ and EAL share ($r = -0.25$). This is the detail behind the body's statement that the culture scores are not close proxies for intake or school type. Selective schools score modestly higher on both culture dimensions ($r = +0.20$ for warmth and $+0.22$ for strictness), and selective admissions is held constant in every specification.

Table 3A.15: Stability of Progress 8 and academisation

*Panel A: within-school variation (2017–18 to 2023–24, $n = 3{,}232$ schools)*

Within-school SD: mean 0.196, median 0.177, 10th percentile 0.086, 90th percentile 0.330

| Adjacent-year pair | $N$ | Moved >0.3 | Moved >0.5 |
|---|---|---|---|
| 2017–18 → 2018–19 | 2,762 | 19.0% | 4.5% |
| 2021–22 → 2022–23 | 3,185 | 21.9% | 5.1% |
| 2022–23 → 2023–24 | 3,126 | 18.5% | 3.1% |
| All adjacent pairs pooled | 9,073 | 19.9% | 4.3% |

*Panel B: mean Progress 8 by event year around academisation*

| Event year | Converter | Sponsor-led |
|---|---|---|
| $t-5$ | −0.08 ($n = 135$) | −0.42 ($n = 71$) |
| $t-4$ | −0.08 ($n = 121$) | −0.45 ($n = 68$) |
| $t-3$ | −0.11 ($n = 165$) | −0.52 ($n = 123$) |
| $t-2$ | −0.12 ($n = 246$) | −0.54 ($n = 197$) |
| $t-1$ | −0.10 ($n = 335$) | −0.53 ($n = 242$) |
| $t+0$ | −0.08 ($n = 359$) | −0.43 ($n = 274$) |
| $t+1$ | −0.03 ($n = 293$) | −0.38 ($n = 226$) |
| $t+2$ | −0.07 ($n = 207$) | −0.33 ($n = 165$) |
| $t+3$ | −0.08 ($n = 167$) | −0.32 ($n = 137$) |
| $t+4$ | −0.06 ($n = 145$) | −0.27 ($n = 136$) |
| $t+5$ | −0.05 ($n = 195$) | −0.27 ($n = 181$) |
| $t+6$ | −0.03 ($n = 196$) | −0.29 ($n = 175$) |
| $t+7$ | −0.03 ($n = 132$) | −0.31 ($n = 107$) |
| $t+8$ | +0.01 ($n = 56$) | −0.21 ($n = 43$) |

*Notes:* Panel A summarises each school's own standard deviation of Progress 8 across the five published years and the share of schools moving by more than 0.3 or 0.5 of a grade between consecutive years. Panel B gives mean Progress 8 by year relative to the year of academy conversion, separately for converter and sponsor-led academies. Panel A uses schools with at least three published years. There is no Progress 8 for 2019–20 or 2020–21 (COVID), so "adjacent" means consecutive academic years only; the 2018–19 to 2021–22 boundary is not counted as a pair. Panel B rests on the predecessor–successor bridging table (converter and sponsor-led academies; cells with fewer than five schools suppressed); these are raw event-time means, not a matched event study.

A final check re-estimates the Stage 1 specification on the schools whose headteacher did not change. The flag compares the headteacher recorded for each school in the Get Information About Schools (GIAS) extract of 1 September 2022 with the extract of 1 July 2025. Names count as the same person when they match after titles are stripped, when one is an initial form of the other, or when a co-headteacher pair is listed in a different order, and a school absent from the 2022 extract is looked up under its predecessor URN. The flag covers all 99 estimation-sample schools: 58 kept the same head and 41 changed. On the 58, warmth is +0.160 (se 0.044) and strictness +0.130 (se 0.052), both significant

Table 3A.16: The SEMH composition mechanism

| | Visited (S) | Visited (W) | National (S) |
|---|---|---|---|
| Strictness ($S_{\text{visit}}$, per SD) | −0.257 | | |
| | (0.280) | | |
| SEMH share 2015–16 (%) | 0.176 | 0.191* | 0.194*** |
| | (0.120) | (0.115) | (0.022) |
| Warmth ($W_{\text{visit}}$, per SD) | | −0.094 | |
| | | (0.240) | |
| Strictness (marking-scheme bands, per SD) | | | −0.114*** |
| | | | (0.038) |
| $N$ | 94 | 94 | 2,594 |
| $R^2$ | 0.559 | 0.547 | 0.484 |

*Notes:* Each column regresses a school's 2023–24 share of pupils with social, emotional and mental-health (SEMH) needs, as a percentage of the roll, on one culture score plus the school's 2015–16 baseline SEMH share and the primary control set: visited-school strictness and warmth ($n = 94$ visited schools), and the national marking-scheme strictness bands ($n = 2{,}594$). Coefficients are percentage points per standard deviation of the score. Nationally, stricter schools have a slightly lower current SEMH share given their 2015–16 share. In the visited sample the sign is the same but the estimate is imprecise. This is consistent with some sorting of pupils by school strictness. Adding the baseline share to the main regression reduces the strictness coefficient by under a tenth. Standard errors (HC3) in parentheses. Outcome: 2023–24 SEMH pupils as a percentage of the roll; baseline is the 2015–16 SEMH count over the current roll. Primary controls; late-entry excluded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3A.17: Pairwise correlations: culture scores and control variables

| Control | $W_{\text{enac}}$ | $S_{\text{enac}}$ | $T_{\text{enac}}$ | $W_{\text{esp}}$ | $S_{\text{esp}}$ |
|---|---|---|---|---|---|
| KS2 | +0.17 | +0.15 | +0.24 | +0.16 | +0.29 |
| FSM% | −0.13 | −0.04 | −0.03 | −0.03 | −0.19 |
| EAL% | −0.25 | +0.00 | −0.26 | +0.09 | −0.06 |
| SEN% | −0.07 | −0.11 | −0.04 | −0.03 | −0.21 |
| Log size | +0.03 | −0.04 | +0.06 | −0.02 | −0.04 |
| Academy | −0.01 | −0.07 | +0.06 | −0.11 | +0.02 |
| Urban | −0.16 | −0.04 | −0.16 | −0.03 | +0.06 |
| Selective | +0.20 | +0.22 | +0.27 | +0.10 | +0.11 |
| Yrs Ofsted | +0.00 | −0.09 | −0.12 | −0.05 | −0.11 |

*Notes:* Pearson correlation coefficients. $W_{\text{enac}}$, $S_{\text{enac}}$ and $T_{\text{enac}}$ are the enacted culture scores built from school visit observations only (mean of $W_1, W_2$; $S_1, S_2$; $T_1$, each on the 1–5 scale). $W_{\text{esp}}$ and $S_{\text{esp}}$ are the espoused scores built from the headteacher interview. The two sets share no component; they are reported side by side rather than averaged, because the two are correlated at only $r = 0.25$ (warmth) and 0.41 (strictness). $N$ varies slightly by column due to missing control values; all correlations use the sample of 103 visited schools.

and close to the full-sample values. On the 41, warmth is +0.028 (se 0.111) and strictness +0.121 (se 0.100), neither significant. A fully interacted comparison cannot reject equal associations across the two groups ($p = 0.27$ for warmth, 0.94 for strictness). This flag replaces an earlier one based on the headteacher named

in the inspection report, which could be determined for only 29 of the 99 schools, because most reports do not name the headteacher.

Table 3A.18 reports the sensitivity measures quoted in the robustness discussion: Oster's $\delta$ and the equal-selection bound for each culture score, computed with `psacalc` under $R_{\max} = 0.81$, and the E-values for the point estimates and for the confidence-interval boundaries.

Table 3A.18: Sensitivity to unmeasured confounding

| | Controlled $\beta$ | No controls $\beta$ | $\delta$ for $\beta = 0$ | $\beta$ at $\delta = 1$ | E-value (point) | E-value (CI bound) |
|---|---|---|---|---|---|---|
| Warmth ($W$) | 0.146 | 0.082 | 1.71 | 0.112 | 1.99 | 1.50 |
| Strictness ($S$) | 0.106 | 0.145 | 1.09 | 0.013 | 1.76 | 1.27 |

*Notes:* Primary specification on overall Progress 8 ($n = 99$); coefficients per standard deviation of the enacted scores. Oster's $\delta$ (Oster, 2019, via `psacalc`) is the strength of selection on unobservables, relative to selection on the full control set, that would drive the coefficient to zero given $R_{\max} = 1.3\tilde{R}^2 = 0.81$; $\beta$ at $\delta = 1$ is the coefficient that survives when unobservables are assumed to matter exactly as much as the controls. Each score is treated as the single treatment with the other score among the controls. The E-value converts the coefficient to an approximate risk ratio using the outcome standard deviation on the estimation sample (0.47) and reports the minimum association a confounder would need with both the score and the outcome to explain the estimate away; the CI-bound column applies the same conversion to the boundary of the 95 per cent confidence interval nearer zero.

# Chapter 4

# The Productivity and Diversity Costs of Constraints to Subject Choice

*Arun Advani, Elaine Drayton and Damian Phelan*[1]

***Abstract.*** *School and university subject choices are an important determinant of career pathway and hence lifetime wages. We show that access to high-return subjects varies systematically across regions and demographic groups of school students in England. Using within-school variation in access, consistent with the arrival or departure of a subject teacher within a school, we show this has a large causal impact on subject studied at university. Using linked education and employment data, we document the lifetime income cost of not having access to certain subjects at school. The cost estimates rest mainly on precisely estimated effects on university attendance and degree choice combined with published lifetime returns by degree subject, while the direct age-28 earnings estimate points the same way but is less precise. Because availability is lowest outside London and the South East, expanding it would also shift the intake of high-return degrees towards under-represented regions. We estimate that the fiscal return to additional teachers in high-return subjects exceeds the cost of recruiting, training and employing them.*

[1]Arun Advani: University of Warwick, CenTax, and Institute for Fiscal Studies. Elaine Drayton: Institute for Fiscal Studies. Damian Phelan: University College London and Institute for Fiscal Studies. The authors thank Jack Britton, Laura van der Erve, Christine Farquharson and Ross Warwick, and seminar participants at the IFS, CEPEO, UCL and the Royal Economic Society Conference 2024. This research was supported by the Economic and Social Research Council (ES/W010224/1), the Discover Economics campaign and the Productivity and Diversity group at the London School of Economics. This work was produced using statistical data from ONS. The use of the ONS statistical data in this work does not imply the endorsement of the ONS or the Statistical Service Agency in relation to the interpretation or analysis of the statistical data. This work uses research datasets which may not exactly reproduce National Statistics aggregates. The Longitudinal Education Outcomes data are owned by the Department for Education and were accessed through the ONS Secure Research Service.

## 4.1 Introduction

Field of study is one of the largest determinants of lifetime earnings. The gap between the highest- and lowest-return university subjects rivals the entire college wage premium (Altonji et al., 2016). Causal estimates confirm that much of this gap reflects the effect of the field itself, not selection (Kirkebøen et al., 2016; Bleemer and Mehta, 2022). Yet access to high-return fields is not equally distributed: who gets to choose depends on institutional constraints that vary across schools, regions, and demographic groups (e.g. Goodman, 2019). Such constraints cause human capital misallocation that can have large aggregate growth costs (Hsieh et al., 2019), but micro-level evidence on which specific constraints matter and what it costs to remove them is scarce. Existing causal estimates of the return to a field of study come from students who just gained or just missed a place, at admission cutoffs or grade thresholds. They do not tell us what happens when the option to choose the subject does not exist at all. We study that second case.

We study the causal effect of taking economics at A-level – the main upper-secondary qualification in England – on university subject choice and lifetime earnings. Our instrument is within-school, across-cohort variation in whether a school offers A-level economics, consistent with the arrival or departure of a suitably qualified teacher. We do not observe teachers directly. Provision is inferred from exam sittings, and we present evidence below that these changes are not preceded by rising student demand for the subject and are not accompanied by changes in the cohorts' prior attainment or composition. This is the pattern we would expect if the changes came from a teacher arriving or leaving. Using population-level linked administrative data that follow students from age 11 through university and into the labour market, we show that taking A-level economics substantially increases the probability of studying economics at university. We also find evidence of substantial earnings gains for the students whose choices are shifted. We estimate the marginal value of public funds (MVPF) of hiring additional economics teachers

and find that the policy more than pays for itself.

England provides an unusually clean setting for studying supply-side constraints on subject choice. At age 16, students select three or four subjects to study at A-level, and these choices largely determine which degree they can pursue at university. A-level economics is the principal gateway to an economics degree: students who take it are 27–29 percentage points more likely, in raw and covariate-adjusted terms, to study economics at university than those who do not.[2] But during our study period fewer than half of state schools offered A-level economics, and whether a school does so depends primarily on whether it employs a teacher qualified to teach the subject.[3]

There is no centralised competitive selection for A-level economics places. Schools may set general sixth-form entry requirements, but conditional on entering the sixth form the subject is not rationed, so this is a binary supply constraint rather than a rationing problem. Either the school has an economics teacher and offers the subject, or it does not.

The principle that supply-side constraints on high-return human capital matter is general; the English institutional setting provides three features that make it possible to measure the effect cleanly. First, students typically remain in the same secondary school, or move to its linked school, from age 11 to 18, so school choice at age 11 is unlikely to reflect anticipated A-level economics availability years later. Second, whether a school offers economics varies across cohorts within the same school, with roughly one-third of schools switching between offering and not offering economics over our sample period. This generates within-school variation that is plausibly driven by staffing changes rather than by shifts in student characteristics. Third, the Longitudinal Education Outcomes (LEO)

[2]In the pooled 2002–2008 sample, 29.5% of A-level economics takers study economics at university, against 0.9% of non-takers: a raw gap of 28.6 percentage points, or 27.3 controlling for attainment and demographics.

[3]Access has improved only modestly since: in 2025, economics was offered at just over half of non-selective state schools in England, against 90% of selective schools and 82% of independent schools, a gap the Bank of England attributes partly to a shortage of qualified economics teachers (Bank of England, 2025).

dataset links the universe of school records, university enrolment, and tax-based earnings data, allowing us to trace the consequences of school-level subject access through to adult labour market outcomes at population scale.

Our sample comprises all students who completed GCSEs (nationally graded exams at age 16) in England between 2002 and 2008 and continued to A-level study, for whom we observe earnings through to age 28. The instrument is an indicator for whether a student's school offered A-level economics to their cohort. We include school fixed effects, to absorb all time-invariant differences between schools – including long-run attainment levels, socioeconomic composition, and resource availability – so that identification comes solely from within-school changes in whether the subject is offered. Cohort fixed effects and rich individual-level controls for prior attainment, demographics, and socioeconomic background address time-varying confounders.

We report three sets of results. First, we document that access to A-level economics is unequally distributed along every dimension we examine – sex, ethnicity, socioeconomic status, and geography. Differences in whether schools offer the subject, rather than differences in student preferences or attainment, explain 36% of the regional take-up gap and between 6% and 82% of the take-up gaps across demographic groups. The regional gap is the largest we observe. The fraction of it explained by differences in availability is about the same in every cohort we examine.

The 82% share applies to the gap by free school meal status, which is small (0.4 percentage points) but almost entirely a matter of access. Students eligible for free school meals (FSM), a standard administrative marker of low socioeconomic family background, are 3.5 percentage points less likely to attend a school that offers economics than their non-FSM peers. Students in London and the South East of England are up to 35 percentage points more likely to have access than students in the Midlands, North East, or South West.[4]

[4]From the full regional breakdown of 2011 access rates: London 82.0%, North East 46.8%, South West 55.8%, East Midlands 56.6%, West Midlands 60.3%. Table 4.2 reports the two-way

Second, we estimate the causal effect of taking A-level economics on earnings. Within-school availability of A-level economics increases the probability of taking the subject by 5.7 percentage points. Some students take A-level economics only because their school offers it. We call these students compliers, and they are the students our causal estimates describe. Compliers are 15.2 percentage points more likely to study economics at university, confirming the gateway role documented in our descriptive analysis.

The instrumental variable (IV) estimate implies that taking A-level economics raises earnings at age 28 by around 14%, though this estimate is significant only at the 10% level.

Combining our estimates with published estimates of lifetime returns by degree subject (Britton et al., 2026), the implied lifetime gain is approximately £190,000 in present value (gross) and £90,000 net of tax and student loan repayments. These lifetime figures are calculated from the effects of A-level economics on university attendance and on degree choice, valued at the published returns. Both effects are estimated precisely, so the lifetime figures do not rest on the imprecise age-28 earnings estimate. Most of the implied lifetime gain comes from the students who switch into economics degrees, which carry among the highest lifetime returns of any subject. Around 15 in every 100 compliers switch into an economics degree, and each of these gains the difference between the net lifetime return to an economics degree, around £560,000 (Britton et al., 2026), and the return to the subjects they would otherwise have studied. Around 14 in every 100 are drawn into university altogether, and each of these gains the average net return to a degree. The switchers account for roughly 85% of the implied £90,000 gain per complier. However, most of the earnings gain we observe at age 28 goes to compliers who do not change their degree subject, so the gains cannot come only through degree choice. Because the degree-based calculation counts only gains of that kind, we also value the lifetime gain a second way, assuming the 14%

London/South East split.

age-28 premium persists over the working life. This gives a similar figure.

Third, we compute the return to expanding access. If all schools offered A-level economics, our estimates imply an additional 4,000 students per cohort would take A-level economics, of whom approximately 610 would go on to study economics at university. The composition of the intake would shift towards regions outside London and the South East, with smaller movements by sex and socioeconomic background.

We estimate that the MVPF of hiring an additional economics teacher is infinite. The fiscal returns from higher lifetime tax revenue exceed the salary and training costs of the teacher, so the policy pays for itself. It would still do so if the true return per complier were only a third of our central estimate. Because the schools that lack provision are concentrated outside London and the South East, expanding provision raises earnings and tax revenue and, at the same time, moves the intake of economics degrees towards regions that are currently under-represented. The shares of women and of FSM students among economics undergraduates would barely change. The case for expansion on fairness grounds is therefore that these groups currently have less access to the subject, not that we can show they gain more from taking it.

Causal estimates from admission lotteries and scoring cutoffs have established that field of study is a major determinant of earnings, comparable in magnitude to the college wage premium itself. Much of this variation reflects the effect of the field, not selection (Kirkebøen et al., 2016; Bleemer and Mehta, 2022; Hastings et al., 2013). For the UK, Britton et al. (2020) document large variation in lifetime returns across degree subjects using the same administrative data we use, and Britton et al. (2026) provide the updated estimates our calculations draw on. The study closest to ours is Dahl et al. (2023), who estimate long-run earnings effects of upper-secondary fields of study in Sweden. They study which field a student is admitted to. We study whether the field is available at all. All of these studies compare students who narrowly gained a place with students who narrowly missed

one. They measure the return for students at an entry cutoff where places are scarce.

We estimate the return at the extensive margin, meaning what happens when the option to study a high-return subject does not exist. This distinction matters for policy. Expanding access at a competitive cutoff displaces other applicants; expanding access at the extensive margin does not. The complier population is also different: rather than students at a scoring threshold, our compliers are students who would study economics if it were available but currently cannot.

A growing body of work argues that barriers preventing talented individuals from entering high-return occupations impose large aggregate costs. Hsieh et al. (2019) attribute 20–40% of US output growth between 1960 and 2010 to improved allocation of talent across occupations, using a calibrated Roy model applied to observed changes in the occupational distribution. Their framework implies that removing specific barriers would generate measurable gains, but does not isolate any individual barrier or estimate the cost of removing it. We provide the type of micro evidence their framework calls for: a concrete, causally identified barrier (the absence of a subject teacher), an estimate of the earnings consequences of removing it, and a cost-benefit calculation showing that removing the barrier returns more to the government in tax than it costs. We compute the marginal value of public funds (Hendren and Sprung-Keyser, 2020) for expanding teacher supply, connecting the misallocation literature to the welfare-evaluation framework that has become standard in applied public economics.

Research on under-representation in economics and other high-return fields has focused predominantly on demand-side interventions: exposing students to role models (Porter and Serra, 2020), redesigning introductory courses (Bayer et al., 2020), providing information about career prospects (Avilova and Goldin, 2018), and mentoring (Blau et al., 2010). These interventions implicitly assume that students have the option to study economics and the question is whether they choose to. We show that for a substantial fraction of students the constraint

is more basic: the subject is not available at their school. Splitting each take-up gap into an access part and a preference part, we find that availability alone accounts for 36% of the regional gap in A-level economics take-up, and between 6% and 82% of the gaps across demographic groups. Because schools serving disadvantaged students are least likely to offer economics, supply-side constraints compound existing inequalities. The causal estimates show that relaxing these constraints shifts students into high-return educational and career paths, and the cost-benefit calculation shows that the fiscal return exceeds the cost of additional teachers. Supply-side interventions are therefore a necessary complement to the demand-side measures that have dominated the policy discussion.

The paper proceeds as follows. Section 4.2 describes the data and institutional context. Section 4.3 documents the unequal distribution of access to A-level economics and decomposes take-up gaps into preference-driven and constraint-driven components. Section 4.4 presents the empirical framework and estimates the causal effect of A-level economics on degree choice, earnings, and lifetime income. Section 4.5 computes the MVPF of expanding teacher supply and presents a counterfactual exercise on universal access. Section 4.6 concludes.

## 4.2 Data and Institutional Context

Students in England follow a structured educational pathway with two critical transition points relevant to this paper. At age 14–16, students study for the General Certificate of Secondary Education (GCSE), a set of nationally graded examinations covering a broad curriculum. GCSE results determine eligibility for further study and serve as the main measure of prior attainment in our analysis. At age 16, students who continue to upper-secondary education select three or four subjects to study at Advanced Level (A-level) over two years. A-level choices largely determine which degree subjects a student can pursue at university. A student who does not take A-level economics, for instance, will find it difficult to gain admission to an economics degree at most universities. Additionally, since

economics is almost never taught at any earlier stage at school, students who do not study it at A-level are unlikely to be aware of it as a subject (Smith et al., 2025). Whether a school offers A-level economics depends primarily on whether it employs a suitably qualified teacher: a supply constraint we exploit for identification.

Most students who continue to A-levels stay in the same secondary school from age 11 to 18, or make a single move at 16 to their school's linked school; together these two groups are 71% of A-level students (Appendix Table 4A.6). The rest transfer at the end of GCSE (age 16) to an unrelated institution, either by choice or because their school does not offer post-16 education. We discuss the implications of school-changing for identification in Section 4.4.

We use the Longitudinal Education Outcomes (LEO) dataset, which links three administrative sources covering the universe of school students in England. The analysis covers state-educated students only, because much of the individual-level information we use (sex, ethnicity and region among it) is missing for privately educated students. The National Pupil Database (NPD) provides individual-level demographic characteristics (sex, ethnicity, free school meal eligibility), school identifiers, and attainment at each Key Stage, including GCSE and A-level results. The Higher Education Statistics Agency (HESA) records provide information on university attendance, institution, and degree subject. His Majesty's Revenue and Customs (HMRC) tax records provide annual earnings from employment and self-employment. The linked dataset allows us to follow each student from their GCSE results at age 16 through A-level and university choices and into the labour market.

We observe school-level provision of A-level economics for the wider window of GCSE cohorts 2002–2013, which the descriptive figures draw on. We restrict the estimation sample to the 2002–2008 GCSE cohorts, for whom we observe earnings at age 28 (the latest year in our earnings data is 2021). Identification in Section 4.4 comes from within-school changes in availability. Schools that offer

economics in every cohort, or in none, contribute no such changes once school fixed effects are included. The estimation samples are therefore restricted to students in the schools that switch between offering and not offering economics over the period. We refer to these three groups as always-offer, never-offer and switcher schools, and classify offer behaviour over the full sample window.

The analysis population is students who continue to Key Stage 5 (A-level study). Students who leave education at 16 take no A-levels in any subject, so the access and take-up rates in Section 4.3 and the estimation samples below are all defined on students who continue. One concern with defining the population this way is that the decision to stay on past 16 might itself depend on whether the school offers economics. If it did, a change in provision would change who enters the analysis population. Section 4.4 checks this in two ways. It tests whether changes in availability are related to students' prior attainment and background. It also tracks the size and composition of the GCSE intake in the years before and after a school starts or stops offering economics. Neither check finds a change in who is in the school. Within the population, whether a student appears in our estimation sample does not depend on whether their school offered economics, as reported below.

The university-outcome regressions use 417,849 switcher-school students and the earnings regressions 368,260.[5] We focus on age-28 earnings as the primary outcome because it is the latest age observed for all cohorts, and combine these estimates with published estimates of lifetime returns by degree subject (Britton et al., 2026) to assess lifetime effects.

We measure A-level economics availability using a school-cohort indicator equal to one if at least two students in the school-year group sat the A-level

[5]Of the 1,449,714 student records in the 2002–2008 cohorts, 1,409,352 have university outcomes observed and 1,231,580 have positive earnings at age 28; the estimation samples are the switcher-school students among these with all controls observed. Key Stage 2 attainment, which the controls include, is missing for around six per cent of students. The levels specification, which retains students with zero earnings, uses 406,352. Regressing an indicator for inclusion in the earnings sample on availability, with school and cohort fixed effects, gives a coefficient of $-0.0004$ (standard error 0.0015), so inclusion in the estimation sample is unrelated to the instrument.

economics examination.[6] This is the instrument in our IV design.

This way of measuring availability makes part of the take-up result automatic. A school-cohort is counted as not offering economics when at most one student took it, so take-up is bound to be higher where economics is counted as offered. That would be true even if offering the subject changed nobody's choices. Earnings and degree choices play no part in the counting, so the way we count offers cannot create a link between availability and either of those outcomes. If earnings or degree choices change when a school starts or stops offering economics, something real has changed. For this reason we place more weight on the earnings and degree estimates than on the take-up estimate. The footnote above and Appendix Table 4A.13 set out the consequences of measuring availability this way.

Across our sample, the share of schools offering A-level economics by this measure rises gradually from 38% to 42% over the estimation window, and reaches 46% by 2011 (Appendix Figure 4A.2). Around one-third of schools switch between offering and not offering economics (or vice versa) at some point in our sample period, generating the within-school variation that drives identification (see also Figure 4A.1).

Table 4.1 presents summary statistics for the 2011 GCSE cohort, the cohort used for the descriptive access analysis, with the final panel describing the 2002–2008 estimation cohorts. Each is shown overall and separately for three categories

[6]We do not observe subject offerings directly, so we infer availability from take-up. Availability is measured from the complete exam records, before any sample restriction. We require at least two students rather than one because a single student taking economics may be doing so at a neighbouring school or by studying at home rather than at their own. Such actions are unusual, and where they occur would represent a substantially higher cost of access. Appendix Table 4A.13 shows the main results using thresholds of one to five students. The instrument is strong and the IV estimate positive at every threshold, and the estimates rise with the threshold, from 0.135 at two students to 0.256 at five. This range reflects genuine uncertainty about the size of the estimate across codings; the baseline threshold of two gives the smallest point estimate, so the headline numbers are the most conservative of the set. Because availability is inferred from take-up, part of the link between availability and take-up is automatic. A cohort counted as not offering economics has at most one taker, because that is how we count it. However, the effects of availability itself on earnings and degrees (the reduced-form effects) are not mechanically linked to the coding. If a switch simply meant a school drifting across the two-student line, take-up would rise by one or two students. Instead it rises by around six students in a typical year group, roughly a dozen across the sixth form's two year groups. A jump of that size is what a new class with a teacher looks like, not a small change in demand.

Table 4.1: Summary statistics by school economics offer behaviour

| | All | Switcher | Always offer | Never offer |
|---|---|---|---|---|
| **Panel A: Student characteristics** | | | | |
| N students | 221,118 | 60,762 | 119,665 | 40,691 |
| % female | 54.7 | 57.2 | 52.4 | 57.5 |
| % White British | 71.8 | 73.2 | 68.6 | 79.3 |
| % Indian | 3.6 | 2.7 | 4.6 | 2.3 |
| % Pakistani/Bangladeshi | 4.3 | 4.2 | 4.2 | 4.8 |
| % Black African/Caribbean | 4.1 | 4.4 | 4.5 | 2.6 |
| % Chinese | 0.8 | 0.7 | 0.9 | 0.5 |
| % Other ethnicity | 10.6 | 11.0 | 11.3 | 7.8 |
| % FSM eligible | 6.4 | 7.1 | 5.6 | 7.5 |
| Mean std. GCSE score | 0.63 | 0.60 | 0.64 | 0.63 |
| **Panel B: Subject choices and outcomes** | | | | |
| % taking A-level economics | 6.8 | 4.2 | 10.4 | 0.1 |
| % attending university | 81.5 | 80.5 | 83.5 | 77.4 |
| % studying economics at UG | 2.6 | 2.0 | 3.5 | 0.9 |
| **Panel C: Estimation cohorts (2002–2008)** | | | | |
| Mean log earnings at 28 | 10.24 | 10.22 | 10.28 | 10.17 |
| N with university outcomes | 1,409,352 | 444,952 | 651,374 | 313,026 |

*Notes:* Panels A and B report means for the 2011 GCSE cohort, the cohort used for the descriptive access analysis in Section 4.3. Panel C describes the 2002–2008 GCSE cohorts used for estimation, for whom earnings at age 28 are observed; it counts students with university outcomes observed, and the estimation samples are subsets of these, described in Section 4.2. School offer behaviour is defined over the full sample period: "Always offer" schools offer A-level economics in every cohort; "Never offer" schools never offer it; "Switcher" schools offer it in some cohorts but not others. A school is defined as offering A-level economics if at least two students in the school-year group sit the exam. Standardised GCSE score has mean zero and unit variance in the full population. Ethnicity shares do not sum to 100% because ethnicity is missing for some students.

of schools defined by their offering pattern across cohorts: always-offer schools, which provide A-level economics in every year they appear in our data; never-offer schools, which never do; and switcher schools, which change status at least once. The first column covers all schools.

The remaining columns reveal systematic differences across school types that motivate our identification strategy. Always-offer schools have higher average GCSE attainment, lower shares of students eligible for free school meals, and higher shares of students from ethnic groups that disproportionately study economics at

university. Never-offer schools are the mirror image. Switcher schools, the source of our identifying variation, lie between the two on most observable dimensions; on attainment and single-sex composition they resemble never-offer schools. These differences underscore the importance of school fixed effects, which absorb all time-invariant differences between schools and ensure that identification comes solely from within-school changes in availability. We discuss the characteristics of switcher schools further in Section 4.4, where we present the empirical framework.

## 4.3 Who gets to study economics?

Whether a student can take A-level economics depends largely on where they go to school. Students in London and the South East are 15.3 percentage points more likely to attend a school that offers economics than students in the rest of England. The gap is too large to reflect differences in student aptitude or interest; it instead reflects which schools employ an economics teacher. Across England in 2011, only 46% of state schools offered A-level economics (Appendix Figure 4A.2). Students in the remaining schools cannot choose a subject with among the highest returns of any A-level, both directly and through its role as the principal pathway to an economics degree.

Table 4.2 documents access rates, take-up conditional on access, and unconditional take-up for the 2011 GCSE cohort, broken down by sex, socioeconomic status, region, and ethnicity. The regional gap is the starkest dimension, but it is not the only one. Students eligible for free school meals are 3.5 percentage points less likely to attend a school offering economics than their non-FSM peers. Women have lower access than men (66.0% versus 70.1%), but are also less likely to take the subject conditional on access (5.6% versus 15.0%), indicating the role of preferences alongside constraints. Appendix Table 4A.4 shows the same pattern by school type: where economics is offered, girls in single-sex schools take it at around twice the rate of girls in mixed schools. Indian and Chinese students have both higher access and conditional take-up than White students. Pakistani,

Bangladeshi and Black students also have higher take-up conditional on access, though by smaller margins.

Table 4.2: Access to and take-up of A-level economics by demographic group, 2011 cohort

| | Access rate | Take-up \| access | Take-up (uncondit.) | Gap from access | Share of gap from access |
|---|---|---|---|---|---|
| **Panel A: Sex** | | | | | |
| Male | 70.1 | 15.0 | 10.5 | | |
| Female | 66.0 | 5.6 | 3.7 | | |
| Gap (M − F) | 4.1 | 9.4 | 6.8 | 0.4 | 6.1 |
| **Panel B: Socioeconomic status** | | | | | |
| Non-FSM | 68.4 | 9.9 | 6.8 | | |
| FSM | 64.9 | 9.8 | 6.4 | | |
| Gap | 3.5 | 0.1 | 0.4 | 0.3 | 82.0 |
| **Panel C: Region** | | | | | |
| London/South East | 77.8 | 12.3 | 9.6 | | |
| Rest of England | 62.5 | 8.3 | 5.2 | | |
| Gap | 15.3 | 3.9 | 4.4 | 1.6 | 36.0 |
| **Panel D: Ethnicity** | | | | | |
| White British | 65.3 | 8.2 | 5.4 | | |
| Indian | 78.8 | 20.2 | 16.0 | | |
| Pakistani/Bangladeshi | 67.8 | 12.8 | 8.7 | | |
| Black African/Caribbean | 77.9 | 14.0 | 11.0 | | |
| Chinese | 75.7 | 21.3 | 16.4 | | |
| Gap: Indian − White | | | 10.6 | 1.9 | 18.1 |
| Gap: Pak./Bang. − White | | | 3.4 | 0.3 | 7.9 |
| Gap: Black Afr./Car. − White | | | 5.7 | 1.4 | 24.7 |
| Gap: Chinese − White | | | 11.1 | 1.5 | 13.9 |

*Notes:* Access rate is the share of students attending a school that offers A-level economics (defined as $\geq 2$ students sitting the exam in the school-cohort). Take-up | access is the share taking A-level economics conditional on the school offering it. The take-up gap between groups $A$ and $B$ is decomposed as $p_A - p_B = \bar{c}(a_A - a_B) + \bar{a}(c_A - c_B)$, where $a$ denotes the access rate, $c$ denotes conditional take-up, and bars denote simple averages across the two groups. "Gap from access" reports the first term: the component of the take-up gap attributable to differential access, holding conditional take-up at the cross-group average. "Share of gap from access" divides this by the unconditional take-up gap. Gap rows and shares are computed from unrounded values, so arithmetic on the rounded cells may differ slightly. For ethnicity, gap directions are chosen so that a positive unconditional gap corresponds to the group listed first having higher take-up. Sample: 2011 GCSE cohort, $N = 221{,}118$.

Table 4.2 also helps distinguish the mechanism behind these gaps. For FSM students, take-up conditional on access is almost identical to that of their non-FSM peers, so their lower unconditional take-up reflects where they go to school rather

than weaker demand for economics. For ethnic groups, by contrast, large differences in conditional take-up remain. Indian and Chinese students are substantially more likely to choose economics than White British students even where both have access, and Pakistani, Bangladeshi and Black African and Caribbean students also choose it at higher rates. This indicates that preferences and demand-side factors shape economics take-up alongside supply constraints. Appendix Table 4A.1 reports regression counterparts of these patterns: the effect of availability on take-up estimated separately for each subgroup.

How much of the overall gap in take-up is explained by supply constraints versus preferences? We decompose the unconditional take-up gap between any two groups into an access component and a preference component using the descriptive statistics in Table 4.2.[7]

The access component measures the mechanical contribution of differential availability: how much smaller the take-up gap would be if both groups had equal access. The preference component captures the residual: differences in take-up conditional on the subject being available, reflecting tastes, information, peer effects, and other demand-side factors.

The final two columns of Table 4.2 report these components for each pairwise comparison. The regional gap is the clearest case. Students in London and the South East have substantially higher access than students elsewhere, but conditional take-up rates differ more moderately (12.3% versus 8.3%), so that 36.0% of the regional take-up gap is attributable to access alone.

The other panels of Table 4.2 show three distinct patterns. For socioeconomic status, take-up where economics is offered is almost the same for FSM and non-FSM students (9.8% against 9.9%). The take-up gap between the two groups is

[7]For groups $A$ and $B$ with access rates $a_A, a_B$ and conditional take-up rates $c_A, c_B$, the unconditional take-up gap decomposes as $p_A - p_B = \bar{c}(a_A - a_B) + \bar{a}(c_A - c_B)$, where $\bar{c}$ and $\bar{a}$ are simple averages across the two groups. The first term is the access component: the share of the gap attributable to differential availability, holding conditional take-up at the cross-group average. The second is the preference component: the share attributable to differential take-up conditional on access, holding access constant. This is an accounting identity, not a causal decomposition; the causal effect of access on take-up is estimated in Section 4.4.

therefore almost entirely a matter of access: the access component accounts for 82% of it, though the gap itself is small, at 0.4 percentage points. For sex, the opposite holds. Women have somewhat lower access than men, but the far larger difference is in take-up where the subject is available (5.6% against 15.0%), so the access component explains only 6.1% of the gap. Supply-side expansion is necessary but not sufficient for closing the gender gap; demand-side interventions matter too (Porter and Serra, 2020; Smith et al., 2025).

For ethnicity, the gaps run the other way: all four minority groups in the table have higher take-up than White British students. For Indian and Chinese students the preference component dominates. These groups have strong demand that is largely met by supply, with access accounting for 18.1% and 13.9% of their take-up gaps. For Black African and Caribbean students a quarter of the gap (24.7%) is attributable to access, and for Pakistani and Bangladeshi students 7.9%. Across all comparisons, the access component explains between 6% and 82% of the take-up gap. These shares come from the rates in Table 4.2 and involve no regression. They are best read as a lower bound on the role of supply constraints, because they count only the mechanical effect of unequal access. If access also changes what students want to take, through peers who take the subject or through better information about it, the full effect of access on take-up is larger. The causal estimate follows in Section 4.4.

To put a scale on the shortfall, we ask what would change if every school offered economics. Applying the estimated take-up effect of availability (Section 4.4) to the students in schools without access in 2011 implies approximately 4,000 additional A-level economics students per cohort: an increase of 27% relative to current numbers. We apply the estimated take-up effect rather than the take-up rates of the schools that already offer economics (Table 4.2). Schools that offer economics tend to be schools whose students want it, which is part of why they offer it, so their take-up rates are higher than a school without economics would see if it started to offer the subject. The demographic composition of A-level

economists would shift correspondingly, with the share from FSM backgrounds rising from 6.4% to 6.6%, and the share from outside London and the South East from 52% to 58%.[8]

The demographic skew in economics degree take-up is even more pronounced than the skew in A-level take-up (Appendix Table 4A.3), which is what we would expect if a shortfall at A-level carries through to the degree, since the A-level is the main route into it (Bayer and Rouse, 2016; Buckles, 2019; Lundberg and Stearns, 2019).[9] The 4,000 additional students counted above are students who would choose economics if the option existed at their school. Whether that choice would generate earnings returns, and at what cost the constraint could be removed, is what we estimate in Sections 4.4 and 4.5.

## 4.4 Effect on degree choice and lifetime earnings

### 4.4.1 Empirical framework

We estimate the causal effect of taking A-level economics on later earnings using two-stage least squares. The first stage relates A-level economics take-up to school-cohort availability:

$$\text{TakeEcon}_{ist} = \pi_0 + \pi_1 \,\text{AvailEcon}_{st} + \lambda_s + \delta_t + X'_{ist}\Gamma + u_{ist}, \tag{4.1}$$

where $\text{TakeEcon}_{ist}$ indicates whether student $i$ in school $s$ and cohort $t$ takes A-level economics, $\text{AvailEcon}_{st}$ is an indicator for whether the school offers A-level economics to that cohort, $\lambda_s$ are school fixed effects, $\delta_t$ are cohort fixed effects, and $X_{ist}$ includes individual-level controls for sex, ethnicity, free school meal eligibility,

[8]The additional students are the take-up effect of availability applied to the 2011 cohort's no-access population; the compositional shifts use the group-specific counts underlying Table 4.2. Table 4.5 Panel B reports the degree-level analogues.

[9]Appendix Table 4A.2 replicates the decomposition for the 2003 and 2007 cohorts. The broad patterns are stable: the access-explained share of the regional gap is stable at 36–40%, while the FSM take-up gap is too small for its access share to be stable. It is 40% in 2003, and in 2007 the gap is essentially zero, so no share can be computed.

Key Stage 2 attainment, and standardised GCSE attainment. The second stage estimates the effect of A-level economics on earnings:

$$Y_{ist} = \beta_0 + \beta_1 \widehat{\text{TakeEcon}}_{ist} + \lambda_s + \delta_t + X'_{ist}\Gamma + \varepsilon_{ist}, \tag{4.2}$$

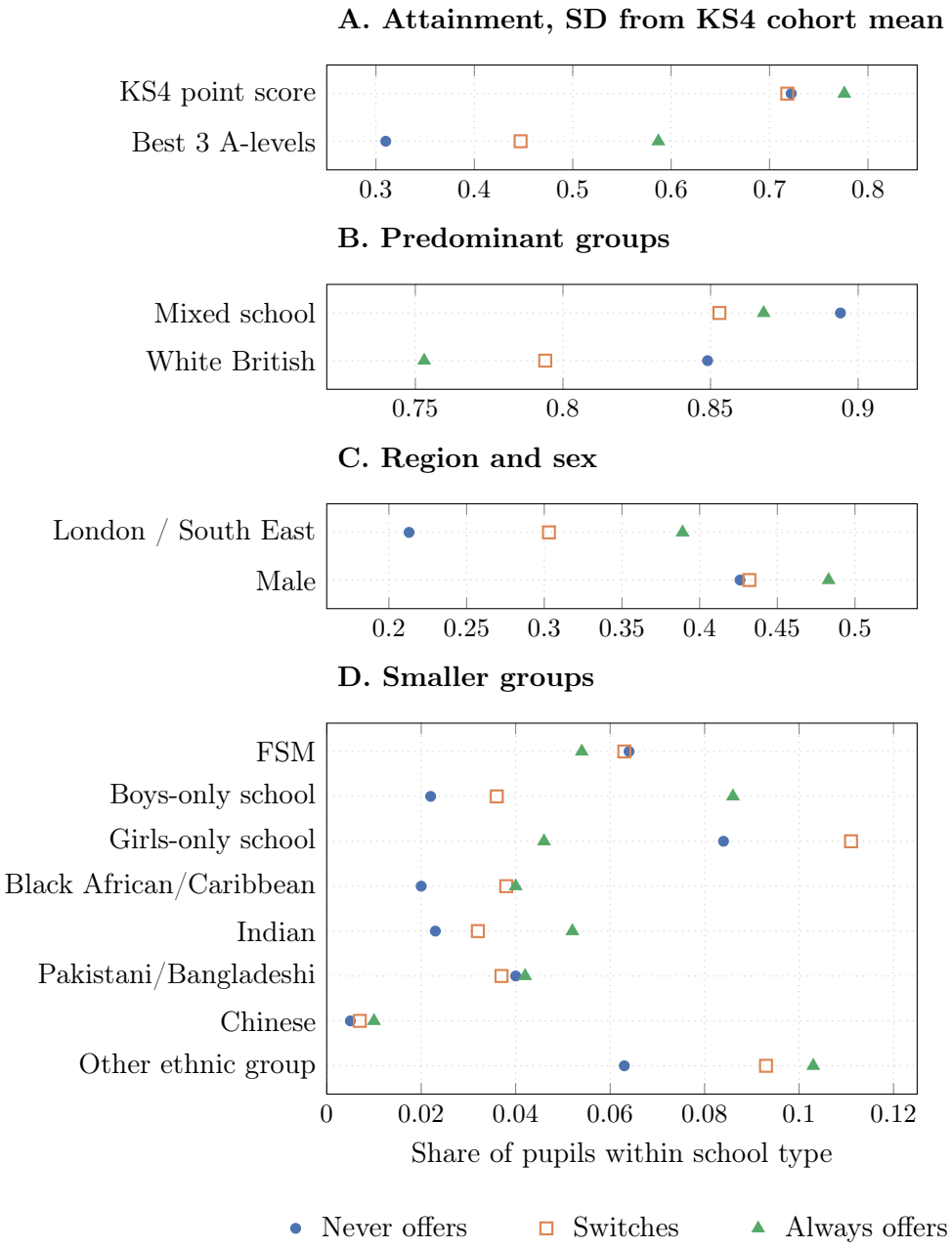
where $Y_{ist}$ is log real earnings at age 28 and $\widehat{\text{TakeEcon}}_{ist}$ is the predicted value from the first stage. The coefficient $\beta_1$ is the local average treatment effect (LATE) of taking A-level economics for students whose take-up is shifted by within-school changes in availability. Standard errors are clustered at the school level throughout.

The IV estimate identifies the causal effect for compliers: students who take A-level economics when it is offered but would not otherwise. This is directly the population a teacher-hiring policy would affect, so within any given school the LATE is the policy-relevant parameter. The harder question is whether the estimate carries over from switcher schools to schools that have never offered economics. Switcher schools sit between always-offer and never-offer schools on most observable dimensions, and on attainment and single-sex composition they resemble never-offer schools (Figure 4.1; Appendix Table 4A.5 reports the underlying school-level averages); school fixed effects absorb these differences for identification. But never-offer schools serve a more disadvantaged population, so projecting returns to universal access requires adjusting for composition. We do this in Section 4.5 using the subgroup-specific estimates from Section 4.4.5, following Angrist and Fernández-Val (2013).

Most students' A-level institution is determined by their choice of secondary school at age 11, whether they remain in the same school or transfer to a linked senior school. In our sample, 55% of students who continue to A-level remain in the same school, and a further one in six moves to its linked school.[10] At least 7% of students have a change in access to economics as a result of moving (see

[10]In the English school system, some areas have a feeder school system, where children move together from a 'feeder school' to its linked school at some point between 11 and 18.

Figure 4.1: Pupil characteristics by school A-level economics provision




*Notes: KS4 cohorts 2002–2013. School types are defined by A-level economics provision across the cohorts in which a school appears in the data: never-offer, always-offer, and switcher schools. Panels B–D report the share of pupils within each school type; note that the horizontal scale differs across panels. Complementary categories (rest of England, not FSM, female) are omitted, as each is one minus the category shown. Panel A reports mean point scores in standard deviations from the KS4 cohort mean, standardised within GCSE cohort year; best 3 A-level points are measured among A-level entrants only. They are determined after the student has been exposed to the school's economics provision, so unlike KS4 scores they are not a pre-treatment characteristic.*

Appendix Table 4A.6).[11]

Availability and the school fixed effects are defined at the school where the student takes A-levels, so students who move at 16 could in principle sort towards schools that offer economics. The take-up evidence indicates that they do not. Among those who move to a school that offers economics from one that does not, 6.3% take A-level economics, a somewhat lower rate than the 7.9% among non-movers at offer schools (Appendix Table 4A.6 Panel B). If moves were driven by demand for economics, movers who gain access would take the subject at a higher rate than students who always had it, not a lower one.

Our identification strategy requires that within-school changes in whether A-level economics is *offered* affect earnings only through the effect on whether students take the subject. It is difficult to construct a channel through which a school *offering* economics could raise the earnings of students who do not take it. It is nonetheless important to verify there are no other correlated school-level changes, such as new leadership, shifts in resources, or changes in cohort composition, that independently affect student outcomes.

We assess this through several checks, reported in the appendix to this chapter. First, we test whether within-school changes in economics availability predict pre-treatment outcomes, including GCSE scores and Key Stage 2 attainment. We find no significant relationship except for standardised GCSE maths attainment (0.025 standard deviations) and the share of White British students (a fall of 0.8 percentage points) (Appendix Table 4A.7). Both imbalances are small, and both variables are controlled directly in the estimating equations, attainment through the standardised GCSE score and ethnicity through the ethnic-group indicators. The controls absorb any imbalance in what they measure, so only the part they miss, such as maths skill not reflected in the overall GCSE score, could still affect

[11]This is a lower bound. A school's economics offer is measured from its own A-level records, so for students leaving a school with no sixth form there is no offer to compare against. These students had no A-level provision at their original school, so those who move into a school offering economics also gain access, but they cannot be counted by this method. The figure is computed as the 46% of students who move, times the 28% of movers whose origin school's offer can be measured, times the 53% of measurable movers whose access changes.

the estimates. Even without the controls, the imbalance could account for at most part of the result. Suppose maths attainment raised earnings by 0.15 log points per standard deviation, a generous figure. The maths imbalance of 0.025 standard deviations would then move earnings by about 0.004, around half of the estimated 0.008 effect of availability, and that is before the controls absorb it.

Second, as a placebo test, we re-run the earnings regression on the students who do not take A-level economics. Availability has no effect on their earnings (Appendix Table 4A.8), which is what we would expect if offering economics affects earnings only through the students who take it. The test has one limit. When a school starts offering economics, the compliers stop being non-takers. The non-takers at schools that offer economics are therefore a slightly different group of students from the non-takers at schools that do not, so the comparison is not exactly like for like. The test therefore supports the assumption but does not prove it. One concrete channel is that economics and business studies are often taught by the same teacher, so a change in economics provision could change business provision or teaching at the same time, affecting the earnings of students who never take economics. The null placebo estimate and the small, insignificant effect of availability on business and finance degrees (Appendix Table 4A.10) give no indication of such a channel. If it operated, part of the earnings effect of availability would come from non-takers. The IV estimate attributes the whole effect to takers, so it would be biased upwards.

Third, we estimate specifications that include school-specific linear time trends and region-by-year fixed effects; the estimates are stable (Appendix Table 4A.9).

Fourth, we estimate an event study around the arrival and departure of economics teachers, examining the number of students taking economics within the school in the years before and after the change. We find no pre-trend in take-up prior to the staffing shock (Appendix Figure 4A.3), supporting the interpretation that the variation in availability is supply-driven rather than reflecting shifts in student demand. We discuss two features of the event studies alongside the figures

in the appendix.

Panel A of Table 4.3 reports the first-stage estimates. Within-school availability of A-level economics increases the probability of taking the subject by 5.7 percentage points. The first-stage $F$-statistic is 843, so weak instruments are not a concern.[12] The saturated specification of column 4, which interacts the instrument with cells defined by sex, ethnicity, FSM status, GCSE attainment tercile, and region, following Blandhol et al. (2026), gives a nearly identical first stage. Section 4.3 showed, as a matter of accounting, that differences in access could explain between 6% and 82% of the take-up gaps between demographic groups. That calculation assumed that access changes what students take. The first stage shows this directly, since take-up moves within a school when it starts or stops offering the subject.

### 4.4.2 Earnings at age 28

Panel B of Table 4.3 reports the main results. The OLS estimate of the effect of taking A-level economics on log earnings at age 28 is 0.174, suggesting that students who take economics earn approximately 19% more than observationally similar students who do not. The IV estimate is smaller: taking A-level economics raises earnings at age 28 by around 14% (0.135, standard error 0.069). The IV estimate is significant at the 10% level. It is evidence of a substantial positive return rather than a firm point estimate.[13]

The OLS estimate compares students who take economics with students who do not. It is too high if takers would have earned more anyway for reasons the controls do not capture. The IV estimate uses only changes in what a school offered, which the students did not choose, so it does not have this problem. The

[12]The Lee et al. (2022) tF procedure replaces the standard critical value of 1.96 with an F-dependent adjusted value. At the conventional threshold of $F = 10$, the correct critical value is 3.43, not 1.96. Our $F$-statistics exceed 104.7, above which the conventional value of 1.96 remains valid, so no weak-instrument adjustment is required for valid 5% inference in just-identified IV.

[13]Re-estimating the IV with earnings in pounds rather than logs gives £3,700 (standard error £3,300). The point estimate is consistent with the log specification, but earnings in levels are noisy, so the estimate is imprecise. The levels sample ($N = 406{,}352$) also retains students with zero earnings, whom the log specification excludes.

Table 4.3: Effect of A-level economics on take-up, earnings, and lifetime income

| | OLS | Reduced form | IV (linear) | IV (saturated) |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A: Takes A-level economics (First stage)** | | | | |
| A-level economics available | | 0.057*** | 0.057*** | 0.058*** |
| | | (0.002) | (0.002) | (0.002) |
| First-stage $F$ | | | 843 | 891 |
| **Panel B: Log earnings at age 28** | | | | |
| Takes A-level economics | 0.174*** | | 0.135* | 0.178** |
| | (0.008) | | (0.069) | (0.072) |
| A-level economics available | | 0.008* | | |
| | | (0.004) | | |
| **Panel C: Implied lifetime returns (gain vs. no university, PV at age 18, £)** | | | | |
| | | | Implied value | |
| Gross lifetime return | | | 190,000 | |
| Net individual return | | | 90,000 | |
| Exchequer return | | | 100,000 | |
| School FE | Yes | Yes | Yes | Yes |
| Cohort FE | Yes | Yes | Yes | Yes |
| Individual controls | Yes | Yes | Yes | Saturated |
| Observations | 368,260 | 368,260 | 368,260 | 367,935 |

*Notes:* Panel A reports the first-stage relationship between A-level economics availability and take-up. Panel B reports the effect of taking A-level economics on log real earnings at age 28 (OLS in column 1; instrumented by availability in columns 3–4); column 2 reports the reduced-form effect of availability on earnings. Panel C reports implied lifetime returns: the effect of taking A-level economics on lifetime returns, calculated from two estimated effects, the effect on attending university at all and the effect on choosing an economics degree, each combined with published lifetime returns by degree subject (Britton et al., 2026). They are calculations rather than regression estimates, so no standard errors are shown. Each outcome is a gain relative to not attending university, in present value at age 18 and 2025/26 prices, discounted at HM Treasury Green Book rates. The net individual return deducts income tax, National Insurance, and Plan 5 student loan repayments; the exchequer return nets the additional tax and loan repayments against fee loans, maintenance loans and teaching grants. Column 4 uses a saturated specification that interacts the instrument with 144 coarse cells defined by sex, ethnicity, FSM status, GCSE attainment tercile, and region (London/South East, rest of England, or missing), following Blandhol et al. (2026); the cells include categories for missing values and replace the linear controls. Standard errors clustered at the school level in parentheses. First-stage $F$-statistics are cluster-robust: with a single instrument, the $F$ equals the square of the instrument's cluster-robust $t$-statistic (the Kleibergen–Paap Wald statistic). Sample: students in switcher schools, 2002–2008 GCSE cohorts, with non-zero earnings at age 28. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

two are close, at 0.135 against 0.174, so any such bias in OLS is small and, if anything, upward. This is consistent with the evidence in Bleemer and Mehta (2022) that selection bias in the return to economics is small.

The saturated estimate in column 4 is somewhat larger (0.178), but its confidence interval comfortably includes the linear estimate. Blandhol et al. (2026) show that when the controls enter linearly, the two-stage least squares (TSLS) estimate is a weighted average of the effects for different groups of students in which some weights can be negative, so the estimate need not lie between the effects it averages. The agreement between the linear and saturated specifications is reassuring on this point. Their result strictly calls for the controls to enter fully flexibly, with an indicator for every combination of student characteristics and school. That is not possible here, because interacting the cells with an indicator for every school would leave too few students in each group. Column 4 therefore makes the student characteristics fully flexible but leaves the school fixed effects unchanged, so it is a partial check on their concern rather than a complete answer.

The return of around 14% is well below the 46% that Bleemer and Mehta (2022) estimate at the GPA threshold for the economics major at UC Santa Cruz. The two populations differ. Their estimate is for students competing at an admissions threshold; ours is an average over the compliers, whose schools did not offer the subject at all. Column 2 reports the reduced form: the effect of availability itself on earnings, without dividing by take-up. It is approximately 0.8%, and it equals the take-up effect multiplied by the IV estimate. It is the average earnings gain per student in the cohort when a school begins to offer economics. This is the policy-relevant parameter for Section 4.5. It is also less affected than the IV estimate by the measurement problem described in Section 4.2. The IV estimate is the reduced form divided by the take-up effect, and Section 4.2 showed that part of the take-up effect is automatic. No part of the reduced form is automatic, because the way we measure availability cannot create a link with earnings.

### 4.4.3 Lifetime earnings

The age-28 estimates capture only the early-career return to A-level economics. To assess lifetime effects we calculate the lifetime returns implied by our estimates in combination with published estimates of the lifetime returns to undergraduate degrees (Britton et al., 2026). For each degree subject, the published estimates measure the gain relative to not attending university. They come in three versions. The gross return is the extra earnings before tax. The net return is what the graduate keeps after income tax, National Insurance and Plan 5 student loan repayments. The exchequer return is what the government receives in those deductions. All three are in 2025/26 prices and discounted to age 18 at the rate HM Treasury's Green Book sets for public appraisal.

Taking A-level economics changes two things. It raises the probability of going to university by 14.0 percentage points (Appendix Table 4A.10), and it raises the probability of studying economics at university by 15.2 percentage points (Table 4.4). The implied net individual return per complier is

$$0.140 \times \bar{R} \; + \; 0.152 \times (R_{\text{econ}} - R_{\text{mix}}) \; \approx \; \text{£}90{,}000, \tag{4.3}$$

where $\bar{R} = \text{£}99{,}000$ is the average net lifetime return to an undergraduate degree; $R_{\text{econ}}$ is the return to an economics degree, £560,000 when the published returns of £650,000 for men and £335,000 for women are weighted by the 72% male share of economics undergraduates in our sample (Appendix Table 4A.3); and $R_{\text{mix}}$ is the return to the subjects compliers would otherwise study, around £75,000 when the counterfactual bundle in Appendix Table 4A.11 Panel B (principally business, English literature, history and psychology) is weighted by the published returns to those subjects.[14]

The first term captures compliers drawn into university altogether; the second

[14]This maps the counterfactual A-level bundle into degree-subject returns, assuming compliers' counterfactual degrees resemble their counterfactual A-levels; the degree-level substitution estimates in Appendix Table 4A.10 point the same way but are too imprecise to construct the mix directly. A better-paid counterfactual mix would shrink the switching term.

captures compliers who switch into economics from another subject.[15]

Repeating the calculation with the exchequer versions of the published returns, meaning the extra tax, National Insurance and loan repayments the government receives less its spending on the degree, gives an exchequer return of roughly £100,000 per complier. The gross return is roughly £190,000. Each extra pound a graduate earns goes either to the graduate after tax or to the government, so the gross return is approximately the sum of the net individual and exchequer returns (Table 4.3 Panel C).

There is a second way to value the lifetime gain, which does not use the degree-level returns at all. If the 14% earnings gain we observe at age 28 lasted over the whole working life, the gross lifetime gain would be roughly £100,000 and the net gain around £65,000. The difference, roughly £35,000, is taken in income tax, National Insurance and loan repayments and accrues to the exchequer. That is an exchequer share of 35%, slightly below the 37% marginal deduction rate (basic-rate income tax, employee National Insurance, and Plan 5 repayments) under the 2025/26 schedules.

We treat the two calculations as bracketing the plausible range, with the degree-based figure at the top. That figure is likely to be generous for a further reason. It values every complier who switches into economics at the published return for the average economics graduate, and the students drawn in by new provision may earn less than that average.

The exchequer return per complier is the benefit side of the cost-benefit calculation in Section 4.5. Section 4.4.5 adjusts it for the composition of the schools where new teachers would be hired.

These figures have two limitations. The first concerns the published returns. They are averages for each subject, so they need not match the returns of the

[15]The two margins overlap: a complier drawn into university who studies economics receives both terms, roughly £584,000, against the £560,000 return to an economics degree. Because the overlap affects at most the 14.0 percentage point attendance margin, and the overstatement per overlapping student is $\bar{R} - R_{\text{mix}} \approx £24{,}000$, treating the margins as nested rather than additive would lower the total by under five per cent, so we retain the simpler additive form.

particular students our policy moves, and they are estimated by comparing graduates with similar observed characteristics rather than from a randomised design. The report's authors note that estimates of this kind are usually assumed to somewhat overstate the true causal returns (Britton et al., 2026).

The second is that the calculation counts only the gains that come through degree choice. Panel B of Table 4.4 shows that the age-28 return is found mostly among compliers who do not change their degree subject, so these figures understate the full lifetime effect. The understatement is potentially large. The age-28 effect for compliers who do not change degree subject is 0.128, and if that premium persisted over the lifecycle it would be worth roughly £95,000 in gross present value per student, valued in the same way as above. That is comparable in size to the degree-channel calculation itself.

The near-zero age-28 return for compliers who switch into economics does not contradict the lifetime value placed on switching, because the two look at different points of working life. The lifetime calculation values degree choices whose earnings differences arrive mostly later in working life. The Panel B split measures earnings at age 28, early in the careers of economics graduates, before those differences have opened up.

### 4.4.4 The role of degree choice

The earnings return to A-level economics could operate through several channels: a shift into economics degrees (which have among the highest lifetime returns of any subject), the development of quantitative and analytical skills, improved financial literacy and career information, or the signalling value of the qualification itself. We do not attempt a formal causal decomposition, but we can assess how much of the return is concentrated among students who go on to study economics at university.

Panel A of Table 4.4 reports the effect of A-level economics on the probability of studying economics at university. The reduced-form estimate shows that students

Table 4.4: The role of degree choice

| | Reduced form (1) | IV (linear) (2) | IV (saturated) (3) |
|---|---|---|---|
| **Panel A: Effect on studying economics at university** | | | |
| Takes A-level economics | | 0.152*** | 0.158*** |
| | | (0.010) | (0.010) |
| A-level economics available | 0.009*** | | |
| | (0.0006) | | |
| Observations | 417,849 | 417,849 | 419,204 |
| **Panel B: Effect on log earnings at age 28, by degree pathway** | | | |
| | All | Econ degree | No econ degree |
| | (1) | (2) | (3) |
| Takes A-level economics | 0.135* | −0.037 | 0.128 |
| | (0.069) | (0.056) | (0.090) |
| Observations | 368,260 | 5,813 | 362,447 |
| Share of compliers | | 15.2 | 84.8 |

*Notes:* Panel A reports the effect of A-level economics take-up on the probability of studying economics at university. Column 1 is the reduced-form effect of availability; columns 2–3 instrument take-up with availability using the linear and saturated specifications. Panel B reports the IV estimate of the effect of taking A-level economics on log real earnings at age 28, for the full sample (column 1, reproducing the linear IV estimate from Table 4.3) and separately for students who do and do not go on to study economics at university (columns 2–3). Panel B splits students by a choice they make after taking A-level economics, so each subgroup estimate may be biased. The two estimates show where the earnings gain is concentrated. They should not be read as separate causal effects. The saturated specification replaces the linear controls with the cell indicators, and the cells include categories for missing values, so its variable requirements differ from the linear specification's and its estimation sample is slightly different. Among compliers, 15.2% shift into an economics degree (Panel A) and 84.8% do not. The subsample estimates need not aggregate exactly to the overall estimate, because each is estimated on a different sample; for the same reason the IV estimate in Panel A differs slightly from the reduced form divided by the first stage. All specifications include school and cohort fixed effects and individual controls. Standard errors clustered at the school level in parentheses. Sample: students in switcher schools, 2002–2008 GCSE cohorts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

in school-cohorts where economics is offered are 0.9 percentage points more likely to study economics at university. Dividing the effect on degree take-up by the take-up effect gives the effect for the students who take A-level economics because it is offered. For these compliers, taking the A-level raises the probability of an economics degree by 15.2 percentage points. A-level economics is the main route into the degree (Advani et al., 2020).[16]

[16]Effects on other educational outcomes, including university attendance, attendance at a

Panel B splits the earnings estimate by whether the student goes on to study economics at university. Among compliers who shift into economics degrees, the estimated effect on log earnings at age 28 is −0.037 and statistically indistinguishable from zero. Among compliers who do not change their degree subject, the effect is 0.128, a return of around 14%.[17]

Panel B suggests that the earnings return is not concentrated in the shift into economics degrees. For students who go on to study economics at university, the estimated return to A-level economics is close to zero; for the far larger group who do not, it is 0.128, close to the overall IV estimate of 0.135. A substantial return therefore exists among compliers who do not change their degree subject, suggesting that A-level economics generates earnings gains through channels beyond the degree pathway alone, such as quantitative skills, career information, or signalling. The estimates in this split are imprecise. The split also divides students by a choice (degree subject) that taking A-level economics itself affects, so the two groups are not fixed and the estimates are not clean causal effects for each group. For both reasons, the split indicates where the gains appear; it is not a formal division of the overall effect into parts. Goodman (2019) and Joensen and Nielsen (2009) treat later outcomes of this kind in the same way, as evidence that helps interpret the main estimate.

### 4.4.5 Who benefits most?

Section 4.3 showed that access to A-level economics is most constrained for students from disadvantaged backgrounds. Students from schools outside London and the South East, those eligible for free school meals, and women all have lower earnings on average. If the returns to A-level economics are also largest for these groups, then expanding access would raise total earnings and narrow gaps between

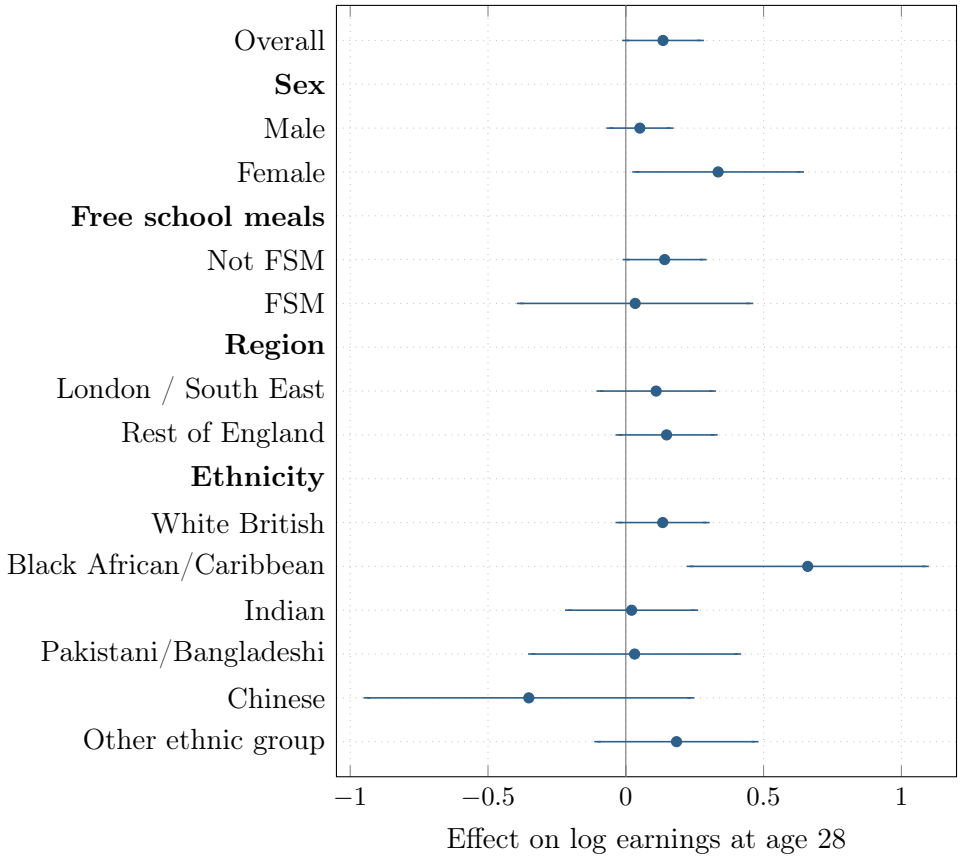
'Russell Group' university (similar to Ivy League for the UK), broader degree subject mix, and degree completion, are reported in Appendix Table 4A.10.

[17]This split conditions on a post-treatment variable (degree choice), so the subgroup estimates are not separately identified as causal effects. However, the complier shares (15.2% shifting into economics degrees, 84.8% not) make the decomposition informative about where in the distribution of compliers the earnings gains are concentrated.

groups at the same time.

Figure 4.2 reports the IV estimate of the effect of A-level economics on log earnings at age 28, estimated separately for subgroups defined by sex, FSM eligibility, region, and ethnicity. Each estimate uses the same specification as the main result, with the sample restricted to the relevant subgroup.

Figure 4.2: Effect of A-level economics on earnings at age 28, by subgroup




*Notes: Each row reports a separate two-stage least squares estimate of the effect of taking A-level economics, instrumented by within-school availability, estimated on the indicated subgroup. All specifications include school and cohort fixed effects and the full set of individual controls. Bars are 95% confidence intervals from standard errors clustered at the school level. The outcome is log real earnings at age 28. The Overall row reproduces the linear IV estimate and confidence interval of Table 4.3.*

Three patterns stand out. First, the point estimate for women (0.335) is much larger than for men (0.051). The women induced into economics by availability are fewer, but those who are induced appear to gain substantially more, which suggests that expanding provision could also narrow the gender gap in earnings. The difference between the two estimates is not itself precisely determined, though, and the female estimate carries a wide interval of its own. Second, the estimate is

larger in the rest of England (0.148) than in London and the South East (0.110): it is not that London students benefit more from economics; they are simply more likely to have the opportunity. Third, the largest estimate is for Black African and Caribbean students (0.660), though with twelve subgroup estimates and wide intervals one large estimate is to be expected by chance, and we place no weight on it. The estimates for Indian and Chinese students are 0.02 and −0.35, and both are very imprecise. Few students in these groups change their take-up when availability changes, because most already attend schools that offer economics, so our estimate of their return is uninformative.

The FSM estimate is small (0.034) but very imprecise: its confidence interval runs from −0.38 to 0.45, so the data cannot distinguish a zero return from a return three times the overall estimate.

The equity case for expansion therefore rests mainly on the access evidence of Section 4.3. The return estimates point the same way where they are informative: for sex and region, the groups with less access show the larger returns, though neither difference is precise. If that pattern holds, expanding access would raise total earnings and narrow gaps between groups at the same time. Section 4.5 quantifies the fiscal return; the equity gain comes through who gains access.

Splitting by demographic group is not the only way to look for differences in returns. Appendix Table 4A.12 instead splits students by how likely they were to take A-level economics, predicted from their characteristics. The estimates are positive in every quartile but precise only in the top quartile, where the first stage is strongest. The top-quartile estimate of 0.152 is close to the pooled estimate, and there is no evidence that returns vary with the propensity to take the subject.

The policy calculation in Section 4.5 needs a return for the schools where new teachers would be hired. Never-offer schools serve a more disadvantaged and more female population than the switcher schools behind our main estimates (Figure 4.1). We therefore recompute the lifetime return using the sex and ethnic mix of never-offer schools. The return for each group is the published lifetime

return for that group in Britton et al. (2026), not our age-28 subgroup estimate. This follows Angrist and Fernández-Val (2013), who adjust an estimate to a new population by reweighting it to that population's observable characteristics. The adjustment is small. Never-offer schools are slightly more female and more White British, and the published returns for these groups are somewhat lower, so the adjusted gross lifetime return is £185,000 against £190,000 unadjusted.

There is a tension here. The age-28 subgroup estimates suggest larger returns for women, while the published lifetime returns used in the adjustment are larger for men. The two are measured at different points of the lifecycle and neither difference is precise, so we do not attempt to reconcile them, and we use the published returns throughout the calculations.

## 4.5 The return on expanding teacher supply

Section 4.4 showed that when a school does not offer economics, its students lose access to a high-return subject, and it estimated what that costs them in earnings. The barrier can be removed by hiring a teacher. We now compute the return to doing so. The policy is to hire an economics teacher at a school that does not currently offer the subject.

Hiring an economics teacher allows a school to start offering A-level economics. The share of students who take the subject when it becomes available is the first-stage effect $\hat{\pi}_{1g}$ (Table 4.3 Panel A), which varies across student subgroups $g$. Each student induced to take economics experiences a private lifetime earnings gain $\hat{\beta}_{1g}^{\text{private}}$ and generates a fiscal return $\hat{\beta}_{1g}^{f}$ through additional tax revenue and reduced student loan write-offs (Table 4.3 Panel C). We use the composition-adjusted implied returns from Section 4.4.5, which reweight the returns to reflect the observable characteristics of students in never-offer schools, where new teachers would be hired. The MVPF is a ratio. The numerator is the total private earnings gain to the students the teacher affects. The denominator is the net cost to the government, meaning the cost of the teacher less the extra tax those students

later pay. A higher ratio means more benefit per pound of public money. With $N_g$ students in subgroup $g$ per cohort and a teacher who serves $T$ cohorts, the MVPF following Hendren and Sprung-Keyser (2020) is

$$\text{MVPF} = \frac{\sum_{t=1}^{T} \frac{1}{(1+r)^{t-1}} \sum_g \hat{\pi}_{1g} \, N_g \, \hat{\beta}_{1g}^{\text{private}}}{C^{\text{sign-on}} + \sum_{t=1}^{T} \frac{1}{(1+r)^{t-1}} \left( C^{\text{annual}} - \sum_g \hat{\pi}_{1g} \, N_g \, \hat{\beta}_{1g}^{f} \right)}, \tag{4.4}$$

where $g$ indexes student subgroups, $\hat{\pi}_{1g}$ is the subgroup-specific first-stage effect, $N_g$ is the number of students in subgroup $g$ per cohort, $\hat{\beta}_{1g}^{\text{private}}$ is the net private lifetime earnings gain, $\hat{\beta}_{1g}^{f}$ is the fiscal return per student induced to take economics, and $r$ is the discount rate (3.5%, the Green Book rate).[18] On the cost side, $C^{\text{sign-on}}$ is the government's share of initial teacher training costs (approximately £25,000, primarily through student loan write-offs), and $C^{\text{annual}}$ is the teacher's annual salary plus employer pension and social security costs. The annual salary is £51,000 (the median classroom teacher salary of £48,892 in the November 2024 School Workforce Census, uprated to 2025/26), and additional employer costs add approximately 42% (the Teachers' Pension Scheme employer contribution of 28.68% plus employer National Insurance contributions of 15% above a minimum threshold), giving $C^{\text{annual}} = £72{,}000$.

The salary is the median for classroom teachers. The marginal recruit into a shortage subject may cost more than the median, while a teacher who also teaches other subjects would justify attributing less than a full salary to economics provision. The biases on the cost side therefore run in both directions. We assume an expected tenure of $T = 10$ years, based on retention data from the NFER Teacher Labour Market Report (McLean et al., 2024). The conclusion does not depend on this choice: with a five-year tenure both sides of the calculation shrink, to an exchequer return of roughly £1.2 million against costs of roughly £360,000.

After adjusting for the composition of never-offer schools, the exchequer return

[18] We value all quantities in 2025 prices. The lifetime earnings estimates from Section 4.4 are uprated from their original price base using cumulative growth in the ONS Average Weekly Earnings index.

per complier is £97,000. The first stage can be composition-adjusted in the same way: the availability effect on take-up is 2.8 percentage points for women and 9.2 for men, and reweighting to the 57.5% female share of never-offer schools lowers the projected take-up effect from 5.7 to 5.5 percentage points. This reduces the discounted exchequer return by around three per cent and the fiscal surplus by around five per cent, leaving the conclusion unchanged.

Using the unadjusted take-up effect of 5.7 percentage points and an average cohort of 46 A-level students in never-offer schools, each school cohort generates an average exchequer return of approximately £254,000. In headcount terms the numbers are small. A take-up effect of 5.7 percentage points in a 46-student cohort is about 2.6 compliers per cohort, or roughly 26 over a ten-year tenure. The surplus arises because each complier carries a large exchequer return, not because many students are affected. Over a ten-year teaching career, the total discounted exchequer return is £2.2 million. The denominator of the MVPF is the cost of the teacher less the tax revenue the policy returns to the government. The cost of the teacher over ten years is £645,000, the £25,000 training cost plus ten annual costs of £72,000, discounted. Here the revenue is larger than that cost, so the net cost to the government is negative, and the convention in the MVPF literature is to report the MVPF as infinite. In plain terms, the policy more than pays for itself, with a fiscal surplus of roughly £1.5 million per teacher (Table 4.5 Panel A). Among the 133 historical US policies evaluated by Hendren and Sprung-Keyser (2020), an infinite MVPF places this intervention alongside the highest-return investments in children's education and health.

The conclusion does not require the central estimate to be precise. We can ask how small the return per complier could be before the policy stopped paying for itself. The discounted tax revenue equals the £645,000 cost when the exchequer return per complier is about £28,000. That is less than a third of the central £97,000, so the policy pays for itself even if the central return is overstated by two-thirds. The fiscal case also rests mainly on the two precisely estimated

education effects, on university attendance (14.0 percentage points, standard error 4.0) and on economics degree take-up (15.2 points, standard error 1.0), combined with published returns, rather than on the less precise age-28 earnings estimate.

This calculation counts only the gains that come through degree choice, and its errors could run in either direction. It leaves out any gains that do not come through degree choice, which Panel B of Table 4.4 suggests are substantial, so on this count it understates the return. It uses published subject-level returns that are likely to overstate the true causal returns, so on that count it overstates the return. And the direct age-28 evidence on the students who switch into economics degrees is too imprecise to settle the size of their gain. Because the central return is more than three times the level at which the policy just covers its cost, this uncertainty does not change the conclusion.

The second calculation of Section 4.4.3, which assumes the age-28 gain persists and does not use the degree-level returns at all, provides an independent check. Its exchequer figure of roughly £35,000 per complier also exceeds the £28,000 at which the policy just covers its cost, and its discounted exchequer return over a ten-year tenure is roughly £0.8 million against costs of £645,000. The policy pays for itself under both calculations. Under the first the return is more than three times the cost. Under the second it exceeds the cost by around 20%.

The calculation also relies on two things we cannot estimate. The first is that a school that hires an economics teacher goes on to offer the subject, year after year. The second is that never-offer schools would respond to new provision in the same way as the switcher schools behind our estimates. On the second, a school that has not offered economics in twelve years of data may be different from a switcher school; it may have little demand for the subject, or a sixth form too small to sustain a class. And because no never-offer school in our data ever starts offering economics, there is nothing to check this assumption against.

Neither assumption needs to hold fully. Suppose the exchequer return per complier is the central £97,000. The policy still pays for itself if a hired teacher

delivers provision in only three of every ten years employed. It also still pays for itself if provision is delivered every year but the take-up effect is as low as 1.7 percentage points, rather than the estimated 5.7. If both happened at once, the policy would no longer pay for itself.

The UK government currently offers training bursaries of up to £30,000 for shortage subjects, principally in science and mathematics, recognising the high value of expanding teacher supply in those fields. Economics does not receive such a bursary. Given the returns documented here, a comparable incentive would be justified: with a £30,000 training bursary, the fiscal surplus per teacher would still be roughly £1.5 million, and the policy would continue to more than pay for itself.

Table 4.5 Panel B shows who would benefit. Schools that do not currently offer economics disproportionately serve students from economically disadvantaged backgrounds. As well as being fiscally beneficial, expansion of access therefore also changes who goes on to study economics at university, though the changes are modest. Universal access would raise the share of economics undergraduates from outside London and the South East from 51% to 53%, still short of the two-thirds of students who attend school outside London and the South East. The shares of women and of FSM students would be roughly unchanged, at 28% and 7%. The female share barely moves because women's take-up response to availability is small; the gender gap is a matter of conditional take-up rather than access, as Section 4.3 showed.

Our estimates take the wages of economics graduates as given. If access expanded everywhere, the supply of economics graduates would rise, and their wages could fall. We think this is a small concern for two reasons. First, Hsieh et al. (2019) model workers choosing occupations and let wages adjust when more workers enter an occupation. In their model the gains from better allocation of talent remain large after wages adjust. Their exercise concerns different barriers, so we take this as suggestive rather than direct evidence for our setting. Second,

Table 4.5: Cost-benefit analysis of expanding economics teacher supply

| **Panel A: Marginal value of public funds** | | | |
|---|---|---|---|
| Parameter | Description | Value | Source |
| $\delta$ | Teacher → sustained provision | 1 | Assumed |
| $\pi_1$ | Availability → A-level | 0.057 | Table 4.3 |
| $\beta_1$ | A-level → gross PV lifetime return | £185,000 | Table 4.3 Panel C, adjusted |
| $\beta_1^p$ | A-level → net private lifetime return | £88,000 | Table 4.3 Panel C, adjusted |
| $\beta_1^f$ | A-level → exchequer return | £97,000 | Table 4.3 Panel C, adjusted |
| $N$ | Cohort size per school | 46 | Data |
| $C$ | PV cost per teacher (salary + training) | £645,000 | DfE; NFER |
| | **MVPF** | Infinite | |
| | **Fiscal surplus per teacher** | £1,540,000 | |
| **Panel B: Universal access counterfactual** | | | |
| | | Current | Universal access |
| | A-level econ students per cohort | 15,026 | 19,018 |
| | Economics undergraduates per cohort | 5,714 | 6,321 |
| | % female among economics undergraduates | 28.2 | 28.3 |
| | % FSM among economics undergraduates | 6.8 | 6.9 |
| | % outside London/SE among econ. undergraduates | 50.6 | 53.3 |
| | Aggregate gross PV earnings gain per cohort | | £740 million |

*Notes:* Panel A reports the components of the MVPF calculation. The teacher-to-provision conversion $\delta$ is assumed to equal one: a hired teacher is taken to translate into sustained provision. Section 4.5 reports the break-even conversion rate. The lifetime return parameters are the composition-adjusted implied values of Section 4.4.5, which reweight the Table 4.3 Panel C returns to the composition of never-offer schools. The numerator of the MVPF is the net private lifetime return, the earnings gain students keep after tax. The denominator is the net cost to the government, the teacher's salary and training cost minus the exchequer return. The gross return is shown for context and approximately equals the net private return plus the exchequer return. An MVPF above 1 means students gain more than one pound for every pound the policy costs the government. If the fiscal return exceeds the cost, the denominator is negative and the MVPF is infinite, meaning that the policy pays for itself. Panel B reports a counterfactual exercise in which all schools are assumed to offer A-level economics. Additional A-level students and economics undergraduates are computed from the first-stage estimates applied to the population of students currently in schools that do not offer economics. The composition rows are computed group by group. For each group, the number of students in schools that do not offer economics is multiplied by the group's take-up effect of availability. The sex row uses the sex-specific effects (2.8 percentage points for women, 9.2 for men, Section 4.5). The FSM and region rows use the pooled effect of 5.7 points. The resulting extra A-level students are converted to economics undergraduates at 15.2%, the effect of A-level economics on studying the subject at university. Shares are computed among students with the relevant characteristic recorded; under the sex-specific effects the implied additions are slightly fewer than the pooled totals in the count rows. The aggregate earnings gain is gross; the net individual analogue, applying the ratio of the net to the gross return in Table 4.3 Panel C, is roughly £350 million. FSM denotes eligibility for free school meals, a standard administrative marker of low socioeconomic family background.

economics graduates enter a broad set of sectors, including finance, consulting, the civil service, technology, and academia, so the relevant labour market is not a narrow occupation with constrained demand.

A further limitation concerns what the teacher would otherwise have done. Our calculation treats the salary as the full cost of hiring the teacher. But someone

recruited to teach economics would otherwise have worked in another job, perhaps teaching another subject, and would have paid tax on those earnings and produced something of value; the calculation counts none of this. On the tax side, what the government loses is only the tax on the difference between the recruit's earnings in the alternative job and their earnings in teaching, because a teacher pays tax as well. A difference of that size is far too small to overturn a surplus of roughly £1.5 million per teacher. The omissions do mean that the reported surplus overstates the true social return, so the cost-benefit figures are best treated as an upper bound. Given the size of the surplus, we do not think the policy conclusion would change if these costs were fully counted.

## 4.6 Conclusion

Whether a school employs an economics teacher determines whether its students can study A-level economics, and this in turn has large consequences for degree choice and lifetime earnings. Differences in school-level availability alone explain 36% of the regional gap in A-level economics take-up, and between 6% and 82% of the gaps across demographic groups. Students induced to take A-level economics by within-school changes in availability earn around 14% more at age 28, though this direct estimate is imprecise. Our two calculations put the net lifetime gain per complier at £65,000 and £90,000. The £90,000 figure comes from the effects on university attendance and degree choice, both precisely estimated, combined with published lifetime returns by subject. The £65,000 figure assumes the age-28 gain lasts for the whole working life, and so inherits the imprecision of that estimate. The fiscal return to hiring an additional economics teacher exceeds the cost under both calculations. The MVPF is infinite, with a fiscal surplus of roughly £1.5 million per teacher under the calculation based on degree choices, and a much smaller but still positive surplus under the calculation that assumes the age-28 gain persists.

Expanding economics teacher supply would generate fiscal returns that more

than cover the cost, even with a training bursary comparable to those offered for science and mathematics. Because schools that do not currently offer economics disproportionately serve disadvantaged students, the same policy also raises the share of economics students who come from under-represented regions. The share of economics undergraduates from outside London and the South East would rise from 51% to 53%, with the shares of women and of students from poorer socioeconomic backgrounds (FSM) roughly unchanged. Supply-side expansion is a necessary precondition for the demand-side interventions that have dominated the diversity discussion: exposing students to role models or redesigning introductory courses cannot help students who cannot access the subject. A natural extension would replace the two-step calculation. The model behind Britton et al. (2026) can predict a lifetime return for each individual student. Using that predicted return as the outcome in our IV regression would estimate the lifetime effect directly, instead of combining our degree effects with published subject averages.

# Appendix to Chapter 4

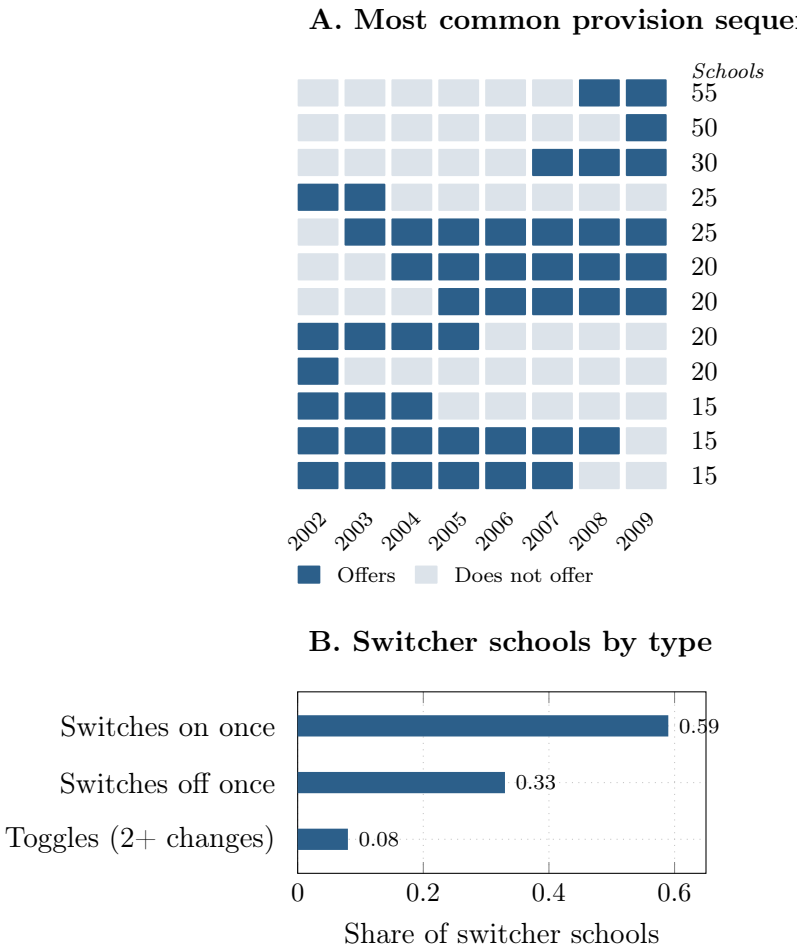
This appendix collects the supporting figures and tables for the chapter. It reports the event studies and the time series of provision discussed in the text, and the tables referenced from each section: the subgroup regressions and the decomposition for earlier cohorts, the composition of economics students, the school-type and school-characteristics comparisons, the evidence on school moves, the balance and placebo tests, the robustness checks including the availability threshold, and the estimates for other outcomes.

## 4A.1 Additional Figures and Tables

Figures 4A.3 and 4A.4 report the event studies discussed in Section 4.4.1. Three features of the figures need explaining. First, in the take-up event study (Figure 4A.3), the effect in the year provision changes is smaller than in later years (1.4 percentage points against around 7). This is because we date provision by the year in which a cohort sat its A-level exams, so the first cohort we count as having access may have gained it only part-way through the two-year course.

Second, the two directions of change behave differently. When a school stops offering economics, the fall in take-up fades within two to three cohorts; when a school starts, the rise persists. This suggests that some of the stops we count are not real stops. A school may simply fall below the two-student threshold for a year, or its students may take economics at a nearby school. Most switcher schools change provision only once (Figure 4A.1 Panel B), and starts outnumber stops, so the estimates rest mainly on starts, which are the more reliably measured changes. If some of the stops we count are not real, the two estimates behind the IV are biased in the same direction: the effect of availability on take-up is smaller than it should be, and so is the effect of availability on earnings. The IV estimate is

Figure 4A.1: Patterns of A-level economics provision among switcher schools

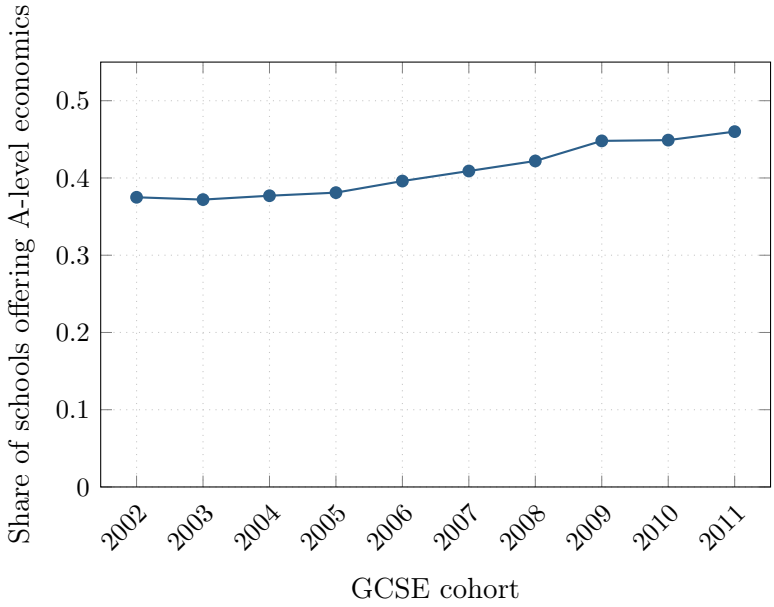



*Notes: Restricted to schools whose A-level economics provision changes at least once across the 2002–2009 GCSE cohorts, and which are observed in every cohort. Panel A shows the twelve most common provision sequences, one row per sequence and one cell per cohort; shaded cells indicate cohorts to which the school offered A-level economics. The right-hand column gives the number of schools following each sequence. Sequences followed by fewer than ten schools are suppressed. Panel B classifies all switcher schools by whether provision changes once or more than once. Provision is measured in a wider extract than the 2002–2008 earnings sample.*

the ratio of the two, and because both shrink by a similar proportion, the ratio changes much less.

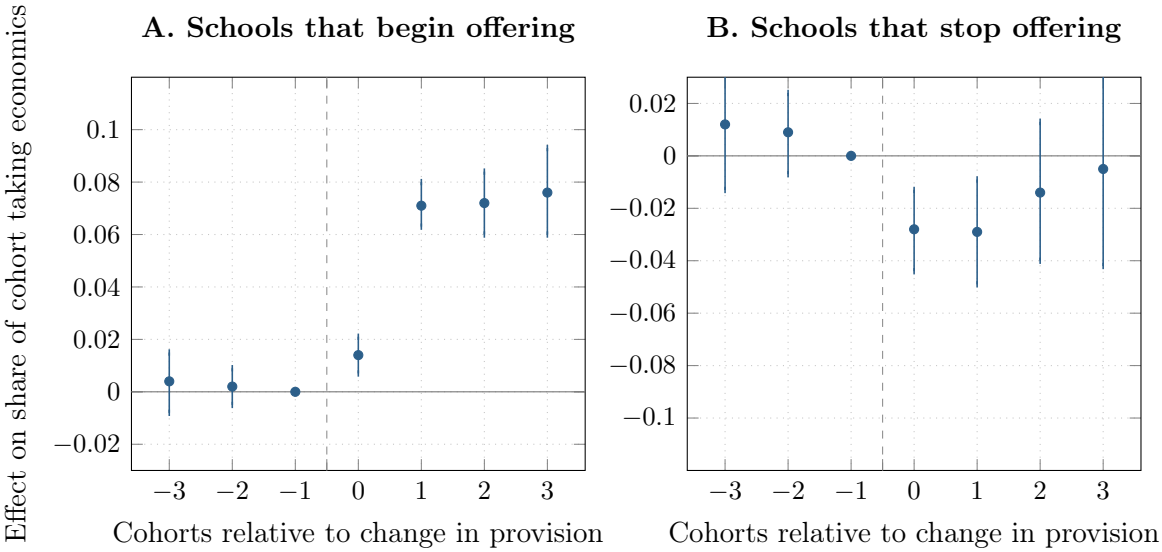
Finally, the rise in take-up after a school starts offering economics partly repeats the definition of the availability measure, since a school counted as offering must have at least two takers. The informative part of the take-up event study is therefore the years before the change, which show no build-up in demand, together with the event study of cohort attainment (Figure 4A.4).

Figure 4A.2: Share of schools offering A-level economics over time



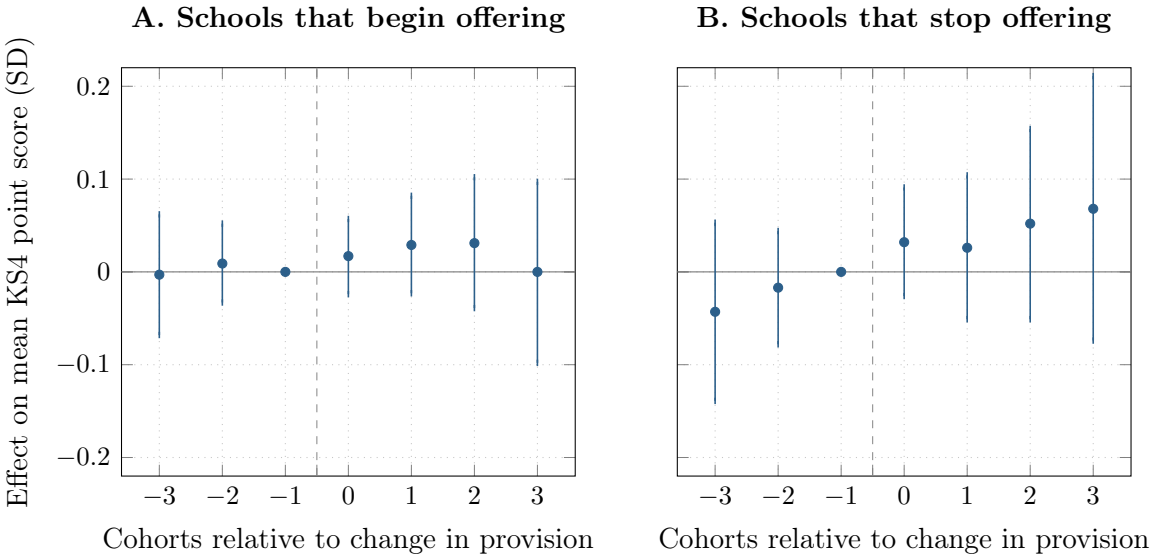

*Notes: A school is counted as offering A-level economics to a cohort if at least two students in that school-year group sat the A-level economics examination. The sample comprises all schools with post-16 provision observed in the given year; the number of schools observed ranges from around 2,120 to 2,670 per year. Provision is measured in a wider extract covering GCSE cohorts 2002–2013, so the series extends beyond the 2002–2008 earnings sample; the figure plots cohorts to 2011.*

Figure 4A.3: Event study: A-level economics take-up around provision changes




*Notes: Each point is the coefficient on an indicator for a given cohort before or after the change in A-level economics provision, relative to the cohort just before the change, estimated on switcher schools only. The outcome is the share of the school-cohort taking A-level economics. Each specification includes school and cohort fixed effects; the cohort immediately before the change ($t = -1$) is the omitted category, and the dashed line marks the last pre-change cohort. Bars are 95% confidence intervals. Panel A covers schools that begin offering economics, Panel B those that stop; schools whose provision changes more than once are excluded. Endpoints are binned, so $t = -3$ and $t = 3$ include all earlier and later cohorts respectively. The number of school-cohorts contributing at each event time is reported in Appendix Table 4A.14. A test that all the pre-change coefficients are jointly zero gives $p = 0.81$ for Panel A and $p = 0.54$ for Panel B, so there is no evidence that take-up was already changing before provision changed.*

Figure 4A.4: Event study: cohort attainment around changes in A-level economics provision




*Notes: Each point is the coefficient on an indicator for a given cohort before or after the change in A-level economics provision, relative to the cohort just before the change, estimated on switcher schools only. The outcome is the mean standardised KS4 point score of the school-cohort, a characteristic of the intake determined before any exposure to A-level provision. Each specification includes school and cohort fixed effects; the cohort immediately before the change ($t = -1$) is the omitted category, and the dashed line marks the last pre-change cohort. Bars are 95% confidence intervals; both panels share a common vertical scale. Panel A covers schools that begin offering economics, Panel B those that stop; schools whose provision changes more than once are excluded. Endpoints are binned, so $t = -3$ and $t = 3$ include all earlier and later cohorts respectively. The number of school-cohorts contributing at each event time is reported in Appendix Table 4A.14. A test that all the pre-change coefficients are jointly zero gives $p = 0.82$ for Panel A and $p = 0.68$ for Panel B, so there is no evidence that the intake was already changing before provision changed.*

Table 4A.1: Effect of availability and school type on A-level economics take-up by subgroup, 2011 cohort

| | $\hat{\beta}$: effect of avail. | Always offer vs. never | Switcher vs. never |
|---|---|---|---|
| All students | 0.078*** | 0.025*** | 0.0009*** |
| | (0.004) | (0.005) | (0.0003) |
| Male | 0.120*** | 0.035*** | 0.0019*** |
| | (0.006) | (0.007) | (0.0005) |
| Female | 0.046*** | 0.012*** | 0.0003 |
| | (0.003) | (0.004) | (0.0003) |
| Non-FSM | 0.078*** | 0.026*** | 0.0009*** |
| | (0.004) | (0.005) | (0.0003) |
| FSM | 0.083*** | 0.019** | 0.0009 |
| | (0.007) | (0.009) | (0.001) |
| White British | 0.064*** | 0.021*** | 0.0004* |
| | (0.004) | (0.005) | (0.0002) |
| Indian | 0.161*** | 0.045** | 0.004 |
| | (0.017) | (0.019) | (0.003) |
| Pakistani/Bangladeshi | 0.105*** | 0.027** | 0.0003 |
| | (0.011) | (0.014) | (0.002) |
| Black African/Caribbean | 0.122*** | 0.019 | 0.005 |
| | (0.010) | (0.012) | (0.003) |
| Chinese | 0.128*** | 0.090*** | 0.006 |
| | (0.027) | (0.030) | (0.011) |
| London/South East | 0.110*** | 0.014* | 0.001 |
| | (0.006) | (0.008) | (0.001) |
| Rest of England | 0.060*** | 0.029*** | 0.0008*** |
| | (0.003) | (0.005) | (0.0003) |
| Observations | | 221,118 | |

*Notes:* Each row reports coefficients from a separate LPM of A-level economics take-up on an indicator for availability and school-type dummies (always-offer, switcher; never-offer omitted), estimated on the indicated subgroup from the 2011 GCSE cohort. The availability coefficient $\hat{\beta}$ is the change in take-up when a school offers the subject. The school-type coefficients compare take-up across always-offer, switcher and never-offer schools at the same level of availability. Differences between them reflect differences in student demand for economics across the three school types. Standard errors clustered at the school level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.2: Access to and take-up of A-level economics by demographic group, 2003 and 2007 cohorts

| | Access rate | Take-up \| access | Take-up (uncondit.) | Gap from access | Share of gap from access |
|---|---|---|---|---|---|
| **2003 cohort** | | | | | |
| *Sex* | | | | | |
| Male | 60.8 | 11.4 | 6.9 | | |
| Female | 55.7 | 3.6 | 2.0 | | |
| Gap (M − F) | 5.1 | 7.7 | 4.9 | 0.4 | 7.8 |
| *Socioeconomic status* | | | | | |
| Non-FSM | 58.5 | 7.3 | 4.3 | | |
| FSM | 55.9 | 6.9 | 3.8 | | |
| Gap | 2.6 | 0.4 | 0.5 | 0.2 | 40.3 |
| *Region* | | | | | |
| London/South East | 67.3 | 8.9 | 6.0 | | |
| Rest of England | 53.3 | 6.3 | 3.4 | | |
| Gap | 14.0 | 2.6 | 2.7 | 1.1 | 39.5 |
| *Ethnicity* | | | | | |
| White British | 55.6 | 6.2 | 3.5 | | |
| Indian | 72.4 | 15.4 | 11.1 | | |
| Pakistani/Bangladeshi | 60.1 | 9.2 | 5.6 | | |
| Black African/Caribbean | 69.6 | 9.0 | 6.3 | | |
| Chinese | 67.2 | 15.3 | 10.3 | | |
| Gap: Indian − White | | | 7.6 | 1.8 | 23.7 |
| Gap: Pak./Bang. − White | | | 2.1 | 0.4 | 16.8 |
| Gap: Black Afr./Car. − White | | | 2.8 | 1.1 | 38.0 |
| Gap: Chinese − White | | | 6.8 | 1.2 | 18.4 |
| *N* | | | 188,957 | | |

Table 4A.2: Access to and take-up of A-level economics by demographic group, 2003 and 2007 cohorts continued

| | Access rate | Take-up \| access | Take-up (uncondit.) | Gap from access | Share of gap from access |
|---|---|---|---|---|---|
| **2007 cohort** | | | | | |
| *Sex* | | | | | |
| Male | 64.6 | 12.6 | 8.2 | | |
| Female | 59.9 | 4.4 | 2.7 | | |
| Gap (M − F) | 4.7 | 8.2 | 5.5 | 0.4 | 7.3 |
| *Socioeconomic status* | | | | | |
| Non-FSM | 62.5 | 8.3 | 5.2 | | |
| FSM | 60.8 | 8.3 | 5.1 | | |
| Gap | 1.7 | −0.1 | 0.1 | 0.1 | — |
| *Region* | | | | | |
| London/South East | 72.0 | 10.2 | 7.4 | | |
| Rest of England | 56.9 | 7.0 | 4.0 | | |
| Gap | 15.1 | 3.2 | 3.4 | 1.3 | 38.3 |
| *Ethnicity* | | | | | |
| White British | 59.6 | 6.8 | 4.1 | | |
| Indian | 73.9 | 18.8 | 14.0 | | |
| Pakistani/Bangladeshi | 63.4 | 12.5 | 8.0 | | |
| Black African/Caribbean | 73.9 | 11.3 | 8.4 | | |
| Chinese | 72.9 | 17.5 | 12.8 | | |
| Gap: Indian − White | | | 9.9 | 1.8 | 18.5 |
| Gap: Pak./Bang. − White | | | 3.9 | 0.4 | 9.3 |
| Gap: Black Afr./Car. − White | | | 4.3 | 1.3 | 30.0 |
| Gap: Chinese − White | | | 8.7 | 1.6 | 18.6 |
| *N* | | | 211,651 | | |

*Notes:* Same structure as main-text Table 4.2, replicated for the 2003 and 2007 GCSE cohorts. Access rate is the share of students attending a school that offers A-level economics (defined as $\geq 2$ students sitting the exam in the school-cohort). Take-up | access is the share taking A-level economics conditional on the school offering it. The take-up gap is decomposed as $p_A - p_B = \bar{c}(a_A - a_B) + \bar{a}(c_A - c_B)$, where $a$ denotes the access rate, $c$ denotes conditional take-up, and bars denote simple averages across the two groups. "Gap from access" is the first term; "Share of gap from access" divides this by the unconditional take-up gap. In the 2007 cohort the FSM take-up gap is essentially zero (FSM take-up is marginally higher), so the share is undefined and not reported.

Table 4A.3: Demographic composition of cohort, university students, and economics undergraduates

| | 2003 | | | 2007 | | | 2011 | | |
|---|---|---|---|---|---|---|---|---|---|
| | Cohort | UGs | Econ UGs | Cohort | UGs | Econ UGs | Cohort | UGs | Econ UGs |
| **Panel A: Sex** | | | | | | | | | |
| Male | 46.8 | 45.9 | 71.5 | 47.5 | 46.5 | 70.8 | 48.0 | 46.2 | 72.0 |
| Female | 53.2 | 54.1 | 28.5 | 52.5 | 53.5 | 29.2 | 52.0 | 53.8 | 28.0 |
| **Panel B: Ethnicity** | | | | | | | | | |
| White British | 75.3 | 72.5 | 53.6 | 76.2 | 72.6 | 54.2 | 73.0 | 67.7 | 50.1 |
| Black African | 1.6 | 2.0 | 4.0 | 2.3 | 2.9 | 5.0 | 3.1 | 4.2 | 7.4 |
| Black Caribbean | 1.2 | 1.4 | 0.9 | 1.3 | 1.5 | 1.0 | 1.4 | 1.6 | 1.5 |
| Indian | 3.7 | 4.6 | 13.9 | 3.1 | 3.9 | 11.9 | 2.9 | 3.8 | 10.1 |
| Pakistani | 2.5 | 2.8 | 3.8 | 2.6 | 3.0 | 3.8 | 3.1 | 3.7 | 4.5 |
| Bangladeshi | 1.0 | 1.1 | 2.1 | 1.1 | 1.2 | 2.7 | 1.4 | 1.7 | 2.4 |
| Chinese | 0.6 | 0.7 | 2.3 | 0.5 | 0.7 | 2.1 | 0.6 | 0.8 | 2.2 |

*Notes:* Shares of each group in the GCSE cohort, among university students, and among economics undergraduates, for three cohorts. Shares may not sum to 100 due to omitted categories.

Table 4A.4: Access to and take-up of A-level economics by school type

| | % offer econ | % offer maths | Take-up \| offer | Conversion to econ degree |
|---|---|---|---|---|
| **Panel A: Boys** | | | | |
| Mixed | 67.8 | 99.1 | 14.1 | 33.4 |
| Single-sex boys | 86.9 | 99.5 | 20.0 | 38.3 |
| **Panel B: Girls** | | | | |
| Mixed | 66.4 | 98.6 | 4.8 | 30.2 |
| Single-sex girls | 63.1 | 99.3 | 10.6 | 33.7 |

*Notes:* The table reports the share of schools offering A-level economics and maths, the take-up rate conditional on the school offering economics, and the conversion rate from A-level economics to an economics degree. Sample: 2011 GCSE cohort.

Table 4A.5: School characteristics by A-level economics offer behaviour

| | 2007 | | | 2011 | | |
|---|---|---|---|---|---|---|
| | Always | Switcher | Never | Always | Switcher | Never |
| Av. KS4 score | 0.82 | 0.75 | 0.75 | 0.64 | 0.60 | 0.63 |
| % FSM | 5.4 | 6.0 | 5.8 | 5.6 | 7.1 | 7.5 |
| % White British | 71 | 76 | 84 | 69 | 73 | 79 |
| % Black African | 2.5 | 2.5 | 1.2 | 3.3 | 3.2 | 1.9 |
| % Indian | 4.8 | 3.2 | 2.1 | 4.6 | 2.7 | 2.3 |
| % Chinese | 0.86 | 0.66 | 0.47 | 0.90 | 0.67 | 0.53 |
| % All boys | 8.5 | 4.0 | 2.4 | 8.2 | 2.7 | 2.4 |
| % All girls | 3.9 | 10.9 | 9.5 | 5.7 | 11.3 | 7.4 |
| N schools | 453 | 800 | 924 | 667 | 764 | 966 |

*Notes:* School-level averages by economics offer behaviour. "Always offer" schools offer A-level economics in every cohort in our sample; "Never offer" schools never offer it; "Switcher" schools offer it in some cohorts but not others. Figures correspond to the main-text Figure 4.1.

Table 4A.6: School transitions at age 16 and A-level economics access

| | KS5 school offer behaviour | | | |
|---|---|---|---|---|
| KS4 school offer behaviour | Always | Switcher | Never | Total |
| **Panel A: Transition matrix (% of KS5 students)** | | | | |
| Same school (no move) | 17.2 | 20.8 | 17.3 | 55.3 |
| Moved from: Always offer | 2.2 | 0.9 | 0.4 | 3.4 |
| Moved from: Switcher | 3.7 | 1.6 | 0.7 | 6.0 |
| Moved from: Never offer | 3.4 | 1.8 | 1.0 | 6.2 |
| Moved from: No 6th form and is feeder school | 16.4 | 4.0 | 1.4 | 21.9 |
| Moved from: No 6th form and not feeder school | 3.4 | 2.5 | 1.4 | 7.3 |
| Total | 46.2 | 31.6 | 22.2 | 100 |
| **Panel B: A-level economics take-up among movers (%)** | | | | |
| Moved: non-offer → offer | 6.3 | | | |
| Moved: offer → non-offer | 0.2 | | | |
| Non-movers at offer schools | 7.9 | | | |
| Non-movers at non-offer schools | 0.1 | | | |
| Observations | 1,409,352 | | | |

*Notes:* Panel A shows the distribution of students across school transition types. "Same school" comprises students who remain at the same institution; transfers from feeder schools are shown separately. The feeder row counts all movers from schools without sixth forms that have a linked school, regardless of destination; the share of students moving to their own linked school is smaller (about 16%), which is the figure quoted in the text. Offer behaviour (Always, Switcher, Never) is defined over the full sample period, as in Table 4.1. In Panel B, offer and non-offer status refer to the school's offer in the relevant cohort year. Panel B compares A-level economics take-up rates across transition types. Movers who gain access take up economics at a slightly lower rate (6.3%) than non-movers at offer schools (7.9%), indicating that school moves are not driven by demand for economics. Sample: GCSE cohorts 2002–2008, students continuing to KS5.

Table 4A.7: Balance tests: effect of A-level economics availability on pre-treatment outcomes

| Dependent variable | Coefficient on availability | SE |
|---|---|---|
| Std. GCSE overall score | −0.007 | (0.008) |
| Std. GCSE maths score | 0.025*** | (0.005) |
| Std. KS2 score (age 11) | 0.001 | (0.004) |
| % female | −0.004 | (0.0024) |
| % White British | −0.008** | (0.004) |
| % FSM eligible | 0.001 | (0.001) |
| % London/South East | 0.0004 | (0.0009) |
| School FE | ✓ | |
| Cohort FE | ✓ | |
| Observations | 1,335,629–1,409,039 | |

*Notes:* Each row reports the coefficient from a separate regression of the indicated pre-treatment variable on an indicator for A-level economics availability, with school and cohort fixed effects. Small coefficients support the assumption that within-school changes in availability are not driven by shifts in student composition; the two statistically significant rows (GCSE maths and the White British share) are discussed in Section 4.4.1, and both variables are controlled directly in the estimating equations. The number of observations differs across rows because some variables are missing for some students. Standard errors clustered at the school level. Sample: GCSE cohorts 2002–2008, all schools. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.8: Placebo test: IV estimates for students who do not take A-level economics

| | (1) Full sample | (2) Non-A-level econ students |
|---|---|---|
| **Panel A: Log earnings at age 28** | | |
| A-level econ available | 0.008* | −0.002 |
| | (0.004) | (0.004) |
| School FE | ✓ | ✓ |
| Cohort FE | ✓ | ✓ |
| Individual controls | ✓ | ✓ |
| Observations | 368,260 | 357,168 |

*Notes:* Column (1) reproduces the reduced-form estimates from the full sample for reference. Column (2) restricts the sample to students who do not take A-level economics. The IV design assumes that offering economics changes earnings only for the students who take it. If availability changed the earnings of non-takers, that assumption would fail. The test has a limit. When a school starts offering economics, the students who take it up leave the non-taker group, so the non-takers at offering and non-offering schools are slightly different sets of students. A null result is reassuring but not conclusive. Standard errors clustered at the school level in parentheses. Sample: students in switcher schools, GCSE cohorts 2002–2008. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.9: Robustness of IV earnings estimates to alternative specifications

| | First stage ($\pi_1$) | IV: log earnings | First-stage $F$ |
|---|---|---|---|
| 1. Baseline (linear controls) | 0.057*** | 0.135* | 843 |
| | (0.002) | (0.069) | |
| 2. Saturated, region as London–SE indicator (141 cells) | 0.058*** | 0.152** | 891 |
| | (0.002) | (0.072) | |
| 3. School-specific linear trends | 0.049*** | 0.125 | 804 |
| | (0.002) | (0.097) | |
| 4. Region × year FE | 0.058*** | 0.120* | 760 |
| | (0.002) | (0.071) | |
| 5. Saturated with KS2 attainment | 0.057*** | 0.145** | 823 |
| | (0.002) | (0.068) | |
| 6. Fine saturated (745 cells) | 0.058*** | 0.159** | 890 |
| | (0.002) | (0.074) | |
| 7. DDML | — | 0.128* | — |
| | | (0.068) | |
| School FE | All specifications | | |
| Cohort FE | All specifications | | |
| Observations | 368,260 | | |

*Notes:* Each row reports the first-stage coefficient, the IV estimate of the effect of A-level economics on log earnings at age 28, and the first-stage $F$-statistic. Row 1 is the baseline with linear individual controls. Row 2 uses the saturated covariate specification with region entered as a binary London–SE indicator (sex × ethnicity group × FSM × GCSE tercile × London–SE); the main-text saturated specification instead uses the three-category region (London/South East, rest of England, missing). Row 3 adds school-specific linear time trends. Row 4 replaces cohort FE with region × year FE. Row 5 replaces GCSE attainment with KS2 (age 11) attainment in the saturated cells. Row 6 divides GCSE attainment into twenty equal groups instead of three. Row 7 uses the double/debiased machine learning method of Chernozhukov et al. (2018). Instead of entering the controls as linear terms, a penalised regression (lasso) predicts both take-up and availability from the controls. To avoid overfitting, the predictions for each student are made from a model fitted on the other students. The effect of take-up on earnings is then estimated from the parts of take-up and availability that the controls do not predict. This method has no single first-stage coefficient on availability, so no first-stage $F$ is reported. Standard errors clustered at the school level in parentheses. Sample: students in switcher schools, GCSE cohorts 2002–2008. The observation count refers to the baseline; the saturated specifications use slightly fewer observations, because combinations of characteristics with too few students cannot be estimated and those students are excluded (cf. Table 4.3 column 4).

All first-stage $F$-statistics exceed 104.7, the threshold above which the conventional critical value of 1.96 remains valid under the $tF$ procedure of Lee et al. (2022). No adjustment is therefore required in any specification. The $F$-statistics are cluster-robust throughout: with a single instrument, each equals the square of the instrument's cluster-robust $t$-statistic.

Row 7 reports no first-stage coefficient or $F$-statistic because DDML does not estimate a first stage in the two-stage least squares sense. There is accordingly no single coefficient on availability to report, and conventional weak-instrument diagnostics do not apply. For reference, the linear first stage estimated on the same sample is reported in row 1.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.10: Effect of A-level economics availability on other educational outcomes

| | Reduced form (availability) | IV (A-level econ) |
|---|---|---|
| Attends university | 0.008*** | 0.140*** |
| | (0.002) | (0.040) |
| Attends Russell Group university | 0.003 | 0.061 |
| | (0.002) | (0.043) |
| Studies economics at UG | 0.009*** | 0.152*** |
| | (0.001) | (0.010) |
| Studies business/finance/management | 0.002 | 0.040 |
| | (0.0014) | (0.025) |
| Studies other STEM | −0.002 | −0.044 |
| | (0.002) | (0.040) |
| Studies humanities/social science | 0.002 | 0.034 |
| | (0.002) | (0.028) |
| Completes degree | 0.006** | 0.110** |
| | (0.003) | (0.050) |
| School FE | ✓ | ✓ |
| Cohort FE | ✓ | ✓ |
| Individual controls | ✓ | ✓ |
| Observations | 417,849 | 417,849 |

*Notes:* Each row reports the coefficient from a separate regression. The first column is the reduced form: effect of school-cohort A-level economics availability on the indicated outcome. The second column instruments A-level economics take-up with availability (linear controls specification). The economics UG row repeats the main-text estimate for comparison. Standard errors clustered at the school level in parentheses. Sample: students in switcher schools, GCSE cohorts 2002–2008. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.11: A-level subject combinations by economics availability

| | Switcher schools | | Always | Never |
|---|---|---|---|---|
| | Econ available | Econ not available | offer | offer |
| **Panel A: Most common subject triples (% of students)** | | | | |
| Maths, Economics, Physics | 0.32 | — | 0.46 | — |
| Maths, Business, Physics | 0.20 | 0.25 | 0.17 | 0.23 |
| Maths, Biology, Chemistry | 3.29 | 3.08 | 3.86 | 2.91 |
| Maths, Geography, Physics | 0.34 | 0.46 | 0.34 | 0.40 |
| Maths, History, Physics | 0.21 | 0.21 | 0.17 | 0.21 |
| Sciences triple | 0.94 | 0.88 | 0.73 | 1.01 |
| Humanities triple | 10.2 | 11.2 | 8.4 | 11.1 |
| Other combinations | 84.5 | 84.0 | 85.9 | 84.1 |
| **Panel B: Compliers' counterfactual subject bundle (%)** | | | | |
| Business/management | 24.7 | | | |
| Mathematics | -7.7 | | | |
| Geography | 1.5 | | | |
| History | 8.0 | | | |
| Sciences | -1.8 | | | |
| English literature | 23.3 | | | |
| Psychology | 7.0 | | | |
| Observations | 104,374 | 117,200 | 321,263 | 138,754 |

*Notes:* Panel A compares the distribution of A-level subject triples across school-cohorts in switcher schools when economics is and is not available, and in always-offer and never-offer schools. Panel B reports the composition of the subjects compliers would have taken instead of economics. Each subject's take-up indicator is regressed on economics availability within switcher schools, with school and cohort fixed effects; every student in this sample takes exactly three A-levels, so each student who takes up economics drops exactly one other subject. The entry is the fall in that subject's take-up divided by the rise in economics take-up, expressed as a percentage. It is the share of compliers who would otherwise have taken that subject. Shares sum to 100 across all subjects; those shown account for 55%, with the remainder spread across smaller subjects. A negative entry means the subject is taken more, not less, when economics is available: mathematics and the sciences are complements to economics rather than substitutes. The counterfactual mix in Section 4.4.3 is weighted by the positive entries. Sample: GCSE cohorts 2002–2008, students taking exactly three A-levels.

Table 4A.12: IV estimates by predicted propensity to take A-level economics

| | (1) Bottom quartile | (2) Second quartile | (3) Third quartile | (4) Top quartile |
|---|---|---|---|---|
| **Panel A: First stage (availability → A-level econ)** | | | | |
| A-level econ available | 0.013*** | 0.033*** | 0.069*** | 0.121*** |
| | (0.0011) | (0.0017) | (0.0027) | (0.0040) |
| *F* | 139.9 | 360.1 | 645.8 | 910.1 |
| **Panel B: IV estimate (A-level econ → log earnings)** | | | | |
| Takes A-level economics | 0.183 | 0.169 | 0.052 | 0.152** |
| | (0.529) | (0.217) | (0.113) | (0.072) |
| School FE | ✓ | ✓ | ✓ | ✓ |
| Cohort FE | ✓ | ✓ | ✓ | ✓ |
| Individual controls | ✓ | ✓ | ✓ | ✓ |
| Observations | 97,253 | 97,915 | 93,740 | 79,352 |

*Notes:* The sample is divided into quartiles of the predicted propensity to take A-level economics, estimated from pre-treatment characteristics (GCSE attainment, sex, ethnicity, FSM status, region, and school-level covariates) using a probit model. The predicted probability uses only characteristics fixed before the school changed its provision. The IV specification instruments A-level economics take-up with school-cohort availability (linear controls). Standard errors clustered at the school level in parentheses. Sample: students in switcher schools, GCSE cohorts 2002–2008. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4A.13: Sensitivity of the main results to the availability threshold

| Threshold (students sitting exam) | First stage | First-stage $F$ | IV estimate |
|---|---|---|---|
| $\geq 1$ | 0.049 | 785 | 0.162 |
| | (0.0018) | | (0.077) |
| $\geq 2$ (baseline) | 0.057 | 843 | 0.135 |
| | (0.0020) | | (0.069) |
| $\geq 3$ | 0.060 | 806 | 0.172 |
| | (0.0021) | | (0.069) |
| $\geq 4$ | 0.060 | 740 | 0.216 |
| | (0.0022) | | (0.064) |
| $\geq 5$ | 0.061 | 737 | 0.256 |
| | (0.0022) | | (0.064) |

*Notes:* Each row re-estimates the first-stage and IV specifications of Table 4.3 under an alternative definition of availability: a school-cohort is coded as offering A-level economics if at least the stated number of students sit the A-level economics exam. The baseline definition in the main text requires two students, and that row reproduces Table 4.3. Standard errors in parentheses. For thresholds other than two, the first-stage $F$ is implied by the square of the ratio of the coefficient to its standard error, computed before rounding. The estimate rises as the threshold rises. A higher threshold counts a school as offering only when several students take the subject, so it picks out schools with settled provision. The sample is that of Table 4.3. The IV estimate is smallest at the two-student threshold used in the main text. The headline estimate is therefore the lowest of the set.

Table 4A.14: School-cohorts contributing at each event time

| Event time | Begin offering | Stop offering |
|---|---|---|
| $-3$ | 445 | 175 |
| $-2$ | 190 | 100 |
| $-1$ | 215 | 125 |
| 0 | 215 | 125 |
| 1 | 165 | 105 |
| 2 | 110 | 90 |
| 3 | 135 | 130 |

*Notes:* The table reports the number of school-cohort observations contributing to each event-time coefficient in Figures 4A.3 and 4A.4. The sample comprises switcher schools observed without gaps whose provision changes exactly once. Endpoints are binned, so $t = -3$ and $t = 3$ include all earlier and later cohorts respectively, and a school can contribute more than one cohort to these cells; at interior event times each school contributes one cohort. Counts are rounded to the nearest five in line with the disclosure rules of the secure research environment.

# Bibliography

Abdulkadiroğlu, A., Angrist, J. D., Dynarski, S. M., Kane, T. J., and Pathak, P. A. (2011). Accountability and flexibility in public schools: Evidence from boston's charters and pilots. *Quarterly Journal of Economics*, 126(2):699–748.

Advani, A., Sen, S., and Warwick, R. (2020). Ethnic diversity in UK economics. Briefing Note BN307, Institute for Fiscal Studies, London.

Allen, J., Gregory, A., Mikami, A., Lun, J., Hamre, B., and Pianta, R. (2013). Observations of effective teacher–student interactions in secondary school classrooms: Predicting student achievement with the classroom assessment scoring system—secondary. *School Psychology Review*, 42(1):76–98.

Altonji, J. G., Arcidiacono, P., and Maurel, A. (2016). The analysis of field choice in college and graduate school: Determinants and wage effects. *Handbook of the Economics of Education*, 5:305–396.

Angrist, J. D., Dynarski, S. M., Kane, T. J., Pathak, P. A., and Walters, C. R. (2010). Inputs and impacts in charter schools: Kipp lynn. *American Economic Review: Papers and Proceedings*, 100(2):239–243.

Angrist, J. D. and Fernández-Val, I. (2013). ExtrapoLATE-ing: External validity and overidentification in the LATE framework. In Acemoglu, D., Arellano, M., and Dekel, E., editors, *Advances in Economics and Econometrics: Theory and Applications, Tenth World Congress*, volume 3, pages 401–434. Cambridge University Press.

Angrist, J. D. and Lavy, V. (1999). Using maimonides' rule to estimate the effect of class size on scholastic achievement. *Quarterly Journal of Economics*, 114(2):533–575.

Avilova, T. and Goldin, C. (2018). What can uwe do for economics? In *AEA Papers and Proceedings*, volume 108, pages 186–190. American Economic Association.

Bank of England (2025). Bank of England teams up with The University of Manchester to improve access to economics in schools. News release, announcing Beynon, K. and Plaister, N., *Understanding Trends in the Study of Economics*, FFT Education Datalab report to the Bank of England.

Baumrind, D. (1971). Current patterns of parental authority. *Developmental Psychology*, 4(1, Pt. 2):1–103.

Bayer, A., Hoover, G. A., and Washington, E. (2020). How you can work to increase the presence and improve the experience of black, latinx, and native american people in the economics profession. *Journal of Economic Perspectives*, 34(3):193–219.

Bayer, A. and Rouse, C. E. (2016). Diversity in the economics profession: A new attack on an old problem. *Journal of Economic Perspectives*, 30(4):221–242.

Blandhol, C., Bonney, J., Mogstad, M., and Torgovitsky, A. (2026). When is TSLS actually LATE? *Review of Economic Studies*. Advance access, published online 14 May 2026.

Blau, F. D., Currie, J. M., Croson, R. T. A., and Ginther, D. K. (2010). Can mentoring help female assistant professors? interim results from a randomized trial. *American Economic Review*, 100(2):348–352.

Bleemer, Z. and Mehta, A. (2022). Will studying economics make you rich? A regression discontinuity analysis of the returns to college major. *American Economic Journal: Applied Economics*, 14(2):1–22.

Britton, J., Dearden, L., van der Erve, L., and Waltmann, B. (2020). The impact of undergraduate degrees on lifetime earnings. IFS Report R167, Institute for Fiscal Studies, London.

Britton, J., Ogden, K., Ornadel, N., and Waltmann, B. (2026). New estimates of the impact of undergraduate degrees on lifetime earnings. Ifs report, Institute for Fiscal Studies, London. Published by the Department for Education as Research Report RR1642. ISBN 978-1-83870-818-4.

Buckles, K. (2019). Fixing the leaky pipeline: Strategies for making economics work for women at every stage. *Journal of Economic Perspectives*, 33(1):43–60.

Chabrier, J., Cohodes, S., and Oreopoulos, P. (2016). What can we learn from charter school lotteries? *Journal of Economic Perspectives*, 30(3):57–84.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.

Chetty, R., Friedman, J. N., and Rockoff, J. E. (2014). Measuring the impacts of teachers ii: Teacher value-added and student outcomes in adulthood. *American Economic Review*, 104(9):2633–2679.

Cohen, J., McCabe, E. M., Michelli, N. M., and Pickeral, T. (2009). School climate: Research, policy, practice, and teacher education. *Teachers College Record*, 111(1):180–213.

Coleman, J. S., Campbell, E. Q., Hobson, C. J., McPartland, J., Mood, A. M., Weinfeld, F. D., and York, R. L. (1966). *Equality of Educational Opportunity*. U.S. Government Printing Office, Washington, DC.

Dahl, G. B., Rooth, D.-O., and Stenberg, A. (2023). High school majors and future earnings. *American Economic Journal: Applied Economics*, 15(1):351–382.

Dobbie, W. and Fryer, Roland G., J. (2013). Getting beneath the veil of effective schools: Evidence from new york city. *American Economic Journal: Applied Economics*, 5(4):28–60.

Edmonds, R. (1979). Effective schools for the urban poor. *Educational Leadership*, 37(1):15–24.

Eloundou, T., Manning, S., Mishkin, P., and Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702):1306–1308.

Epple, D., Romano, R., and Zimmer, R. (2016). Charter schools: A survey of research on their characteristics and effectiveness. In Hanushek, E. A., Machin, S., and Woessmann, L., editors, *Handbook of the Economics of Education, Vol. 5*, pages 139–208. Elsevier, Amsterdam.

Eyles, A., Hupkau, C., and Machin, S. (2016). Academies, charter and free schools: Do new school types deliver better outcomes? *Economic Policy*, 31(87):453–501.

Eyles, A., Machin, S., and Silva, O. (2018). Academies 2: The new batch – the changing nature of academy schools in england. *Fiscal Studies*, 39(1):121–158.

Fryer, Roland G., J. (2014). Injecting charter school best practices into traditional public schools: Evidence from field experiments. *Quarterly Journal of Economics*, 129(3):1355–1407.

Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3):535–574.

Gilardi, F., Alizadeh, M., and Kubli, M. (2023). Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120.

Gillen, B., Snowberg, E., and Yariv, L. (2019). Experimenting with measurement error: Techniques with applications to the Caltech Cohort Study. *Journal of Political Economy*, 127(4):1826–1863.

Goodman, J. (2019). The labor of division: Returns to compulsory high school math coursework. *Journal of Labor Economics*, 37(4):1141–1182.

Gorard, S. (2010). Serious doubts about school effectiveness. *British Educational Research Journal*, 36(5):745–766.

Gorard, S., Hordosy, R., and Siddiqui, N. (2013). How unstable are 'school effects' assessed by a value-added technique? *International Education Studies*, 6(1):1–9.

Gottfredson, G. D., Gottfredson, D. C., Payne, A. A., and Gottfredson, N. C. (2005). School climate predictors of school disorder: Results from a national study of delinquency prevention in schools. *Journal of Research in Crime and Delinquency*, 42(4):412–444.

Gregory, A., Cornell, D., and Fan, X. (2011). The relationship of school structure and support to suspension rates for black and white high school students. *American Educational Research Journal*, 48(4):904–934.

Gregory, A., Cornell, D., Fan, X., Sheras, P., Shih, T.-H., and Huang, F. (2010). Authoritative school discipline: High school practices associated with lower bullying and victimization. *Journal of Educational Psychology*, 102(2):483–496.

Hanushek, E. A. (1997). Assessing the effects of school resources on student performance: An update. *Educational Evaluation and Policy Analysis*, 19(2):141–164.

Hastings, J. S., Neilson, C. A., and Zimmerman, S. D. (2013). Are some degrees worth more than others? Evidence from college admission cutoffs in Chile. Working Paper 19241, National Bureau of Economic Research.

Hendren, N. and Sprung-Keyser, B. (2020). A unified welfare analysis of government policies. *Quarterly Journal of Economics*, 135(3):1209–1318.

Holmlund, H., McNally, S., and Viarengo, M. (2010). Does money matter for schools? *Economics of Education Review*, 29(6):1154–1164.

Hoy, W. K. and Miskel, C. G. (2001). *Educational Administration: Theory, Research, and Practice*. McGraw-Hill, New York, 6th edition.

Hsieh, C.-T., Hurst, E., Jones, C. I., and Klenow, P. J. (2019). The allocation of talent and us economic growth. *Econometrica*, 87(5):1439–1474.

Jenkins, A., Levačić, R., and Vignoles, A. (2006). Estimating the relationship between school resources and pupil attainment at gcse. Research Report RR727, Department for Education and Skills, London.

Joensen, J. S. and Nielsen, H. S. (2009). Is there a causal effect of high school math on labor market outcomes? *Journal of Human Resources*, 44(1):171–198.

Kirkebøen, L. J., Leuven, E., and Mogstad, M. (2016). Field of study, earnings, and self-selection. *Quarterly Journal of Economics*, 131(3):1057–1111.

Krueger, A. B. (1999). Experimental estimates of education production functions. *Quarterly Journal of Economics*, 114(2):497–532.

Krueger, A. B. and Whitmore, D. M. (2001). The effect of attending a small class in the early grades on college-test taking and middle school test results: Evidence from project star. *Economic Journal*, 111(468):1–28.

Lee, D. S., McCrary, J., Moreira, M. J., and Porter, J. (2022). Valid $t$-ratio inference for IV. *American Economic Review*, 112(10):3260–3290.

Levačić, R., Jenkins, A., Vignoles, A., Steele, F., and Allen, R. (2005). Estimating the relationship between school resources and pupil attainment at Key Stage 3. Research Report RR679, Department for Education and Skills, London.

Ludwig, J., Mullainathan, S., and Rambachan, A. (2026). Large language models: An applied econometric framework. *Annual Review of Economics*, 18:283–316.

Lundberg, S. and Stearns, J. (2019). Women in economics: Stalled progress. *Journal of Economic Perspectives*, 33(1):3–22.

Maccoby, E. E. and Martin, J. A. (1983). Socialization in the context of the family: Parent–child interaction. In Hetherington, E. M., editor, *Handbook of*

*Child Psychology, Vol. 4: Socialization, Personality and Social Development*, pages 1–101. Wiley, New York.

Machin, S., McNally, S., and Meghir, C. (2010). Resources and standards in urban schools. *Journal of Human Capital*, 4(4):365–393.

Machin, S. and Sandi, M. (2020). Autonomous schools and strategic pupil exclusion. *Economic Journal*, 130(625):125–159.

Machin, S. and Vernoit, J. (2011). Changing school autonomy: Academy schools and their introduction to england's education. CEE Discussion Paper 123, Centre for the Economics of Education, London School of Economics.

Majumder, M. A. (2016). The impact of parenting style on children's educational outcomes in the united states. *Journal of Family and Economic Issues*, 37(1):89–98.

McLean, D., Worth, J., and Smith, A. (2024). Teacher labour market in England: Annual report 2024. Technical report, National Foundation for Educational Research.

Murphy, R. and Machin, S. (2011). Improving the impact of teachers on pupil achievement in the uk: Interim findings. Technical report, The Sutton Trust, London.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Pellerin, L. A. (2005). Applying baumrind's parenting typology to high schools: Toward a middle-range theory of authoritative socialization. *Social Science Research*, 34(2):283–303.

Pianta, R. C., Hamre, B. K., and Mintz, S. L. (2012). *Classroom Assessment Scoring System (CLASS) Manual, Secondary*. Teachstone.

Porter, C. and Serra, D. (2020). Gender differences in the choice of major: The importance of female role models. *American Economic Journal: Applied Economics*, 12(3):226–254.

Rivkin, S. G., Hanushek, E. A., and Kain, J. F. (2005). Teachers, schools, and academic achievement. *Econometrica*, 73(2):417–458.

Rutter, M., Maughan, B., Mortimore, P., and Ouston, J. (1979). *Fifteen Thousand Hours: Secondary Schools and Their Effects on Children*. Open Books, London.

Sammons, P., Hillman, J., and Mortimore, P. (1995). *Key Characteristics of Effective Schools: A Review of School Effectiveness Research*. Institute of Education, University of London, London. Report commissioned by the Office for Standards in Education.

Sammons, P., Sylva, K., Melhuish, E., Siraj-Blatchford, I., Taggart, B., Barreau, S., and Grabbe, Y. (2008). The influence of school and teaching quality on children's progress in primary school. Research Report DCSF-RR028, Department for Children, Schools and Families, Nottingham.

Slater, H., Davies, N. M., and Burgess, S. (2012). Do teachers matter? measuring the variation in teacher effectiveness in england. *Oxford Bulletin of Economics and Statistics*, 74(5):629–645.

Smith, S., Advani, A., and Harvey, L. (2025). Going back to the high schools: Challenging stereotypes about economics. CEPR Discussion Paper 20483, Centre for Economic Policy Research.

Teddlie, C. and Reynolds, D., editors (2000). *The International Handbook of School Effectiveness Research*. Falmer Press, London.

Thapa, A., Cohen, J., Guffey, S., and Higgins-D'Alessandro, A. (2013). A review of school climate research. *Review of Educational Research*, 83(3):357–385.

Van Houtte, M. (2005). Climate or culture? a plea for conceptual clarity in school effectiveness research. *School Effectiveness and School Improvement*, 16(1):71–89.

Zhang, W., Wei, X., Ji, L., Chen, L., and Deater-Deckard, K. (2017). Reconsidering parenting in chinese culture: Subtypes, stability, and change of maternal parenting style during early adolescence. *Journal of Youth and Adolescence*, 46(5):1117–1136.